

Australian Household Consumption, Wealth and Credit Conditions: An Updated Single-Equation LIVES Estimate

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1 Australian Household Consumption, Wealth and Credit Conditions: An Updated Single-Equation LIVES Estimate

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1.1 Abstract

We estimate a single-equation Muellbauer–Williams LIVES consumption model for Australia over 1988Q3–2024Q4, extending Williams (2010) and Muellbauer and Williams (2012) by sixteen years of post-GFC data and applying contemporary identification methods. Wealth is disaggregated into housing, illiquid financial (equities plus superannuation), and net liquid assets (deposits net of total household debt), each entered as an asset/annualised-income ratio, and we adopt the direct (single-regression) forecast of the discounted future-income aggregate used in the Italian implementation of De Bonis, Liberati, Muellbauer and Rondinelli (2020). The cross-equation restriction $\gamma_{LA} + \gamma_{LOANS} = 0$ is accepted in every disaggregated specification and sample window we estimate, validating the net-liquid aggregation used in that Italian convention.

Our central methodological finding is that the *form* of the equation is decisive. A conventional disaggregated error-correction model that enters wealth as constant-marginal-propensity levels — the specification most of this literature, and an earlier draft of this paper, treated as the LIVES equation — is not in fact the LIVES equation. In Williams' framework credit conditions *scale* the housing-wealth effect: there is no classical housing wealth effect, and the marginal propensity is unlocked only as the credit-conditions index (CCI) rises. Estimating a standalone housing-wealth coefficient therefore tests a parameter the theory predicts to be approximately zero; reading its insignificance as a failed housing-wealth effect is a category error. When the equation is instead specified faithfully — housing wealth entering only through its CCI interaction, the autonomous-consumption CCI intercept restored, and illiquid financial assets combined — the error-correction and core wealth structure come alive (the faithful LIVES specification, Spec 11), estimated on $n = 146$ rather than the $n = 86$ to which the conventional credit term binds the model. We anchor the headline on the COVID-controlled estimates: the speed of adjustment is $\lambda \approx -0.25$ and tightly clustered across the pre-COVID, COVID-excluded and COVID-rich-dummy variants (-0.266 , -0.248 and -0.242 , with t -ratios of -4.8 to -6.7 ; `australia_spec11_variants.csv`), within about 13 per cent of Williams' published -0.286 ; the full-sample $\lambda = -0.448$ ($t = -3.6$) is inflated by the COVID quarters and we do not headline it. The credit-invariant wealth structure is right-signed and significant at 5 per cent in the full-sample and COVID-dummy variants, weakening to 10 per cent in the pre-COVID subsample: the structural net-liquid MPC is 0.060 (95 per cent CI $[0.022, 0.098]$) and the illiquid financial MPC 0.035 ($[0.012, 0.057]$), the latter spanning Williams' calibrated 0.022. The estimates are now precise enough to *reject* parts of Williams' profile rather than merely fail to distinguish them: his net-liquid MPC of 0.159 and his peak housing-collateral MPC of 0.0488 lie outside our intervals, while the housing-collateral channel itself is right-signed but insignificant (structural 0.0055, CI $[-0.010, 0.021]$). The permanent-income response is strong (OLS $+0.30$ to $+0.46$, $t \approx 4-6$), but its structural value of 1.0–1.1 exceeds the theoretical admissibility bound $\psi \leq 1 - \eta \approx 0.95$, a tension we disclose and show is not an artefact of the GFC learning-weight applied to the series. The conventional constant-MPC specification (Spec 6) delivers $\lambda = -0.239$ ($t = -2.6$) on a sample less than two-thirds the length. Much of the apparent weakness of single-equation LIVES estimates for Australia is thus a specification artefact, not an economic result. (Permanent income is the full-sample Italian direct-forecast measure; its non-causal construction and a causal real-time variant — $\lambda = -0.159$, with the permanent-income coefficient reversing sign — are discussed in §7.4.)

The LIVES *structure* transfers to Australia, but Williams' Australian *calibrations* do not. Imposing his calibrated permanent-income gearing ($\psi_0 = 0.20$, $\psi_1 = 0.93$) collapses the error-correction to $\lambda = -0.030$ ($t = -0.7$; the Williams-calibration-imposed specification, Spec 12, which flips sign pre-COVID), independently reproduced by Spec 10 ($\lambda = -0.048$): the Australian data freely estimate a permanent-income gearing several times Williams', and forcing his smaller value destroys the equilibrium. This reconciles a puzzle in our companion work, where a Wald test fails to reject Williams' joint calibration ($\chi^2(6) = 7.55$, $p = 0.27$) — the free estimates are too imprecise to reject his values jointly, yet imposing them still ruins the fit; low power is not the same as good fit. The credit-conditions *interactions* are themselves only weakly identified: the CCI-scaled regressors are 0.66–0.97 mutually correlated in absolute value on this sample (australia_cci_interaction_corr.csv) because each is approximately proportional to the latent index, so they cannot be separately estimated from a single equation. This is exactly why Williams calibrates and estimates a four-equation FIML system; on contemporary Australian data the single-equation calibration shortcut is empirically closed, leaving joint estimation and pre-1988 back-extension as the only routes to sharpen the credit channels.

The deployed CCI itself is disclosed in full rather than implied: only four of fifteen candidate knots survive the sign-prior reduction, all post-2007 (2007Q3, 2009Q1, 2019Q1, 2020Q2), so the index is identically zero from 1988 to mid-2007, plateaus at its peak over 2010–2018, and turns negative after 2019 (range -2.1 to 1 ; australia_cci_williams_path.png). The credit channels are therefore identified off roughly seventy post-2007 quarters, not the full $n = 146$, and the housing-collateral MPC implied by γ_1 -CCI is negative in the post-2019 regime — a property Williams' $[0, 0.8]$ index cannot produce and a caveat on any structural reading of the interactions. Placebo evidence is split and reported both ways: the literal Williams 4-knot spline sits at the placebo median (45th adjusted- R^2 percentile over 200 random-knot draws), but the *deployed* construction — the maximal candidate set under the same iterated sign-prior reduction used in estimation — beats 84 per cent of random draws (australia_williams_knot_placebo_deployed_verdict.csv), moderate support for institutional knot placement. A two-equation SUR of consumption and house prices delivers negligible cross-equation residual correlation ($\hat{\rho} = -0.013$) and essentially no efficiency gain, and a joint cross-equation CCI identification retains only one knot across all four equations. We read these findings as indicating that the structural identification Williams (2010) delivers comes from cross-equation parameter restrictions in his four-equation FIML system rather than from sample length, knot count, or sign-prior structure.

We assemble a back-extended master dataset to 1976Q3 — using a TRYM long-run house-price series, RBA D03 monetary aggregates, RBA D02 total credit, historical labour-force compilations, and documented aggregate and disaggregated wealth proxies anchored at 1988Q3, when ABS sectoral balance-sheet data begin — to test whether sample length is the binding constraint on tighter agreement with Williams. Refitting the disaggregated no-CCI specification (Spec 4) on the back-extended sample moves λ about 12 per cent toward Williams ($-0.182 \rightarrow -0.203$) but the net-liquid coefficient collapses toward zero — sample length is not the binding constraint. The remaining honest negatives are reported as substantive findings: the structural specifications beat a random walk with drift out of sample at the one-quarter horizon but lose at four and eight quarters; the automated selection screen (which caps $|\lambda|$ at 0.30) prefers a net-worth specification while the Bayesian information criterion and LIVES theory now both point to Spec 11; no specification passes the Engle–Granger cointegration

screen at MacKinnon critical values; and the full-sample permanent-income measure partially reverses under a causal real-time forecaster. The paper includes a structured robustness suite covering instrumental variables with first-stage diagnostics, joint SUR estimation, Chow tests at multiple break dates, the Drehmann (2017) amortising-mortgage adjustment, the Williams smoothed-step credit-conditions spline with maximal-GETS reduction, a Kalman state-space credit factor, sectional sign-prior alternatives, rolling-window estimation, out-of-sample forecast validation against random-walk and AR(1) benchmarks, and back-extension robustness on Spec 1 and Spec 4 – with the headline Spec 11 included in every exercise. A full reproducibility kit accompanies the paper, along with a multi-equation LIVES scaffold that lays the foundations for a companion paper.

1.2 1. Introduction

1.2.1 1.1 Motivation

Australian household consumption poses a set of policy and analytical questions that the standard New Keynesian DSGE workhorse – with its representative consumer, exogenous wealth process, and assumed-frictionless credit access – is structurally ill-equipped to answer. Among them: how much of the post-2008 moderation in consumption growth is attributable to the macroprudential tightening of 2014 and 2017, and how much to post-GFC household balance-sheet repair? How sensitive is consumption to housing wealth at different points in the credit cycle, when households are or are not able to extract home equity? Did the COVID shock and JobKeeper income support move permanent-income expectations, or only the short-run quarter? And how should a central bank think about the wealth channel of monetary policy when much of household wealth is housing wealth, mortgage debt is at near-historic levels relative to income, and the credit conditions in which households operate have evolved markedly since financial deregulation in the 1980s?

The Muellbauer–Williams "LIVES" framework – Latent Interactive Variable Equation System, in the flow-of-funds tradition of Tobin and of Duca, Muellbauer and Murphy (2013) – was developed to answer exactly these questions. It augments the standard credit-augmented life-cycle consumption function (Friedman 1957; Ando and Modigliani 1963; Tobin and Dolde 1971) with three features. First, wealth is disaggregated into net liquid assets (NLA, liquid assets less total household debt), illiquid financial assets (IFA, equities plus superannuation) and housing assets (HA), each entered as a ratio to annualised income and each carrying a different marginal propensity to consume (Backus and Purvis 1980). Second, a latent credit-conditions index (CCI) interacts with the long-run equilibrium so that key channels are switched on only as credit eases. Third – and central to Williams' (2010) Australian estimate – the latent CCI is identified *jointly* across a four-equation system (consumption, house prices, mortgage stock and home-equity withdrawal) under cross-equation parameter restrictions estimated by full-information maximum likelihood (FIML).

The theoretically decisive feature for this paper is how housing enters. In the canonical LIVES equation (Williams' Eq (7); his Table 1, column 1) **there is no classical housing-wealth effect**. Housing wealth appears *only* through its interaction with credit conditions, as $\gamma_1 \cdot \text{CCI} \cdot (\text{HA}/4y)$. The housing marginal propensity to consume is therefore zero when credit is fully constrained ($\text{CCI} = 0$) and is unlocked as credit eases, reaching its peak when CCI is at its maximum. By contrast, illiquid financial and net liquid assets enter as plain, credit-

invariant marginal propensities (γ_2 and γ_3). This functional form has an immediate implication for empirical work: a specification that enters a standalone housing-wealth ratio (a credit-invariant ha_y term) is *not* the LIVES equation, and reading an insignificant standalone housing coefficient as evidence of a failed housing-wealth effect is a **category error** — the theory predicts that coefficient to be approximately zero in the absence of the CCI interaction.

Williams (2010) applied the LIVES framework to Australia for 1978–2008, producing the canonical Australian estimate. Sixteen years of post-GFC and post-COVID data have since accumulated, and a contemporary central-bank-quality update is warranted. This paper provides that update. Its pivot, set out below, is that an earlier reading of single-equation Australian LIVES estimates as "weak" was in substantial part a specification artefact: when the equation is estimated in its *faithful* form — housing entered only through the credit interaction, the autonomous CCI intercept restored, and illiquid financial assets combined — the error-correction and core wealth structure come alive in a way the conventional constant-MPC disaggregated specification cannot deliver.

1.2.2 1.2 Contribution

This paper makes five contributions to the Australian household consumption literature.

(i) A faithful single-equation LIVES estimate, with the form correction as the lead methodological result. We show that the functional *form* of the LIVES equation, not the fit of any one regressor, is what identifies it. Estimating the faithful form — housing entered only through ha_x_cci , the de-meaned CCI·(HA/4y) interaction; the autonomous-consumption CCI intercept ($cci_williams$, the ζ_c loading) restored; and illiquid financial assets combined into a single $ilfa_y = eq_y + super_y$ term — yields our headline specification, Spec 11. It recovers a correctly signed and significant error-correction speed and a right-signed wealth structure (§1.3). The same data fitted to the conventional constant-MPC disaggregated error-correction model (Spec 6), which carries plain $ha_y / eq_y / super_y / nla_y$ terms with no credit scaling and admits CCI only as a short-run regressor, delivers an insignificant speed of adjustment. The contrast is one of theoretical form, not of estimation luck.

(ii) The finding that the LIVES structure transfers to Australia but Williams' Australian calibrations do not. Imposing Williams' Australian permanent-income gearing ($\psi_0 = 0.20$, $\psi_1 = 0.93$) and his illiquid financial MPC ($\gamma_{IFA} = 0.022$) — the Williams-calibration-imposed specification, Spec 12 — *collapses* the equilibrium: the speed of adjustment falls to $\lambda = -0.030$ ($t = -0.74$, full sample), statistically indistinguishable from zero, and flips sign pre-COVID. An independent calibration route (Spec 10, Williams-prior) reproduces the collapse ($\lambda = -0.048$, $t = -0.78$). The reason is that Australia freely estimates a structural permanent-income gearing of order one — several times Williams' 0.20 — so forcing his value breaks the long-run fixed point. This reconciles the companion paper's Wald non-rejection of Williams' joint calibration ($\chi^2(6) = 7.55$, $p = 0.27$): the free Spec 6 estimates are too imprecise to *reject* Williams' values jointly — although the sharper Spec 11 intervals now reject his net-liquid and peak-housing magnitudes individually (§7.3.1) — yet imposing his calibration nonetheless wrecks the fit. Low power is not the same as good fit.

(iii) An interaction-collinearity diagnosis explaining why Williams needs FIML. The CCI-interacted regressors that constitute the credit channels have absolute pairwise correlations between roughly 0.66 and 0.97 on the post-1988 Australian sample (*australia_cci_interaction_corr.csv*), because each is approximately proportional to the same

latent CCI — which itself has no variation before 2007Q3 (§5.1.1), so the credit channels are identified off roughly seventy quarters. They cannot, therefore, be separately free-estimated off a single equation. This is the structural reason Williams identifies the credit channels through cross-equation FIML restrictions rather than within the consumption equation alone. The corollary is that single-equation calibration is empirically closed: sharpening the credit channels requires either the four-equation FIML system or a longer (pre-1988) sample that spans the financial-liberalisation episode.

(iv) A back-extended master dataset (1976Q3–2024Q4, $n = 194$ quarters) and a direct sample-length test. We construct documented growth-rate splices for house prices (Treasury TRYM historical compilation), the M3 monetary aggregate (RBA D03), total credit (RBA D02) and labour force, with disaggregated wealth proxies anchored at 1988Q3 — the quarter in which ABS sectoral household balance-sheet data begin. Because the financial-liberalisation episode that identifies the credit channels largely predates 1988Q3, this back-extension is the natural route to longer credit variation. Refitting the disaggregated no-CCI specification (Spec 4) on the extended sample moves λ about 12 per cent toward Williams (-0.182 to -0.203), but individual wealth coefficients shrink rather than strengthen and the net liquid MPC collapses — so sample length is *not* the binding constraint on tighter agreement with Williams.

(v) A structured robustness and placebo suite, and a set of honest negative results reported as substantive findings. Mirroring the De Bonis, Liberati, Muellbauer and Rondinelli (2020) Italian methodology, we run instrumental variables, a Zellner SUR, multi-window Chow and Bai–Perron break tests, the Drehmann amortisation adjustment, an AR/Italy direct-forecast permanent-income comparison with a real-time column, a permanent-income filter sensitivity grid, rolling-window estimation, and out-of-sample validation against random-walk and AR(1) benchmarks. We retain — and report prominently, without apology — the negative results that a freely estimated single-equation framework permits: the literal Williams 4-knot spline sits at the placebo median, although the *deployed* maximal-GETS construction beats 84 per cent of random draws under its own iterated reduction protocol; the SUR delivers essentially no efficiency gain (negligible cross-equation residual correlation); the structural specifications beat a random walk with drift out of sample at one quarter but lose at four and eight; the automated selection screen prefers a net-worth specification while the BIC and LIVES theory now both point to the faithful Spec 11; no specification passes the Engle–Granger cointegration screen; and the full-sample permanent-income measure is partly reversed under a causal real-time projection. Each negative is diagnostic: it points to why FIML and back-extension are the routes forward.

1.2.3 1.3 Headline result

The faithful LIVES specification (Spec 11) is the headline, and we anchor it on the COVID-controlled estimates. The speed of adjustment is tightly clustered at $\lambda \approx -0.25$ across the three COVID-robust variants — -0.266 ($t = -4.85$) on the pre-COVID sample (1988Q3–2019Q4, $n = 126$), -0.248 ($t = -6.66$) dropping 2020Q1–2021Q4, and -0.242 ($t = -6.25$) with quarterly 2020–21 dummies — within about 13 per cent of Williams' (2010) published -0.286 (his ϕ_c). The full-sample estimate (1988Q3–2024Q4, $n = 146$) is $\lambda = -0.448$ (Newey–West $t = -3.57$); it is inflated by the COVID quarters — the three pulse dummies the full-sample specification carries are not sufficient — and it fails the upper-bound screen on $|\lambda|$ in the selection rubric (§6), so we report it but do not headline it. The full Spec 11 coefficient vector

under all four sample treatments is committed (australia_spec11_variants.csv) so this choice is checkable. [Source: australia_spec11_variants.csv; australia_lambda_robustness.csv.]

The credit-invariant wealth structure is right-signed and significant at 5 per cent in the full-sample and COVID-dummy variants, weakening to 10 per cent in the pre-COVID subsample. The net liquid MPC is $\mathbf{nla_y = +0.027}$ ($t = 3.75$) on the full sample, $+0.017$ ($t = 2.11$) with the COVID quarters dropped and $+0.014$ ($t = 3.10$) under quarterly COVID dummies, implying a structural MPC $\gamma_3 = \text{OLS}/|\lambda|$ of about **0.060** with a 95 per cent delta-method interval of $[0.022, 0.098]$. The illiquid financial MPC is $\mathbf{ilfa_y = +0.015}$ ($t = 3.09$) full-sample and significant at 5 per cent under the COVID-dummy variants, weakening to 10 per cent on the pre-COVID subsample ($t = 1.74$), implying structural $\gamma_2 \approx \mathbf{0.035}$ $[0.012, 0.057]$ – an interval that spans Williams' calibrated 0.022. The housing-collateral channel enters with the right sign but is not significant in any variant: $\mathbf{ha_x_cci = +0.0025}$ ($t = 0.71$ full sample), implying a structural peak housing-collateral MPC of $\gamma_1 \approx 0.0055$ $[-0.010, 0.021]$. [Source: australia_spec11_variants.csv; australia_gamma_inference.csv.]

The gamma-inference exercise now cuts both ways, which is a substantive upgrade on an earlier draft in which every comparison was vacuous. Spec 11's intervals are tight enough to *reject* parts of Williams' profile: his net-liquid MPC (0.159) lies outside $[0.022, 0.098]$, and his peak housing MPC (0.0488) lies outside $[-0.010, 0.021]$; his illiquid financial MPC (0.022) is comfortably inside ours. The honest reading is agreement on the *form* and on the illiquid-financial channel, disagreement on magnitudes for the net-liquid channel, and an underpowered housing-collateral channel whose interval contains zero. [Source: australia_gamma_inference.csv.]

Permanent income enters strongly: $\mathbf{ln_yp_over_y = +0.459}$ ($t = 4.04$) on the full sample and $+0.298$ ($t = 5.81$) pre-COVID. Applying the paper's own structural-recovery rule (§4.2), however, the implied gearing is $\text{OLS}/|\lambda| \approx \mathbf{1.02 \text{ full-sample and } 1.12\text{--}1.13 \text{ in the COVID-controlled variants} \text{ -- above the theoretical admissibility bound } \psi \leq 1 - \eta \approx 0.95}$. We flag this openly rather than re-scale it away: the breach is not an artefact of the GFC learning-weight applied to the permanent-income series (re-estimating without the ogive gives $\lambda = -0.574$ with structural gearing ≈ 1.05 ; australia_spec11_ogive_robustness.csv), and §7.0 discusses the candidate explanations, including remaining COVID leverage and the non-causal construction of the measure itself. Comparisons with Williams' $\psi_0 = 0.20$ must be made on a consistent scale: the correct statements are "structural $\approx 1.0\text{--}1.1$ versus 0.20" or "OLS $+0.46$ versus $\phi \cdot \psi_0 \approx 0.057$ ", not the OLS-to-structural mix an earlier draft used. [Source: australia_spec11_variants.csv; australia_spec11_ogive_robustness.csv.]

The contrast with the calibration route is sharp. Imposing Williams' Australian calibrations (Spec 12: $\psi_0 = 0.20$, $\psi_1 = 0.93$, $\gamma_{\text{IFA}} = 0.022$) collapses the error-correction term to $\lambda = -0.030$ ($t = -0.74$) on the full sample and flips its sign on the pre-COVID sample ($\lambda = +0.041$, $t = 2.03$, significant at 5 per cent); the independent Williams-prior route (Spec 10) reproduces the collapse at $\lambda = -0.048$ ($t = -0.78$). The LIVES *structure* transfers to Australia; Williams' Australian *calibrations* do not. Williams' rate, affordability and autonomous-consumption loadings cannot even be imposed at their published magnitudes – his raw rate loading ($\alpha_r = -0.871$) is roughly thirty times too large on the repository's percent real-rate by unit-normalised-CCI scaling, and diverges the long-run fixed point. [Source: australia_all_results.csv; australia_lambda_robustness.csv.]

By contrast, the conventional constant-MPC disaggregated ECM (Spec 6) — the specification an earlier draft, and prior Australian work, treated as the LIVES equation — delivers $\lambda = -0.239$ ($t = -2.55$) on the $n = 86$ sample to which its credit term binds it, with an insignificant standalone housing coefficient ($ha_y = +0.0022$, $t = 0.30$). We retain Spec 6 as the conventional baseline against which the faithful form is the alternative, but we no longer read its insignificant standalone ha_y as a failed housing-wealth effect: under LIVES theory that coefficient is expected to be approximately zero absent the credit interaction. [Source: `australia_all_results.csv`.]

Finally, the credit-conditions identification limits are reported as findings rather than hidden, starting with the deployed index itself: only four of fifteen candidate knots survive, all post-2007, so the CCI is identically zero from 1988 to mid-2007 and negative after 2019 (§5.1.1 and `australia_cci_williams_path.png`) — the credit channels are identified off roughly seventy quarters, not the nominal $n = 146$. The placebo evidence is split: the literal Williams 4-knot sits at the placebo median (45th adjusted- R^2 percentile), while the deployed maximal-GETS construction under its own iterated reduction protocol beats 84 per cent of random draws — moderate support, short of strong identification. The CCI-scaled regressors are mutually collinear ($|\rho| 0.66-0.97$; `australia_cci_interaction_corr.csv`) and so cannot be separately identified off one equation; the SUR of consumption and house prices shows negligible residual correlation ($\hat{\rho} = -0.013$) and no efficiency gain; and the structural specifications beat a random walk with drift out of sample at one quarter but lose at four and eight. We do not promote the free-interaction specification (Spec 8, $\lambda = -0.458$) as a credit-conditions success: it re-allocates identification across regressors rather than closing the gap with Williams. [Source: `australia_williams_knot_placebo_verdict.csv`; `australia_williams_knot_placebo_deployed_verdict.csv`; `australia_oos_rmse.csv`; `australia_all_results.csv`.]

1.2.4 1.4 Roadmap

Section 2 surveys the LIVES literature, the Australian aggregate-wealth-effect literature and permanent-income measurement, and frames the LIVES claim that there is no classical housing-wealth effect — so that standalone-wealth specifications mis-test the theory. Section 3 documents data construction, including the disaggregated wealth ratios, the Italian direct-forecast permanent-income measure with its look-ahead versus real-time caveat, and the 1976Q3 back-extension sources and proxies. Section 4 presents the canonical LIVES equation (Eq (7)) and distinguishes the faithful form (Spec 11) from the conventional constant-MPC ECM (Spec 6) as a matter of theory. Section 5 develops the identification of credit conditions, including the interaction-collinearity result, the placebo battery, the near-zero SUR residual correlation and the joint knot-survival test. Section 6 sets out the full specification ladder (Specs 1–12, now including the faithful Spec 11 and the calibration-imposed Spec 12) and the selection rubric, and reports the selector divergence honestly. Section 7 presents the faithful-LIVES headline results and the calibration-collapse finding, with Spec 6 reframed as the conventional baseline. Section 8 runs the structured robustness suite, preserving the honest negatives. Section 9 reconciles our estimates with Williams (2010, 2012) around the structure-transfers / calibrations-don't thesis. Section 10 presents the long-run decomposition and policy implications, based on the faithful Spec 11 channels. Section 11 concludes.

1.3 2. Literature review

This paper sits at the intersection of three literatures: the Muellbauer-Williams "LIVES" tradition that integrates wealth, credit conditions and life-cycle behaviour into consumption equations; the Australian empirical consumption literature, which has historically focused on aggregate or constant-marginal-propensity wealth effects without the explicit credit-conditions machinery; and the small but growing body of work that disciplines permanent-income measurement using forecasting approaches more robust than the standard AR(p) recipe. We review each in turn before placing the present contribution.

The thread that runs through the review, and that anticipates the paper's central finding, is a point about functional form. The LIVES framework makes a specific theoretical claim — that there is *no classical housing wealth effect*: the marginal propensity to consume out of housing wealth is zero when credit conditions are tight and is unlocked only as credit conditions ease. A consumption equation that enters housing wealth as a standalone level term, with a constant marginal propensity, does not test this claim — it tests a different, incompatible one. Reading an insignificant standalone housing-wealth coefficient as evidence of "no housing wealth effect" is therefore a *category error*: theory predicts that coefficient to be near zero in the absence of the credit-conditions interaction. This distinction — between the *faithful* LIVES form, in which housing enters only through its interaction with the credit-conditions index, and the *conventional constant-MPC* disaggregated error-correction model that prior work and an earlier draft of this paper treated as the LIVES equation — is the organising idea of the literature that follows and the lead result of the paper (§7.0).

1.3.1 2.1 Theoretical foundations

The empirical specification estimated below has its origins in the Davidson-Hendry-Srba-Yeo (1978) error-correction consumption function and the canonical permanent-income hypothesis (PIH) of Friedman (1957) and Hall (1978). The DHSY approach embeds a long-run cointegrating relationship between consumption, income and wealth into a short-run dynamic equation in growth rates, with an error-correction term governing the speed at which the consumption ratio reverts to its long-run determinants. Engle and Granger (1987) provide the formal econometric framework, with Hendry and Krolzig (2005) and Doornik (2009) refining the general-to-specific reduction methodology that disciplines the choice of short-run regressors.

A long line of empirical work has documented systematic departures from strict PIH behaviour. Campbell and Mankiw (1989, 1991) show that roughly half of US consumption tracks current rather than permanent income, consistent with rule-of-thumb behaviour or liquidity constraints. Carroll and Kimball (1996) establish the concavity of the consumption function under prudence, providing a microeconomic basis for differential marginal propensities to consume across wealth classes. Carroll (2001) and Deaton (1992) develop the buffer-stock interpretation that motivates the distinction between liquid and illiquid wealth in the Muellbauer specifications used here. The empirical implication — that wealth components with different liquidity, transactions costs, and ownership concentration should enter the consumption equation with different marginal propensities — is the cornerstone of the disaggregated specifications reported in §7. In the faithful LIVES form, this implication is sharpened further: liquidity matters not only through a level distinction but through how each component interacts with the prevailing state of credit. Net liquid assets and illiquid financial assets enter

as plain marginal propensities, but housing wealth is collateral whose consumability depends on the borrowing technology — and so enters only through its credit-conditions interaction.

1.3.2 2.2 The LIVES framework

Muellbauer (2007), originally prepared for the Federal Reserve Bank of Kansas City Jackson Hole symposium, sets out the integrated approach to modelling consumption with wealth, housing collateral and credit conditions. The paper's central insight is that credit-market liberalisation interacts with the wealth and interest-rate channels: when credit is tight, housing wealth collateralises borrowing only weakly and the down-payment hurdle dampens consumption; when credit is loose, housing wealth and current consumption become more tightly linked. This insight requires a credit-conditions index (CCI) entering the long-run relationship not as a single additive term but as a multiplicative shifter across several channels simultaneously.

The functional form is decisive, and worth stating precisely because it is what distinguishes the framework from a conventional wealth error-correction model. In the canonical Williams (2010) consumption equation — his Eq (7), reproduced in §4 — the credit-conditions index multiplies *six* channels jointly inside the long-run bracket: the autonomous-consumption intercept ($\zeta_c \cdot \text{CCI}$), the real-interest-rate term ($\alpha_1 \cdot r \cdot \text{CCI}$), the housing-collateral term ($\gamma_1 \cdot \text{CCI} \cdot (\text{HA}/4y)$), the permanent-income gearing ($\psi(\text{CCI}) \cdot \log(y^p/y)$, with $\psi(\text{CCI}) = \psi_0 + \psi_1 \cdot \text{CCI}$), and the affordability term ($\alpha_4 \cdot (1 - \omega \cdot \text{CCI}) \cdot \log(p^h/y)$, with the affordability multiplier $\omega = 1.2$). Critically, housing wealth enters *only* through $\gamma_1 \cdot \text{CCI} \cdot (\text{HA}/4y)$ — there is no standalone housing wealth level in the long-run relationship. The housing marginal propensity is therefore zero at $\text{CCI} = 0$ and rises as credit conditions ease, reaching Williams' implied peak housing MPC of 0.0488 (Williams' Table 1; `williams_comparison.csv`). The illiquid financial asset term (IFA = equities + superannuation) and the net liquid asset term (NLA = liquid assets – total household debt) enter as plain marginal propensities (Williams' calibrated $\gamma_{\text{IFA}} = 0.022$ and estimated $\gamma_{\text{NLA}} = 0.159$ respectively). Income enters with a *unit* coefficient, so that the error-correction term is the (log) consumption-to-income ratio, and all wealth terms enter as asset/annualised-income ratios, $x_{\{t-1\}}/4y$.

It follows that a specification entering housing wealth as a standalone constant-MPC level term — alongside equities, superannuation and net liquid assets, with credit conditions relegated to a short-run dynamic regressor — is not the LIVES equation at all, but a generic wealth error-correction model. Under the LIVES theory, the standalone housing coefficient in such a model is expected to be near zero, because the operative channel (the CCI interaction) has been omitted. This is the sense in which reading an insignificant standalone housing coefficient as a failed housing wealth effect is a category error, and it is the reason §7 distinguishes the *faithful* LIVES specification from the *conventional constant-MPC* baseline as a matter of theory, not fit.

The framework was operationalised in a series of country studies. Aron, Duca, Muellbauer, Murata and Murphy (2012) jointly estimate the framework on Japan, the United Kingdom and the United States, finding consistent positive long-run housing wealth effects in the UK and US (where home equity withdrawal is institutionally available) and a much smaller effect in Japan (where it is not) — precisely the cross-country pattern that the credit-interaction form predicts, since the housing channel should be muted where the collateral-borrowing technology is weak. Duca, Muellbauer and Murphy (2010) apply the framework to the global financial crisis, demonstrating that the abrupt tightening in CCI from 2007 quantitatively

rationalises the sharp consumption pullback observed in heavy-MEW economies. Duca and Muellbauer (2013), in *European Economy* Discussion Paper 14, formalise what they label the "LIVES" approach — the Latent Interactive Variable Equation System — emphasising the joint determination of consumption, house prices, mortgage debt and home equity withdrawal in a four-equation system identified by common factors and cross-equation sign restrictions.

Two parallel implementations frame the present paper. De Bonis, Liberati, Muellbauer and Rondinelli (2020) estimate a single-equation Italian adaptation that imposes the cross-equation restriction $\gamma_{LA} + \gamma_{LOANS} = 0$ (deposits and household debt enter with equal-and-opposite coefficients, so that net liquid assets is the operative quantity), adopts a direct single-regression forecast of the discounted future-income aggregate as its permanent-income measure (their Appendix A.2), applies a Drehmann (2017) amortising-mortgage adjustment to the real mortgage rate, and validates the single-equation OLS estimator against joint SUR. Chauvin and Muellbauer (2018), in the Banque de France working-paper series with an associated online complement, undertake a similar France adaptation, with particular attention to the institutional differences — limited home equity withdrawal, a larger social housing sector — that shape the housing-wealth channel. Both papers explicitly take Williams' Australian work as a methodological precedent; the present paper closes the loop by applying the Italian methodology back to the Australian data on which the original LIVES Australia estimation was performed.

1.3.3 2.3 The Australian application and prior Australian consumption work

The Australian application was developed in two companion papers by David Williams during his Oxford doctorate. Williams (2009), Oxford Economics Series Working Paper 432, focuses on the house-price equation and its identification under financial liberalisation; the paper develops the spline-based credit-conditions index from STAMP (Koopman et al. 2000) unobserved-components estimation, anchors the index at the four institutional turning points 1979/1992/1998/2007, and constructs the non-property household disposable income measure that we replicate in §3. Williams (2010), Oxford Economics Series Working Paper 492, estimates the four-equation LIVES system (consumption, house prices, mortgage stock, home equity withdrawal) jointly by full-information maximum likelihood (FIML) on Australian data 1978Q1–2008Q2, with the house-price loading on credit conditions normalised ($\zeta_h = 1$) to fix the scale of the latent index. The published version of this work, Muellbauer and Williams (2012) "Credit conditions and the real economy: the elephant in the room", appears as the lead chapter of *BIS Papers* No. 64 and constitutes our primary benchmark.

Australian consumption modelling outside the LIVES tradition has a long history, almost all of it in the standalone-wealth-effect form that the LIVES theory regards as a mis-test of the housing channel. Tan and Voss (2000), in RBA Research Discussion Paper 2000-09, estimate aggregate-wealth effects on Australian consumption using ABS National Accounts and RBA balance-sheet data, finding significant positive effects of both housing and financial wealth. Dvornak and Kohler (2003), in RBA RDP 2003-07, use a state-level panel to identify wealth effects from cross-state variation, finding larger marginal propensities to consume out of stock-market wealth than out of housing wealth, in apparent contrast to the time-series evidence; their findings are partly reconciled by the Muellbauer-Williams framework once credit-conditions interactions are introduced, since the unconditional housing MPC averages over tight- and loose-credit regimes and so understates the loose-credit collateral channel.

The methodological point is that none of these specifications interacts housing wealth with credit conditions, and so each estimates an unconditional housing MPC whose magnitude the LIVES form predicts will be small and unstable — a property to keep in mind when comparing coefficient magnitudes across studies.

The Reserve Bank of Australia's macroeconometric model MARTIN, introduced in Cusbert and Kendall (2018) in the RBA Bulletin and documented in Ballantyne et al. (2019) in RBA RDP 2019-07, includes a household consumption block that incorporates wealth effects and credit conditions in a more reduced-form way than the LIVES specification. The MARTIN consumption equation imposes calibrated elasticities for several channels rather than estimating the full long-run cointegrating vector, and abstracts from the explicit CCI spline; its calibrated net-wealth elasticity is 0.17 (`martin_nesting.csv`). The present paper complements MARTIN by providing a freely estimated benchmark against which calibrated coefficients can be evaluated — our unrestricted net-wealth elasticity is 0.1155 (`martin_nesting.csv`), with long-run homogeneity rejected ($\chi^2 = 16.41$, $p = 0.0001$) — and by surfacing the identification choices that drive the estimated speed of adjustment. As §10 discusses, our wealth-elasticity estimate is too imprecise to discipline MARTIN's calibration; the appropriate reading is to import the qualitative structure rather than the point estimates.

A separate strand of Australian work has examined the cyclical co-movement of consumption with credit and housing conditions through reduced-form approaches: Bayesian VAR frameworks, RBA Bulletin analyses of mortgage payments and household debt burdens, and the housing-leverage and household-balance-sheet treatments in the broader RBA Research Discussion Paper series. These approaches identify shorter-run dynamics and stress-test sensitivities but do not deliver the long-run cointegrating vector that the LIVES specification produces, and so are complementary to the present analysis rather than substitutes for it.

1.3.4 2.4 Identification of credit conditions

The credit-conditions index is the most contested ingredient in the LIVES framework, and the heart of the identification problem this paper documents. Williams (2010) constructs CCI as a latent variable identified by a spline of smoothed-step dummies (SDMMA series — five-quarter moving averages of four-quarter moving averages of step dummies), $\Sigma a_s \cdot \text{SDMMA}_s$, at four institutional turning points: 1979 (Campbell Committee, removal of interest-rate ceilings on bank deposits), 1992 (banking distress and the entry of the first mortgage originator, Aussie Home Loans), 1998 (the rise of non-bank financial institutions and securitisation), and 2007 (the global financial crisis tightening). The institutional chronology of Australian financial deregulation underpinning these choices is documented in Battellino and McMillan (1989) and Edey and Gray (1996); Bayoumi (1993) provides a cross-country analysis of the consumption response to financial liberalisation, including Australia, that quantitatively validates a structural CCI shift in the early 1980s.

The decisive feature of Williams' identification is that it is *joint*. In the system-estimation context of Muellbauer and Williams (2012), each spline coefficient is identified by being a *common factor* across the four equations: the same CCI value enters consumption, house prices, the mortgage stock and home equity withdrawal, with different loadings, and the house-price loading is normalised ($\zeta_h = 1$) to fix the index scale. This common-factor, FIML identification is the central methodological contribution of the LIVES family — and, as this paper argues, a structural necessity rather than a stylistic preference. The reason is collinear-

ity. The six CCI-interacted regressors are each approximately proportional to CCI, and on the post-1988 Australian sample they are 0.66–0.97 mutually correlated in absolute value (`australia_cci_interaction_corr.csv`; §5.5). Six near-collinear interaction regressors cannot be separately freely estimated off a single equation; in single-equation OLS the spline can be proxied or calibrated, but the individual credit channels cannot be jointly identified. This is the structural reason Williams uses FIML, and it is why this paper treats four-equation FIML and pre-1988 back-extension — not single-equation refinement — as the only routes to sharpen the credit channels. We adopt the spline approach under sign-prior restrictions enforced by general-to-specific reduction (Hendry-Krolzig 2005) as a single-equation robustness exercise (§5), but with the understanding that single-equation calibration is empirically closed.

The alternative, observable proxy for CCI — the ratio of housing credit flow to disposable income, in logs — has the advantage of being measured directly but the disadvantage of being available only from the early 2000s (the ABS 5601.0 lending series begins 2002Q3), after the most informative deregulation episodes have already occurred. This timing problem is fundamental rather than incidental: ABS sectoral balance-sheet data begin only in 1988Q3, so the financial-liberalisation episode that identifies the credit channels largely predates the data on which the consumption equation can be estimated. We use the observable proxy as a short-run regressor in the conventional-baseline specification, and the spline approach for the long-run identification in the credit-interaction robustness specification.

A separate identification approach uses survey-based credit-conditions indices. The Federal Reserve's Senior Loan Officer Opinion Survey is the prototype; the European Central Bank's Bank Lending Survey serves the same function in the euro area. The Reserve Bank of Australia operates a qualitative Liaison Programme but does not publish a long-running numerical index of credit conditions for households. The unavailability of such a survey-based Australian series is the practical reason for adopting Williams' spline approach in the LIVES tradition.

1.3.5 2.5 Permanent-income forecasting

The empirical operationalisation of permanent income — the discounted expected weighted average of expected future income — requires either explicit forecasts or a parametric assumption about the income process. The standard practice in the consumption literature has been to assume an AR(p) process for log income, fit it on the available sample, and aggregate the multi-step-ahead forecasts using exponentially declining weights with discount factor δ . This recipe descends directly from the PIH literature of Hall (1978) and the consumption-Euler-equation tests of Campbell and Mankiw (1989). In the LIVES form, permanent income enters with a credit-conditions-dependent gearing, $\psi(\text{CCI}) = \psi_0 + \psi_1 \cdot \text{CCI}$, which Williams calibrates rather than estimates ($\psi_0 = 0.20$, $\psi_1 = 0.93$); §4 and §7 take up whether that calibration transfers to Australia.

An alternative to the AR(p) recipe is to forecast the discounted aggregate *directly*: compute, for each quarter whose forecast horizon is observable, the discounted weighted average of realised future log income, and regress that pre-aggregated target on predictors observable at time t in a single equation. This *direct forecast* of the horizon-aggregate — the method of De Bonis, Liberati, Muellbauer and Rondinelli (2020, Appendix A.2), and related in spirit to (though distinct from) Jordà's (2005) direct multi-horizon projections, which fit one regression per horizon — sidesteps the compounding of AR misspecification across forecast horizons to which the standard recipe is vulnerable, and admits a richer predictor set than is feasible

in a parsimonious AR(p). (An earlier draft of this paper labelled the method a "Jordà local projection"; we retain the De Bonis et al. attribution throughout, since the implementation is a single full-sample regression of the pre-aggregated discounted target, not a per-horizon projection.)

De Bonis et al. (2020) adopt this direct-forecast approach for their Italian permanent-income series, reporting that the choice "captures much of the slow-down of permanent income growth in the early 1990s" — a structural feature of Italian growth that the AR-based forecaster missed. The Italian forecasting regression includes a $\log(\text{labour_force} / \text{population})$ predictor, which slowly trends with demographic change and is a natural input to long-horizon income forecasting. We adopt the same predictor in our Italy-style permanent-income helper, with a discounted ($\eta = 0.05$, $k = 40$ quarters) weighted average of expected future income (§3, §4), and find a quantitatively similar role: the permanent-income series implied by the direct forecast in Australia diverges materially from the AR-based series in the early 1990s and after the 2008 GFC. Substantively, the implied long-run coefficient on $\log(y^p/y)$ moves from slightly negative and economically negligible under the AR forecaster — the "Australian permanent-income puzzle" — to positive under the full-sample direct-forecast *measure*. As §8 documents, this reversal reflects the measure's full-sample (non-causal, look-ahead) construction and does not survive a causal real-time projection, which sharply shrinks the speed of adjustment (see §7.4 and `australia_pi_realtime_robustness.csv`). The look-ahead versus real-time distinction is therefore flagged as a measurement caveat throughout: the full-sample permanent-income measure is the headline input, with the real-time variant reported as the operational robustness column.

1.3.6 2.6 Where this paper sits

The contribution of this paper is best understood as a faithful single-equation revisit and extension of Williams (2010), and it has five distinct components.

(i) The faithful single-equation LIVES estimate, and the form correction as the lead methodological result. The paper's central finding is that the functional *form* of the LIVES equation is what identifies it. When housing wealth is entered faithfully — only through its credit-conditions interaction (ha_x_cci , the de-measured CCI(HA/4y) term), with no standalone housing-wealth level — the autonomous-consumption CCI intercept restored ($\zeta_c\text{-CCI}$), and illiquid financial assets combined ($ilfa_y = \text{equities} + \text{superannuation}$), the error-correction and core wealth structure come alive. The faithful specification (the headline of §7.0) delivers a COVID-controlled speed of adjustment of $\lambda \approx -0.25$ (pre-COVID -0.266 , $t = -4.85$), within about 13 per cent of Williams' -0.286 , on a sample of $n = 146$ (full) / 126 (pre-COVID); correctly signed and significant net-liquid and illiquid-financial marginal propensities (structural MPCs of 0.060 and 0.035 respectively, full sample, relative to $|\lambda| = 0.448$); a strong permanent-income coefficient ($+0.46$, $t = 4.0$); and a housing-collateral coefficient (γ_1) that is right-signed but insignificant (implied structural MPC 0.0055 , $t = 0.71$, against Williams' peak 0.0488). By contrast, the conventional constant-MPC disaggregated error-correction model that prior work and an earlier draft of this paper treated as the LIVES equation is *not* the LIVES equation and delivers a weaker, sample-fragile equilibrium ($\lambda = -0.239$, $t = -2.55$, on its $n = 86$ sample, collapsing to -0.087 pre-COVID). Reading its insignificant standalone housing coefficient as a failed housing wealth effect is the category error described above; much of the apparent weakness of Australian single-equation LIVES estimates is a specification artefact.

(ii) The structure-transfers-but-calibrations-do-not finding. The LIVES *structure* transfers to Australia, but Williams' Australian *calibrations* do not. Imposing his permanent-income gearing ($\psi_0 = 0.20$, $\psi_1 = 0.93$) and illiquid-financial MPC ($\gamma_{\text{IFA}} = 0.022$) collapses the equilibrium to $\lambda \approx -0.030$ ($t = -0.74$), independently reproduced by a Williams-prior calibrated specification ($\lambda = -0.048$), because Australia freely estimates a structural permanent-income gearing of order one ($\hat{\psi} = \text{OLS}/|\lambda| \approx 1.0\text{--}1.1$) — roughly five times Williams' 0.20 on the consistent structural scale (equivalently, the OLS coefficient +0.46 against his implied $\phi \cdot \psi_0 \approx 0.057$). This reconciles the companion paper's Wald non-rejection of the joint calibration ($\chi^2(6) = 7.55$, $p = 0.27$): the free estimates are too imprecise to reject Williams' values, but imposing those values wrecks the fit — low power is not the same as good fit. Williams' rate and affordability loadings cannot even be imposed at published magnitudes: his raw $\alpha_r = -0.871$ is roughly thirty times too large on the repository's percent real-rate $\times$ unit-normalised CCI scaling, and diverges the fixed point.

(iii) The interaction-collinearity diagnosis. The six CCI-interacted regressors are 0.66–0.97 mutually correlated in absolute value on this sample (`australia_cci_interaction_corr.csv`) because each is approximately proportional to CCI, and so cannot be separately freely estimated off a single equation. This is the structural reason Williams uses FIML, and it explains why single-equation calibration is empirically closed and why four-equation FIML plus pre-1988 back-extension are the only routes to sharpen the credit channels (§5).

(iv) The 1976Q3 back-extended master dataset and a direct sample-length test. The paper extends the sample back to 1976Q3. The public-data backbone consists of a Treasury TRYM long-run house-price series (1959Q3+), the RBA D03 M3 monetary aggregate (1959Q3+), the RBA D02 total-credit splice (1976Q3+), and a historical labour-force compilation (1964Q3+). For the 1976Q3–1988Q2 window where ABS sectoral household balance-sheet data are unavailable, we construct aggregate and disaggregated wealth proxies anchored at 1988Q3 (§3). Refitting the disaggregated no-CCI specification on the back-extended sample moves the speed of adjustment about 12% toward Williams (λ from -0.182 to -0.203 ; `lambda_robustness.csv` / `spec46_extended_comparison.csv`), but the individual wealth coefficients shrink and the net-liquid-asset coefficient collapses (–95%) — which we read as evidence that the binding constraint is the single-equation framing itself, not the post-1988 sample window. Sample length is not the binding constraint.

(v) The structured robustness and placebo suite, and the honest negative results. Adopting the Italian methodology of single-equation OLS with Newey-West HAC standard errors, the paper runs a structured robustness suite — instrumental variables, joint SUR, multi-window Chow tests, scaled-income and Drehmann real-rate alternatives, a Kalman state-space CCI, and a Williams-style spline credit-conditions column. The paper is honest throughout about the limitations of single-equation LIVES estimation on post-deregulation Australian data, and reports the negative results prominently as substantive findings that a freely-estimated single-equation framework permits (where imposed restrictions would hide them): the knot-placebo evidence on the credit-conditions index is split — the literal Williams four-knot construction sits at the placebo median (45th adjusted- R^2 percentile) and below it on the extended variants (36th–48th), while the deployed maximal-GETS protocol beats 84 per cent of random draws; a two-equation consumption + house-price SUR finds negligible cross-equation residual correlation and so no efficiency gain (SUR $\hat{\rho} = -0.013$ against OLS -0.011 ; `lives_sur_2eq_resid_corr.csv`); out-of-sample, the random walk with drift beats every

structural specification at horizons $h = 4$ and $h = 8$ (though Specs 8 and 11 beat it at $h = 1$); the automated specification selector falls back to a non-LIVES net-worth form (Spec 3) even though BIC and theory agree on Spec 11; and the look-ahead permanent-income reversal does not survive real-time construction. Each negative is read diagnostically — it points to why FIML and back-extension, rather than further single-equation refinement, are the way forward.

The headline paper does not estimate the multi-equation LIVES system itself, though we scaffold the multi-equation build in a separate companion directory. Williams (2010) estimates four equations jointly by FIML; the headline of the present paper estimates the consumption equation alone. The Italian experience (De Bonis et al. 2020) suggests that single-equation OLS produces consumption-equation coefficients close to joint SUR estimation, and we replicate that finding (§8): on our back-extended sample a two-equation consumption + house-prices SUR finds negligible residual correlation, so joint estimation gives no efficiency gain at the quarterly frequency. The case for the full multi-equation build therefore rests on cross-equation parameter restrictions — the common-factor identification of the credit channels — rather than on efficiency. The full multi-equation extension is left to a companion paper, with the scaffolding documented separately.

1.4 3. Data and measurement

The dataset assembles quarterly Australian macroeconomic and household-sector observations from **1976Q3 to 2024Q4** ($n = 194$). The public-data backbone for the pre-1980 window is built from a Treasury TRYM long-run house-price series, the RBA D03 M3 monetary aggregate, the RBA D02 total-credit splice, and a historical labour-force compilation. The disaggregated wealth components — the housing, illiquid-financial and net-liquid ratios that carry the faithful LIVES specification (Spec 11) — remain bounded at 1988Q3 by their primary source (ABS Cat 5232.0 Household Balance Sheet); for the 1976Q3–1988Q2 window we construct proxies (§3.13 below) that growth-rate-splice each component onto its 1988Q3 official value via the most relevant available aggregate. It bears emphasis at the outset that 1988Q3 is when the ABS sectoral balance sheet begins, so the financial-liberalisation episode that identifies Williams' credit channels — the 1980s deregulation — largely *predates* the modern disaggregated data. This timing, rather than estimator choice, is the deepest single-equation constraint on identifying the credit-conditions interactions, and motivates the back-extension to 1976Q3 documented below (and, ultimately, the four-equation FIML route discussed in §5 and §9).

Estimation is performed on the largest contiguous subset for which all variables in a given specification are observed:

- **Spec 1–3 (aggregate net worth)** with the back-extension proxy fit on $n=190$ (1977Q3–2024Q4; the binding constraint is `real_rate`, which needs a 4-quarter CPI lag for inflation computation). On the official `networth_y` (1988Q3+ ABS), Specs 1 and 3 fit on $n=146$; Spec 2 carries the short-run CCI term and so binds at the 2002Q3 loan-flow constraint ($n=86$ full / $n=66$ pre-COVID).
- **Spec 4 (disaggregated, no CCI) and the faithful LIVES specification (Spec 11)** use the official disaggregated series and fit on $n=146$ (1988Q3–2024Q4), with a **pre-COVID** subsample of $n=126$ (1988Q3–2019Q4) treated as the identified window throughout the paper.

- **Spec 5 (full disaggregated) and Spec 6 (conventional constant-MPC disaggregated ECM — the baseline, not the headline)** are both bounded at 2002Q3+ because each carries the short-run CCI variable ($d2_logcci_lag2$), which depends on $cci_ratio = \log(housing_loan_flow / income)$ from ABS Cat 5601.0; that loan-flow series only starts 2002Q3. Both bind at **n=86** (full) / **n=66** (pre-COVID) in both samples.
- **Spec 8 (free CCI interactions)** and the **Williams-calibrated Spec 12** fit on $n=146$, bounded by the disaggregated wealth and the $cci_williams$ smoothed-step spline (§3.5), which is constructed across the full 1988Q3+ window and does not require the 2002Q3 loan-flow series. **Spec 10 (Williams-prior reproduction)**, by contrast, carries the short-run $d2_logcci_lag2$ term and so binds at the 2002Q3 loan-flow constraint, fitting on **n=86** (full) / **n=66** (pre-COVID).
- **Spec 9 (Kalman state-space CCI)** fits on the same $n=146$ window as the disaggregated specs; the Kalman extraction itself is a separate state-space step.

The asymmetry in sample length is itself substantive. The faithful LIVES specification (Spec 11) is estimable on $n=146$ precisely because it proxies the credit channels through the $cci_williams$ spline rather than the 2002Q3-bounded loan-flow ratio, whereas the conventional Spec 6 baseline binds at $n=86$. Comparisons between the two should keep this sample difference in view; the larger $n=146$ window is one reason the faithful form delivers a sharper error-correction estimate (§7).

1.4.1 3.1 Aggregate consumption and income

Real per capita consumption ($cons_real_pc$) is constructed from ABS Cat 5206.0 Table 8 (Household Final Consumption Expenditure, chain volume measures, seasonally adjusted) divided by the civilian population aged 15 years and over (ABS series A84423091W, monthly, averaged to quarterly arithmetic mean; sample 1978Q2–2024Q4, sourced directly from the ABS historical series workbook). Following Williams (2010, 2009), we do not sum the single-year-of-age cohorts in ABS Cat 3101.0; in current ABS vintages, the Persons series only extends to age 47 in the disaggregated file, producing a population total approximately 35 per cent below the true Estimated Resident Population.

Real per capita household disposable income (ydi_real_pc) is the quarterly seasonally adjusted nominal series from ABS Cat 5206.0 Table 20 (Household Income Account), deflated by the consumption deflator implied by Table 8 (chain volume and current prices), and divided by the same 15+ population denominator. Following standard practice (Blinder–Deaton 1985), and as the headline empirical specification, we use gross disposable income; we report a non-property income (NPY) alternative constructed per Williams (2009) §4.2.1 in §3.6 below.

In the canonical LIVES equation (§4) income enters the long run with a **unit coefficient** — the error-correction term is the consumption-to-income ratio $\log y - \log c_{t-1}$, not a freely estimated income elasticity. The disposable-income series above is therefore load-bearing for the *level* of the equilibrium, not merely as a scaling variable, and the choice of income measure (gross vs non-property) is one of the channels through which our estimates and Williams' diverge (§3.6, §9).

1.4.2 3.2 Household balance sheet

Household-sector balance sheet stocks are sourced from ABS Cat 5232.0 Table 35 (Household Balance Sheet Aggregates, current prices, \$ billion), quarterly from 1988Q3:

- Currency and deposits (*fin_deposits*)
- Shares and other equity (*fin_equities*)
- Superannuation reserves (*fin_super*)
- Total household liabilities (loans and placements; *fin_loans*)
- Residential land and dwellings (*housing_wealth*)
- Closing net worth (ABS series A83722648X)

All balance-sheet stocks are converted to real values by dividing by the normalised consumption deflator and to per-capita-15+ values by dividing by the 15+ population.

Following Williams' Eq (7), wealth enters the long run as **asset/annualised-income ratios**. In our implementation the ratio is *contemporaneous*: the end-of-quarter (closing) stock divided by four times current quarterly income, $x_t/4y$. The wealth-to-annualised-income ratios used in the long-run consumption equation are constructed as:

- $ha_y = \text{housing wealth} / (4 \times \text{quarterly nominal disposable income})$
- $eq_y = \text{equities (ex-super)} / (4 \times \text{quarterly nominal disposable income})$
- $super_y = \text{superannuation reserves} / (4 \times \text{quarterly nominal disposable income})$
- $ilfa_y = (\text{equities} + \text{superannuation}) / (4 \times \text{quarterly nominal disposable income})$
- $nla_y = (\text{deposits} - \text{total household debt}) / (4 \times \text{quarterly nominal disposable income})$
- $debt_y = \text{total household debt} / (4 \times \text{quarterly nominal disposable income})$
- $networth_y = \text{closing net worth} / (4 \times \text{quarterly nominal disposable income})$

Williams' bracket dates the stock at $t-1$ ($x_{t-1}/4y$), and an earlier draft described our ratios that way; the implementation in fact uses the current quarter's *closing* stocks throughout, $x_t/4y$. Because ABS 5232 stocks are closing values, the contemporaneous ratio embeds within-quarter revaluations, which weakens the predeterminedness defence of OLS for the wealth terms; §4.6 discloses this timing convention and §8.1 reports the corresponding IV robustness check.

1.4.2.1 3.2.1 Illiquid financial assets are combined in the faithful LIVES specification

The LIVES theory distinguishes three wealth classes by their marginal propensity to consume: housing (a collateral asset, with an MPC that is *zero* at zero credit conditions and unlocked only as CCI rises), **illiquid financial assets** (IFA = equities plus superannuation, with a low plain MPC), and net liquid assets (NLA = deposits net of debt, with the highest plain MPC). Williams (2010) and the Italian implementation (De Bonis, Liberati, Muellbauer & Rondinelli 2020) both enter equities and superannuation as a single illiquid-financial aggregate, calibrating $\gamma_{IFA} = 0.022$.

The faithful LIVES specification (Spec 11) follows this convention: it uses the **combined** $ilfa_y = eq_y + super_y$ rather than entering equities and superannuation separately. This is not a cosmetic aggregation. Separating the two financial classes (as the conventional Spec 6 baseline does, with plain eq_y and $super_y$) repeatedly delivers a wrong-signed, insignificant equities coefficient on post-deregulation Australian data (e.g. Spec 6 full-sample $eq_y = -0.016$, $t = -0.30$; Spec 9 $eq_y = -0.010$, ns), reflecting the high collinearity between the two series and the short modern sample rather than a negative MPC on equities. Combining them into $ilfa_y$ recovers a correctly signed and significant illiquid-financial loading in the faithful form (Spec 11 full-sample $ilfa_y = +0.0155$, $t = 3.09$, ***; implied structural MPC = 0.035), the same

order of magnitude as Williams' calibrated 0.022 though somewhat larger. The aggregation is therefore both theoretically faithful and empirically necessary.

1.4.2.2 3.2.2 Net liquid assets: deposits net of debt

We adopt the convention of the Italian implementation (De Bonis, Liberati, Muellbauer & Rondinelli 2020, eq. 2.5; Table 3 column 3) of defining **net liquid assets as deposits net of total household debt**:

$$nla_y = (fin_deposits - fin_loans) / (4 \times \text{quarterly nominal disposable income})$$

This embeds the cross-equation restriction $\gamma_{LA} + \gamma_{LOANS} = 0$ (equal and opposite MPCs on liquid assets and on debt) implicitly by construction, in place of estimating a separate liquid-asset and a separate debt MPC. We test this restriction formally in §8: the netting restriction is **accepted at the 5 per cent level in every disaggregated specification and sample window** (australia_nla_restriction_test.csv; e.g. Spec 4 full sum = +0.032, $t = 0.73$, $p = 0.46$; Spec 6 full sum = +0.020, $t = 0.20$, $p = 0.84$). The honest reading is that the modern sample cannot statistically distinguish separate liquid-asset and debt MPCs, which is consistent with — but does not by itself confirm — the NLA aggregation; netting is a defensible economy of parameters rather than a sharply identified result.

In the faithful LIVES specification, NLA carries the largest and most robust wealth loading: Spec 11 full-sample $nla_y = +0.0269$ ($t = 3.75$, ***), an implied structural MPC of 0.060, against Williams' calibrated 0.159.

1.4.3 3.3 Mortgage interest rate

The nominal mortgage rate (mortgage_rate) is the RBA Standard Variable Owner-Occupier Rate (RBA Bulletin Table F6 housing lending rates, series FILRHLBVS), monthly from January 1959, averaged to quarterly. We source the historical series from the published RBA archive rather than the live readrba API, to ensure a stable vintage; the rate peaks at 17.0 per cent in 1989Q3, consistent with the Hawke–Keating recession.

The real mortgage rate (real_rate) is the nominal rate less the 4-quarter-ended percentage change in the consumption deflator. In the faithful LIVES specification the rate enters only through its credit interaction, $r_x_cci = real_rate \times CCI$ (Williams' α_1 channel). We note here a measurement point that becomes load-bearing in §9: our *real_rate* is expressed in *percentage* units (e.g. 3.0 for 3 per cent), and CCI is *unit-normalised*, so Williams' published raw rate loading ($\alpha_r = -0.871$) is roughly thirty times too large to impose on this scaling and diverges the iterative fixed point — one of the calibrations that, by construction, cannot transfer. We do not adopt the Italian Drehmann (2017) amortising-mortgage adjustment in the headline specifications but report it as a robustness check (§8).

1.4.4 3.4 House prices

The headline house price index (hpi) is constructed by chain-linking **four** sources, with the deepest segment extending to 1959Q3:

Layer	Source		Coverage	Splice convention
TRYM	Treasury	Macroeco- nomic Model historical database (house_price_history_long.csv)	1959Q3–2018Q2	growth-rate, anchored at 1986Q2 (first overlap with the legacy layer)
Legacy	houseprice_old.csv (privately compiled dwelling-price index, quarterly)		1986Q2–2005Q2	growth-rate, anchored at 2003Q3 (first overlap with the bridge)
Bridge	ABS Cat 6416.0 Residen- tial Property Price In- dex, 8-capital-cities ("old method")		2003Q3–2021Q4	growth-rate, anchored at 2011Q3 (first overlap with the current layer)
Current	ABS Cat 6432.0 To- tal Value of Dwellings, mean price		2011Q3–2024Q4 (sample end; the raw release ex- tends beyond)	(the modern overlay)

The TRYM source supersedes the BIS Shrapnel / REIA chain used in Williams (2010): the TRYM historical compilation already incorporates the same BIS Shrapnel (pre-1978), REIA (1978–1986) and ABS (post-1986) segments that Williams used, pre-chained into a single coherent 236-quarter series (1959Q3–2018Q2). The earliest binding observation for hpi is therefore **1959Q3**, three years deeper than even Williams' fullest sample.

In the faithful LIVES specification (§4), the house price enters the long run *only* through the affordability interaction $hp_x_1_minus_cci = (1 - 1.2 \cdot CCI) \cdot \log(ph/y)$ (Williams' α_4 channel, with affordability multiplier $\omega = 1.2$), not as a standalone level. There is no separate classical house-price level term in the faithful form.

1.4.4.1 3.4.1 Splice methodology — pure growth-rate chain-linking

For each adjacent pair of layers, the splice anchors the level at the first quarter where both series are non-NA, then back-casts via the base series' own QoQ growth rates:

```

chained[t] = overlay[t_anchor] × (base[t] / base[t_anchor])   for t < t_anchor
chained[t] = overlay[t]                                       for t >=
t_anchor

```

This standard ABS chain-linking convention preserves the base series' growth rates exactly while pinning the level to the overlay at the join. By construction there is no level discontinuity at any join quarter. An earlier implementation used $\text{mean}(\text{overlay}/\text{base})$ over the full overlap, which produced step jumps at join quarters where the overlap ratio drifts (notably a -17% step at 1986Q2 under the long TRYM↔legacy overlap and a $+10\%$ step at 2011Q2→Q3 under the bridge↔current overlap with mismatched units — the bridge is an index while current is a \$-value mean). Both artefacts were eliminated by the growth-rate convention.

The relative house-price-to-income ratio used in estimation is

$$\ln_hp_over_y = \log(hpi \times pop_millions / ydi_ann_nom)$$

i.e. the log of the nominal house-price index divided by nominal annualised disposable income per capita. Because numerator and denominator are both nominal, the consumption

deflator cancels exactly and the ratio is identical to the real house price over real income per capita: $(\text{hpi}/\text{defl}) / (\text{ydi_ann_nom}/\text{pop}/\text{defl}) = \text{hpi} \cdot \text{pop_millions}/\text{ydi_ann_nom}$ (`australia_data_download.R`).

We disclose a correction here. An earlier draft of this paper divided the *nominal* hpi by *real* income per capita, leaving the economy-wide price level inside the ratio — the contaminated variable had a correlation of 0.98 with the consumption deflator and was measuring inflation, not relative house-price pressure. Correcting the ratio to the consistent nominal/nominal (equivalently real/real) form changed the surviving CCI knot set (§3.5, §5.1.1) and, with it, every coefficient estimate in the paper; all numbers reported in this draft are from the corrected pipeline.

1.4.5 3.5 Credit conditions index

The credit conditions index (CCI) is the most contested input. Williams (2010, 2012) constructs CCI as the latent factor identifying credit liberalisation episodes: a spline $\sum a_s \cdot \text{SDMMA}_s$ of smoothed-step dummies at the institutional turning points 1979 (Campbell Committee, end of interest ceilings), 1992 (banking distress, Aussie Home Loans), 1998 (NBFI expansion), and 2007 (GFC), estimated **jointly** across the four-equation FIML system (consumption, house-price, mortgage-stock and home-equity-withdrawal equations) under sign priors, with the house-price-equation loading $\zeta_h = 1$ normalised. The structural reason for the joint estimation is identification: CCI enters six channels of the consumption equation, and (as §5 develops) the five CCI-carrying regressors that implement those channels are between 0.66 and 0.97 mutually correlated in absolute value on this sample (`australia_cci_interaction_corr.csv`) — each is approximately proportional to CCI — so they cannot be separately identified off the consumption equation alone.

A single equation cannot reproduce the FIML identification, so we proxy or calibrate CCI by two routes:

- **Default observable proxy** (`cci_ratio`): the log of housing-credit flow (ABS Cat 5601.0 New Loan Commitments Value, total housing) divided by the eight-quarter moving average of nominal disposable income. Available 2002Q3 onward ($n=90$); used only as a *short-run* regressor (`d2_logcci_lag2`) in the conventional Spec 6 baseline and Spec 2. This is the binding 2002Q3 constraint on Spec 6 (§3 opening). We do not back-extend it with a mortgage-spread proxy; that option is retained behind a feature flag but disabled.
- **Williams-style smoothed-step spline** (`cci_williams`): smoothed-step (SDMMA) dummies at the Australian institutional turning points, reduced by general-to-specific search with sign priors enforced by drop-on-violation (in the spirit of Hendry–Krolzig 2005). This is the CCI proxy used in the faithful LIVES specification (Spec 11), where it also restores the autonomous-consumption loading $\zeta_c \cdot \text{CCI}$ (`cci_williams`) inside the bracket. On our 1988Q3-onward sample, **four** of fifteen candidate knots survive the iterated sign-prior reduction (`australia_williams_cci_knots.csv`): `sdmma_2007_09` (–), `sdmma_2009_01` (+), `sdmma_2019_01` (–) and `sdmma_2020_04` (+); the 1979 and 1986 knots are aliased/constant within the window and the remaining nine candidates violate their priors. Only one of Williams’ four canonical knots (2007) survives on a post-1988 sample. The fitted series is committed (`australia_cci_williams_series.csv`; path figure `australia_cci_williams_path.png`) and its path is disclosed in §5.1.1: identically zero before 2007Q4, a plateau at its peak of 1 over 2010Q4–2018Q4, and strongly negative (trough

-2.12, settling at ≈ -1.63) after 2019. We report a full placebo battery on the spline in §5.2: the literal Williams 4-knot construction sits at the 45th adjusted- R^2 percentile of random knot draws — the detrending critique is vindicated for that construction — while the deployed iterated-reduction protocol reaches the 84th percentile, moderate support that we report with its protocol dependence disclosed.

The first-home-buyer share ($\text{fhb_share} = \text{fhb_loans} / \text{total_new_loans}$) is also constructed from ABS Cat 5601.0 from 2002Q3, and enters Spec 7.

1.4.6 3.6 Williams (2009) non-property income

The Williams (2009) §4.2.1 non-property income measure (np_real_pc) adjusts gross disposable income by removing imputed property income and a corresponding share of property-related taxation. Following Williams (2009 p. 10):

$$\begin{aligned} \text{np_rec} &= \text{total_income_rec} - \text{GOS_dwellings} - \text{prop_inc_rec} \\ \text{property_tax_share} &= (\text{GOS_dwellings} + \text{prop_inc_rec}) / \text{total_income_rec} \\ \text{np_pay} &= \text{total_income_pay} - \text{prop_inc_pay} - \text{property_tax_share} \times \text{income_tax_payable} \\ \text{NPY} &= \text{np_rec} - \text{np_pay} \end{aligned}$$

The input components are sourced from ABS Cat 5206.0 Table 20: compensation of employees, gross operating surplus on dwellings, property income receivable, social assistance benefits, property income payable, total income receivable, total income payable, and income tax payable.

The implied non-property income share of disposable income averages 0.84 over 2010–2024, consistent with Williams' implicit ~ 0.85 weighting. We use NPY as a robustness column (§8): in the current vintage, substituting NPY for gross income leaves the estimated speed of adjustment essentially unchanged (Spec 3 $\lambda = -0.191 \rightarrow -0.186$; `australia_williams_income_robustness.csv`), with the long-run permanent-income loading shifting modestly ($+0.196 \rightarrow +0.164$). The income measure remains one of the channels through which our estimates and Williams' can diverge, and we treat it as a methodological caveat rather than a resolved choice.

1.4.7 3.7 Permanent income

Permanent income (yp) is the forward-looking object in Williams' ratio $\log(\text{yp}/y)$. We construct it as a **discounted weighted average of expected future income**, following the Italian implementation's direct (single-regression) forecast of the discounted future-income aggregate (De Bonis, Liberati, Muellbauer & Rondinelli 2020, Appendix A.2). The baseline uses discount $\eta = 0.05$ (an annual discount factor $\delta = 0.95$) over a horizon $k = 40$ quarters.

Three properties of the headline measure, set out in full in §4.4, are flagged here because they are measurement issues:

1. **Look-ahead and tail extrapolation.** The forecasting regression is estimated once on the full sample, so yp_t embeds information dated after t and is non-causal. Moreover, the realised 40-quarter-ahead target is computable only up to **2014Q4**, so the training sample ends there and the final forty quarters of yp — about 27 per cent of the estimation sample, including the entire COVID period — are out-of-training extrapolations from 2014-vintage coefficients. The real-time variant (re-estimating the forecaster on an expanding window) shrinks the speed of adjustment materially and reverses the $\log(\text{yp}/y)$ coefficient (`australia_pi_realtime_robustness.csv`; in the Spec 6 frame, $\lambda = -0.239$ under the full-sample

Italy measure against -0.159 real-time Italy and -0.095 real-time AR). We flag the full-sample measure as the headline and the real-time column as the operational robustness check; headline results that depend on the look-ahead measure are identified as such in §7.

2. **GFC learning ogive.** The headline series is multiplied by an ogive that declines from 1 to 0.5 over 2008Q3–2012Q2, so the regressor is *half* the raw discounted-gap measure over the post-2012 two-thirds of the sample. A no-ogive re-estimate of the headline specification (australia_spec11_ogive_robustness.csv) leaves the structural conclusions unchanged (§4.4).
3. **Forecaster choice and the Australian permanent-income puzzle.** Under an AR forecaster the $\log(y_p/y)$ coefficient is *negative* (the "Australian permanent-income puzzle"); under the Italy-style direct forecaster it is positive, and the Italy forecaster fits better (australia_pi_method_meta.csv: AR adj-R² = 0.696, Italy adj-R² = 0.731). The faithful LIVES specification uses the Italy-style measure and recovers a strong, correctly signed permanent-income loading (Spec 11 full-sample $\ln_{yp_over_y} = +0.459$, $t = 4.04$, ***).

On the discount and horizon settings, the δ/k sensitivity grid (australia_permanent_income_sensitivity.csv; $\delta \in \{0.90, 0.95, 0.97\}$, $k \in \{20, 40, 60\}$, ogive on/off) is run under the **AR constructor** in the Spec 2 frame, where λ moves only at the third decimal. The grid does **not** cover the headline Italy-style method; the headline-method sensitivities are the no-ogive column and the real-time variant of point 1 above. The forecaster *method* (Italy vs AR, full-sample vs real-time), not the discount calibration, is the material choice.

In the canonical LIVES bracket the permanent-income weight is itself credit-dependent, $\psi(\text{CCI}) = \psi_0 + \psi_1 \cdot \text{CCI}$ with $\psi_0 = 0.20$ and $\psi_1 = 0.93$ **calibrated** (not freely estimated) by Williams. As §7 and §9 develop, this is the calibration that most clearly does not transfer: the freely estimated Australian gearing is far above Williams' $\psi_0 = 0.20$ (at or above unity in structural terms, §4.4), and imposing Williams' values collapses the error-correction term. The data section flags only that $\psi(\text{CCI})$ is calibrated in Williams' original — the estimation consequences are reported in the results.

1.4.8 3.8 Demographics and dummies

The prime-working-age share (prime_age_share = age 25–54 share of total ERP) is constructed from ABS Cat 3101.0 single-year-of-age cohorts (Male + Female, summed; ratios are robust to the truncation discussed in §3.1). Annual data are interpolated to quarterly via cubic spline.

The canonical LIVES equation also contains two demographic terms in the long-run bracket — D4 DEMFTB (the first-home-buyer / demographic cohort) and D4 WAPOP (working-age population growth) — for which Williams reports $\alpha_2 = -0.138$ and $\alpha_3 = -0.069$. These are not directly constructed in the single-equation specifications; prime_age_share (Spec 7) and the first-home-buyer share (fhb_share, Spec 7) stand in for the cohort channel, and we treat the full demographic block as a target for the FIML system rather than the single equation.

Australia-specific narrative dummies enter the default dummy set:

- d_neg_gearing_8587: 1985Q3–1987Q3 negative-gearing tax restriction
- d_recession_1991: 1991Q2 ("recession we had to have")
- d_apra_2014: 2014Q4 macroprudential investor-loan-growth cap (smoothed-step ogive)

- `d_apra_2017`: 2017Q2 macroprudential interest-only-loan cap (smoothed-step ogive)
- `d_jobkeeper_2020`: 2020Q2–2021Q1 JobKeeper income support

Together with the standard set (`d2000_gst`, `d2008_gfc`, `d2020_covid`, `d2020_rebound`), these constitute the full dummy set; zero-variance dummies are silently dropped per specification. The COVID quarters carry a very large dummy (`d2020_covid` ≈ -0.155 in Spec 11), and the COVID-inflated full sample fails the upper-bound λ screen; we therefore treat the **pre-COVID** sub-sample ($n=126$) as the identified window for the speed of adjustment (§7).

1.4.9 3.9 Coverage and reproducibility

The full reproducibility kit (R 4.5.3, renv-pinned dependencies, raw ABS workbooks, project-supplied CSVs, master quarterly dataset as CSV and RDS, full estimation pipeline with three execution modes, GitHub Actions CI, testthat suite) accompanies this paper. The master dataset has **194 quarters** $\times$ **~ 120 columns** and is available as a portable CSV for hand-editing or off-line replay. See Appendix A for each variable's source identifier, vintage, and splicing recipe.

1.4.10 3.10 RBA D-tables

Three RBA historical statistical tables support the pre-1988 portion of the sample.

1.4.10.1 3.10.1 RBA D03 — Monetary aggregates (M3)

The M3 monetary aggregate (`m3_aggregate`) is loaded from `d03hist.xlsx`, series `DMAM3N` (M3, original/not-seasonally-adjusted, \$ billion). M3 is total economy-wide currency plus transaction deposits plus all other deposits at ADIs, plus certificates of deposit issued by ADIs. Coverage: monthly, **1959Q3–2026Q1**, aggregated to quarterly by mean of the three monthly observations.

An earlier draft described the series as continuous with no breaks. That is **incorrect**: `DMAM3N` carries a definitional series break at **August 1976**, when the level jumps by +14.25 log per cent month-on-month (computed directly from `d03hist.xlsx`, against a mean monthly log change of 0.8 per cent over the full history). The break falls inside the opening quarters of the 1976Q3+ spine: it makes the 1976Q4 quarterly log-difference of M3 (+9.5 log per cent) an outlier of roughly 4.8 standard deviations of the series' quarterly growth distribution over the spine window, and the outlier propagates into every M3-based back-extension proxy (`m3_household_proxy`, `fin_deposits_proxy`, `nla_y_proxy`, `networth_y_proxy`) at exactly the deepest end of the back-extended sample. Back-extension results that lean on the first spine quarters should be read with this break in mind.

M3 is the headline liquid-asset proxy for the pre-1988 portion of the sample where ABS sectoral household-balance-sheet deposits are not available. Williams (2010) used the same M3 series, multiplied by the household share of factor income, for his pre-1988 splice.

1.4.10.2 3.10.2 RBA D02 — Lending and credit aggregates (total credit)

Total credit (`credit_total_d02`) is constructed by growth-rate-splicing two RBA D02 series across the July 2019 RBA conceptual reform:

- `DLCACN` (Total credit, original, \$ billion) for 1976Q3–2019Q2
- `DLCACSFN` (Total credit including select financial businesses, the post-2019 successor) for 2019Q3+

The two series have no quarterly overlap (DLCACN ends 2019-06, DLCACSFN starts 2019-07). The splice anchors levels at the boundary ($\text{first_post} \times \text{pre}[t] / \text{last_pre}$) so the join is continuous in level; the implicit growth rate at the boundary is exactly zero (no overlap to estimate it from). For analyses that hinge on the 2019Q2→Q3 quarter specifically, treat with caution. Coverage 1976Q3+.

This is total credit, not housing-specific. The housing-specific series in D02 (DLCAC0HN, DLCACIHN) only goes back to 1990Q1 in the current vintage and so cannot extend the housing-credit history pre-1990.

1.4.10.3 3.10.3 RBA D01 — Growth in selected financial aggregates

Downloaded as `d01hist.xlsx` (monthly growth rates of the same aggregates whose levels are in D02). Williams (2010) used D01 housing-credit growth rates to back-cast D02 levels pre-1976; in the current vintage D02 already extends to 1976Q3, so D01 is not strictly needed for back-extension. Retained for future use.

1.4.11 3.11 Pre-1978 labour force (`labour_force_historic.csv`)

A user-supplied CSV (`labour_force_historic.csv`, 188 quarterly rows 1964Q3–2011Q2) provides:

Column	Definition	Units
<code>pop_total</code>	Total resident population (annual + interpolation)	thousands
<code>pop_15_64</code>	Working-age population (15–64)	thousands
<code>labour_force</code>	Civilian labour force	thousands
<code>unemployed</code>	Unemployed persons (level)	thousands

Provenance: the source compilation pulls together pre-1978 series from ABS Cat 6204.0 (historical labour force, archived), ABS Year Book Australia, the Foster (1996) *Australian Economic Statistics 1949–50 to 1996–97* compilation, and RBA Occasional Paper No 8. Conceptually consistent with the current ABS Cat 6202.0 series (which begins Feb 1978) at the join.

The historic series are growth-rate-spliced onto the modern (1978+) series at 1978Q1:

- `pop_15_64` → `pop_millions` (the working-age population denominator used in per-capita normalisation)
- `labour_force` → `master labour_force`
- `unemployed/labour_force × 100` → `master unemp_rate` (level replacement before 1978Q1)

A side-effect of this splice is that the six quarters 1976Q3–1977Q4 now have non-NA values for all per-capita and labour-force-derived variables (`cons_real_pc`, `ydi_real_pc`, `npv_real_pc`, `lf_share`, etc.). On the data download, `cons_real`, `ydi_nom`, `mortgage_rate`, `hpi`, `M3`, `total credit`, and `prime_age_share` all extend to 1976Q3 already; the labour-force splice was the binding remaining constraint. The 1976Q3 back-extension — enabling the direct sample-length test reported in §8 — is one of the contributions of this paper.

`pop_total` is also exposed in the master as `pop_total_thousands` (historic-only 1964Q3–2011Q2 because there is no modern 1978+ counterpart in ABS 6202). It can be used as an alternative per-capita denominator (total resident vs civilian 15+) for sensitivity analyses.

1.4.12 3.12 Household-allocated M3 (m3_household_proxy)

The pre-1988 liquid-asset proxy is constructed by allocating M3 to the household sector via the wage share of GDP:

$$m3_household_proxy = m3_aggregate \times wage_share / 100$$

Williams (2010) uses the household factor-income share for this allocation. The wage share alone (compensation of employees / GDP) is a defensible approximation: it captures most of household factor income (wages dominate Australian household income; mixed income + property income receivable add ~10 percentage points and track wage share over time). **This is a documented simplification for our phase-1 back-extension**; for a fully Williams-faithful splice, replace wage_share with the broader (compensation + mixed income + property income receivable) / GDP series. We flag it explicitly here because the back-extension proxies should be read as approximations adequate for the direct sample-length test in §8, not as substitutes for the official 1988Q3+ series.

wage_share itself is loaded from a user-supplied CSV (household_income.csv) which extracts ABS Cat 5206.0 Table 24 analytical series A2302604K (compensation of employees, % of GDP), 1959Q3–2024Q4. Range over the sample: 49–62%, with values ~60% in the 1970s falling to ~50% today (capital-share rise).

Coverage of m3_household_proxy: 1976Q3+ (limited by wage_share in the supplied CSV), values \$22 billion (1976Q3) to \$1,673 billion (2024Q4).

1.4.13 3.13 Aggregate net-worth proxy (networth_y_proxy)

To enable Spec 1–3 fits on the back-extended sample, we construct an aggregate net-worth proxy that:

1. Uses M3-allocated-to-households (m3_household_proxy) plus a hpi × pop_millions back-cast of housing wealth as the raw wealth aggregate.
2. Growth-rate-splices that raw aggregate ratio onto the official networth_y at 1988Q3 (so the proxy equals the official series from 1988Q3 onwards and back-casts smoothly through 1976Q3).

The housing-wealth back-cast anchors at the first available official housing_wealth observation (1988Q3) and back-casts via housing_wealth × hpi × pop_millions growth, holding dwellings per capita constant:

$$\begin{aligned} \text{housing_wealth_proxy}[t] &= \text{housing_wealth}[1988Q3] \\ &\quad \times (\text{hpi}[t] / \text{hpi}[1988Q3]) \\ &\quad \times (\text{pop_millions}[t] / \text{pop_millions}[1988Q3]) \\ &\quad \text{for } t < 1988Q3 \\ &= \text{housing_wealth}[t] \text{ for } t \geq 1988Q3 \end{aligned}$$

The aggregate proxy is then:

$$\begin{aligned} \text{raw_proxy}[t] &= (m3_household_proxy[t] \times 1000 + \text{housing_wealth_proxy}[t]) / \\ \text{ydi_ann_nom}[t] \\ \text{scale} &= \text{networth_y}[1988Q3] / \text{raw_proxy}[1988Q3] \\ \text{networth_y_proxy}[t] &= \text{networth_y}[t] \quad \text{for } t \geq 1988Q3 \\ &= \text{raw_proxy}[t] \times \text{scale} \quad \text{for } t < 1988Q3 \end{aligned}$$

The $\times 1000$ term is a units conversion the code applies before the sum: `m3_household_proxy` is carried in \$ billion (RBA D03 units, never rescaled), whereas `housing_wealth_proxy` and `ydi_ann_nom` are in \$ million; without the conversion the M3 term would contribute ~0.01 per cent of the numerator instead of its intended ~12 per cent and the aggregate proxy would collapse to a housing-only back-cast.

Caveats explicitly documented in the data appendix:

- The back-cast omits equities and super (quantitatively small in 1976–1988 — Australian super pre-Superannuation Guarantee 1992 was a negligible household asset class).
- The back-cast omits debt netting (mortgage debt was a much smaller share of household balance sheets in the 1970s than today).
- Use only for back-extension exercises; never as a substitute for `networth_y` on the modern sample where the official series exists.

The proxy values across key dates: $5.05 \times$ annual income (1976Q3), 4.71 (1980Q1), 4.78 (1985Q1), 5.37 (1988Q3 — anchored to official), 10.19 (2024Q4). The shape is consistent with the historical Australian wealth-to-income trend (flat through the 1970s/early 80s, sharply rising post-1985 financial deregulation).

1.4.14 3.14 Disaggregated wealth proxies

Four additional proxies extend the disaggregated wealth components to the back-extended 1976Q3+ sample. Each equals the official series for $t \geq 1988Q3$ by construction; for $t < 1988Q3$ it back-casts via the most relevant available aggregate. These proxies enable the direct sample-length test of §8 (back-extension on Spec 1/4), where the honest finding is that lengthening the sample moves λ partway towards Williams but does *not* sharpen the individual wealth channels — sample length is not the binding constraint; the binding constraints are the post-deregulation data window and the interaction collinearity.

ha_y_proxy = `housing_wealth_proxy` / `ydi_ann_nom`. Uses the `hpi×pop` back-cast described in §3.13. Values: 2.68 (1976Q3) → 2.83 (1988Q3) → 6.41 (2024Q4).

fin_deposits_proxy anchors `fin_deposits[1988Q3]` and grows by `m3_household_proxy`:

```
fin_deposits_proxy[t] = fin_deposits[1988Q3]
                      × m3_household_proxy[t] / m3_household_proxy[1988Q3]
                      for t < 1988Q3
```

fin_loans_proxy anchors `fin_loans[1988Q3]` and grows by RBA total credit:

```
fin_loans_proxy[t] = fin_loans[1988Q3]
                   × credit_total_d02[t] / credit_total_d02[1988Q3]
                   for t < 1988Q3
```

nla_y_proxy = (`fin_deposits_proxy` - `fin_loans_proxy`) / `ydi_ann_nom`, again on the deposits-net-of-debt convention of §3.2.2. Values: +0.20 (1976Q3, households are net liquid creditors) → -0.05 (1988Q3) → -0.72 (2024Q4, modern net-debtor position). The sign-flip around 1988 captures the post-deregulation debt build-up — and is precisely the variation that the credit-conditions channel is meant to explain but which the modern data window mostly excludes.

eq_y_proxy: held constant at the 1988Q3 value pre-1988 (Option B in our methodology — Australian household equity holdings were a small wealth share in the late 1970s/early 80s;

the constant assumption introduces little level error and is straightforward to upgrade to an ASX All-Ordinaries back-cast). Value: 0.60 (constant) → 0.92 (2024Q4).

super_y_proxy: linear ramp from 0.1× the 1988Q3 value at 1976Q3 to the 1988Q3 value, then official thereafter. The 0.1 anchor matches Williams (2010) Table A.1 ballpark for the pre-Superannuation-Guarantee era (SGC mandate 1992). Values: 0.07 (1976Q3) → 0.66 (1988Q3) → 2.44 (2024Q4).

We flag these back-extension proxies honestly: the equities proxy is a constant-share assumption, the superannuation proxy is a linear ramp, and both the deposits and debt proxies are aggregate-growth back-casts rather than household-sector measurements. They are adequate for testing whether sample length is the binding constraint (it is not — §8) but they should not be over-interpreted as reconstructions of the 1970s household balance sheet. For the faithful LIVES specification (Spec 11) and all headline results we use the **official** 1988Q3+ disaggregated series and the n=146 / n=126 windows; the proxies are confined to the back-extension robustness exercises.

1.4.14.1 3.14.1 Coherence check — disagg sum vs aggregate proxy

Both `networth_y_proxy` (M3-allocated + housing back-cast, growth-rate-spliced onto the official broad `networth`) and the **sum-of-disaggregated** `networth_y_disagg_proxy` = `ha_y_proxy` + `nla_y_proxy` + `eq_y_proxy` + `super_y_proxy` are exposed in the master.

At 1988Q3 (the boundary):

- official `networth_y` (broad, ABS A83722648X closing net worth): **5.37**
- `networth_y_proxy` (aggregate, anchored to broad): **5.37** by construction
- `networth_y_disagg_proxy` (narrow, sum of components): **4.04**

The roughly 25 per cent gap between the aggregate and the disaggregated sum is the "other wealth" component of ABS closing net worth (life-office reserves, unincorporated business equity, etc.) absent from the narrow definition. For Spec 4–7 and the faithful LIVES specification (Spec 11), which use the disaggregated components individually, the disagg sum is the implicit reference; for Spec 1–3 the aggregate proxy is used.

1.4.15 3.15 Master variable coverage tiers (under 1976Q3+ spine)

After all splicing, master variable coverage falls into the following tiers (`australia_model_dataset.csv`):

First non-NA	n	Variables (selected)
1976Q3	~56	cons, ydi, hpi, mortgage_rate, M3, total credit, prime_age_share, all dummies, m3_household_proxy, ha_y_proxy, nla_y_proxy, eq_y_proxy, super_y_proxy, networth_y_proxy, ln_networth_y_proxy, cci_kalman, cons_real_pc, ydi_real_pc, npy_real_pc, labour_force, unemp_rate, lf_share, pop_millions, ln_cons_real_pc, ln_ydi_real_pc, ln_hp_over_y, ecm_lag (1976Q4)
1977Q3	2	real_rate, hicp_4q_ann (4-quarter CPI lag)
1978Q2	1	ydi_ann_8qma (8-quarter MA)
1988Q3	~21	OFFICIAL disaggregated wealth (ha_y, eq_y, super_y, ilfa_y = eq_y + super_y , nla_y, networth_y, debt_y), housing_wealth, fin_deposits/equities/super/loans (RAW from ABS 5232) — n=146
2002Q3	5	cci_ratio, fhb_share, housing_loan_flow, fhb_loans, non_fhb_loans — n=90
2009Q1	2	mortgage_interest_burden_rba, mortgage_payment_burden_rba (RBA E13) — n=64

The 1988Q3 tier is the binding tier for the faithful LIVES specification (n=146 full; n=126 pre-COVID), and the 2002Q3 tier binds the conventional Spec 6 baseline (n=86 full; n=66 pre-COVID). The combined illiquid financial ratio *ilfa_y* first becomes available at 1988Q3, alongside its *eq_y* and *super_y* constituents.

1.5 4. Model

1.5.1 4.1 The canonical LIVES consumption equation

The decisive contribution of this paper is that the *functional form* of the Muellbauer-Williams LIVES (life-cycle / inter-temporal / credit- conditions) equation is what identifies it. We therefore present the faithful form of Williams' (2010) eq. (7) as the model, and present the generic constant-MPC error-correction model that an earlier draft and much of the Australian literature treat as "the LIVES equation" as a nested baseline (Spec 6, §4.5) that the theory predicts should *not* behave like LIVES.

In its canonical form the LIVES consumption equation is

$$\Delta \ln c_t = \varphi \cdot [\zeta_c \cdot CCI_t + \alpha_1 \cdot r_t \cdot CCI_t + \gamma_1 \cdot CCI_t \cdot (HA/4y)\{t-1\} + \gamma_2 \cdot (IFA/4y)\{t-1\} + \gamma_3 \cdot (NLA/4y)\{t-1\} + \psi(CCI_t) \cdot \ln(y^h p_t / y_t) + \alpha_2 \cdot \Delta_4 DEMFTB_t + \alpha_3 \cdot \Delta_4 WAPOP_t + \alpha_4 \cdot (1 - \omega \cdot CCI_t) \cdot \ln(p^h / y)\{t-1\} + \ln y_t - \ln c_{t-1}] + \beta_1 \cdot DSRISK_t + \beta_2 \cdot (1 - \omega \cdot CCI_t) \cdot \Delta_8 \ln ue_t + \beta_3 \cdot \Delta_4 \ln c_{t-1} + \sum_k \delta_k D_{kt} + \varepsilon_t$$

where

- c_t is real per capita household consumption and y_t is real per capita household disposable income;
- CCI_t is the credit conditions index (a spline of smoothed-step dummies under sign-prior GETS reduction, identified jointly across the four-equation FIML system; see §5);

- $(HA/4y)$, $(IFA/4y)$, $(NLA/4y)$ are housing, illiquid financial and net liquid wealth, each scaled by *annualised* income (the prior quarter's stock divided by $4 \cdot y$; see §4.6);
- r_t is the ex post real mortgage rate;
- p^h_t is the real house price index, so $\ln(p^h_t/y)$ is the affordability / down-payment ratio;
- y^p_t is permanent income — the discounted, expected-future-income measure of §4.4 — so $\ln(y^p_t/y)$ is the permanent-to-current income gap;
- DEMFTB, WAPOP, DSRISK and $\Delta_4 \ln c_{t-1}$ are the demographic first-home-buyer cohort, working-age population, downside-risk and durables-habit terms of Williams' eq. (7);
- $\phi < 0$ is the speed of adjustment (the coefficient λ in our reporting convention, §4.3), and ζ_c , α_1 , γ_1 , γ_2 , γ_3 , α_2 , α_3 , α_4 and the function $\psi(\cdot)$ are the long-run "structural" parameters;
- D_{kt} are the narrative dummies of §3.7.

Three features of this form are load-bearing and distinguish it sharply from the conventional wealth-effect ECM.

(i) Credit conditions multiply six channels jointly. CCI enters the long-run bracket six times — as the autonomous-consumption loading $\zeta_c \cdot CCI$, the rate interaction $\alpha_1 \cdot r \cdot CCI$, the housing-collateral interaction $\gamma_1 \cdot CCI \cdot (HA/4y)$, inside the permanent-income gearing $\psi(CCI)$, and inside the affordability term through $(1-\phi \cdot CCI)$. The credit channel is therefore not an additive shifter but a set of *interactions* that switch the long-run wealth and income channels on and off as credit conditions ease or tighten.

(ii) There is no classical housing wealth effect. Housing wealth enters *only* through $\gamma_1 \cdot CCI \cdot (HA/4y)$. The marginal propensity to consume out of housing wealth is identically zero when $CCI = 0$ and is unlocked as credit conditions ease: it is a collateral / housing-equity- withdrawal channel, not a pure wealth effect. This is the theoretical core of the form-is-decisive result. A specification that enters a standalone $\gamma_{HA} \cdot (HA/4y)$ level term and reads its coefficient as the housing wealth effect is mis-testing the theory: theory predicts that standalone coefficient is approximately zero in the absence of the CCI interaction, so reading an insignificant standalone ha_y as a *failed* housing wealth effect is a category error. Williams' own peak housing MPC (γ_1 evaluated at the credit-loose regime) is 0.0488 — a number that can only be recovered through the interaction, never through a level term.

(iii) Illiquid financial and net liquid wealth enter as plain MPCs; income enters with a unit coefficient. Illiquid financial wealth (IFA, equities plus superannuation) and net liquid wealth (NLA, liquid assets minus total household debt) enter the long-run bracket as *un-interacted* marginal propensities γ_2 and γ_3 (Williams' calibrated 0.022 and estimated 0.159 respectively). Current income $\ln y_t$ enters the bracket with a coefficient restricted to unity, against $-\ln c_{t-1}$, so the error-correction term is the log consumption-to-income ratio $\ln(c_{t-1}/y_t)$ (equivalently $\ln(c_{t-1}) - \ln(y_t)$). This unit restriction is what makes the equilibrium a stationary consumption-to-income ratio in the LIVES tradition, rather than a free cointegrating vector.

The permanent-income gearing is itself credit-dependent:

$$\psi(CCI_t) = \psi_0 + \psi_1 \cdot CCI_t$$

so that the weight on the permanent-to-current income gap rises with credit availability. Williams *calibrates* $\psi_0 = 0.20$ and $\psi_1 = 0.93$ (the cap $\psi(\text{CCI}) \leq 0.95$ binds at the credit-loose extreme); these are not free parameters in his system. The affordability multiplier is $\varpi = 1.2$, also fixed. We treat the distinction between *structure* (the interactions and the unit income restriction) and *calibration* (the specific values $\psi_0, \psi_1, \gamma_2, \alpha_1, \varpi$) as central, and test it directly in §4.5 and the results.

1.5.2 4.2 Recovery of structural coefficients

We estimate the equation by OLS — $\Delta \ln c$ on the contemporaneous and lagged right-hand-side terms — with Newey-West HAC standard errors throughout (heteroskedasticity is structural in every full-sample specification; see §7.2). Because the entire long-run bracket is multiplied by ϕ , the OLS coefficient on each long-run regressor is ϕ times its structural value. We therefore recover each structural parameter as

$$\text{structural } \gamma_i = \text{OLS coefficient on regressor } i / |\lambda|$$

and report both forms throughout, so that the speed-of-adjustment channel (λ) and the long-run-coefficient channel are separable. For example, the faithful specification's full-sample net-liquid OLS coefficient $+0.0269$ and $\lambda = -0.448$ imply a structural NLA marginal propensity of $0.0269 / 0.448 = 0.060$ (§7.0). The same rule applies to the permanent-income term — a discipline §7.0 enforces when comparing the gearing with Williams' calibrated ψ_0 . The structural-recovery identity also makes precise *why* imposing Williams' calibrations can collapse the equation: fixing several γ_i while iterating to the fixed-point implied by the unit income restriction over-determines the bracket, and the only free margin left — λ — adjusts toward zero (§4.5, §7.0.1).

1.5.3 4.3 Reporting and sign conventions

Throughout, λ denotes the `ecm_lag` coefficient (the OLS coefficient on $\ln(c_{t-1}) - \ln y_t$), with $\lambda < 0$ indicating stable error correction. We report the faithful specification's speed of adjustment as $\lambda = -0.448$ ($t = -3.57$) on the full sample and $\lambda = -0.266$ ($t = -4.85$) on the pre-COVID sample (`australia_all_results.csv`). We treat the COVID-controlled value of about -0.25 as the *identified* speed of adjustment: it is nearly invariant to the COVID treatment (-0.266 pre-COVID, -0.248 excluding 2020Q1–2021Q4, -0.242 under quarterly 2020–21 dummies; `australia_spec11_variants.csv`), whereas the full-sample estimate is inflated by the COVID quarters — the three pulse dummies the full-sample specification carries are demonstrably insufficient, since replacing them with quarterly dummies halves $|\lambda|$ — and fails the upper-bound $|\lambda|$ screen of §6.2. The identified -0.25 sits close to Williams' (2010) estimated speed of $\phi_c = -0.286$. Where a headline number depends on the full-sample, non-causal permanent-income measure (§4.4) we say so explicitly and point to the real-time robustness column (§8).

1.5.4 4.4 Permanent-income forecasting

Permanent income y^p_t is the discounted weighted average of *expected* future log income over a 40-quarter horizon ($k = 40q$) at quarterly discount factor $\delta_q = 0.95^{(1/4)}$ — i.e. an annual discount factor $\delta = 0.95$, equivalently an annual discount rate $\eta = 0.05$:

$$\ln(\hat{y}_t^p / y_t) = E_t [\sum_{h=1}^{40} w_h \ln(y_{t+h})] - \ln(y_t), \text{ with } w_h = \delta_q^{(h-1)} / \sum_{h=1}^{40} \delta_q^{(h-1)}.$$

We implement two forecasters. The headline measure is the Italy-style direct forecast of the discounted aggregate (De Bonis, Liberati, Muellbauer and Rondinelli 2020, Appendix A.2); the rolling AR(8) forecaster is reported as a methodology-robustness column (§8).

- **Italy direct forecast (headline measure).** For each t whose 40-quarter-ahead horizon is observable, the discounted weighted average of future log income is computed and regressed, in a single full-sample equation, on a richer predictor set including $\log(\text{lf_share})$ (the labour-force-participation term that captures slow-moving demographic effects on trend income), a trend, a post-2008 split trend, the four-quarter moving average of log income, the unemployment rate, and four-quarter-difference dynamics. The permanent-income series is the fitted value of this single regression. Three properties of the construction are disclosed explicitly. (i) *Look-ahead*: because the coefficients are estimated over the whole sample, $\hat{y}_t^p$ embeds information dated after t and is non-causal — a two-sided *measurement* of permanent income rather than a real-time forecast. (ii) *Tail extrapolation*: the realised 40-quarter target is computable only up to 2014Q4, so the training sample ends there and the final forty quarters of $\hat{y}^p$ — about 27 per cent of the estimation sample, including the entire COVID period — are out-of-training predictions from 2014-vintage coefficients, driven mainly by the deterministic trend terms. (iii) *GFC learning weight*: the series is multiplied by an ogive that declines from 1 to 0.5 over 2008Q3–2012Q2 (mirroring the treatment in the AR forecaster), so the headline regressor is HALF the raw discounted-gap measure over the post-2012 two-thirds of the sample; a no-ogive re-estimate of the headline specification is reported as a robustness column (`australia_spec11_ogive_robustness.csv`) and leaves the structural conclusions unchanged. We carry the full-sample measure as the headline but report a causal, expanding-window variant (re-fitting at each t only on observations whose full horizon is realised by t) as the operational robustness column. The real-time variant shrinks $|\lambda|$ materially and reverses the permanent-income coefficient (§7.4): in the Spec 6 frame the full-sample Italy measure delivers $\lambda = -0.239$ against $\lambda = -0.159$ real-time and $\lambda = -0.095$ under the real-time AR forecaster (`australia_pi_realtime_robustness.csv`). Any forecasting use of the equation — embedding it in MARTIN, in particular — requires the real-time variant or the AR forecaster, not the full-sample measure.
- **Rolling AR(8) (robustness).** A rolling AR(8) regression of log income on eight own lags plus a linear trend, a post-2008Q3 step dummy, and a trend-break interaction; forecasts are aggregated over the 40 horizons using the discount weights, with optional `unemp_rate`, `log_oil`, `log_reer`, `log_stocks` predictors and the same 2008Q3 ogive learning weight. Under this forecaster the long-run permanent-income coefficient turns negative — the "Australian permanent-income puzzle" of §8 — a sign reversal absent under the Italy measure.

The permanent-income discount and horizon settings are not load-bearing for the speed of adjustment: across $\delta \in \{0.90, 0.95, 0.97\}$, $k \in \{20, 40, 60\}$ and the ogive on/off toggle, λ in the AR/Spec 2 frame moves only at the third decimal (`australia_permanent_income_sensitivity.csv`; the grid is run under the AR constructor — the headline-method sensitivities are the ogive column above and the real-time column of

§7.4). The forecaster *method* (Italy vs AR, full-sample vs real-time), not the discount calibration, is the material choice.

In the faithful specification the permanent-income channel is strong and correctly signed: $\ln(y^p/y)$ enters with OLS level coefficient $+0.459$ ($t = 4.04$) on the full sample and $+0.298$ ($t = 5.81$) pre-COVID (`australia_all_results.csv`). Two scale points discipline the comparison with Williams. First, by the structural-recovery rule of §4.2 the implied gearing is $OLS/|\lambda| \approx 1.02$ (full) to 1.13 (COVID-controlled) — **above the theoretical admissibility bound $\psi \leq 1$ — $\eta \approx 0.95$** . We do not hide this: it survives removing the ogive (structural ≈ 1.05 ; `australia_spec11_ogive_robustness.csv`), and §7.0 discusses candidate explanations, including the measure's tail extrapolation and the unit-income restriction forcing the permanent-income gap to absorb low-frequency consumption-to-income drift. Second, any comparison with Williams' calibrated $\psi_0 = 0.20$ must be scale-consistent: structural-to-structural the Australian gearing is roughly five times his; OLS-to-OLS-equivalent the comparison is $+0.46$ against $\phi \cdot \psi_0 \approx 0.057$. Spec 11 does not separately estimate the ψ_0/ψ_1 split: the credit-g geared component enters through a separate yp_x_cci interaction whose full-sample coefficient (-0.510 , $t = -1.51$) is insignificant and wrong-signed relative to Williams' calibrated $\psi_1 = +0.93$ — though it turns right-signed ($+0.19$, $t = 1.80$) on the pre-COVID sample — so the data identify the level of the permanent-income gearing but not its credit slope on a single equation. We return to this in §7.0.1: imposing Williams' ψ_0 , ψ_1 and γ_2 calibrations is what collapses the equilibrium.

1.5.5 4.5 The conventional constant-MPC ECM as nested baseline (Spec 6)

The conventional disaggregated wealth-effect error-correction model — which prior Australian work and an earlier draft of this paper treated as the LIVES equation — replaces the credit interactions with plain, constant marginal propensities:

$$\Delta \ln c_t = \lambda \cdot [\alpha_0 + \gamma_{HA} \cdot (HA/4y)\{t-1\} + \gamma_{eq} \cdot (eq/4y)\{t-1\} + \gamma_{super} \cdot (super/4y)\{t-1\} + \gamma_{NLA} \cdot (NLA/4y)\{t-1\} + \gamma_{HP} \cdot \ln(p^h/y)\{t-1\} + \alpha_r \cdot r_t + \psi \cdot \ln(y^p/y)_t + \ln y_t - \ln c_{t-1}] + \theta \cdot \Delta_2 \ln CCI_{t-2} + \sum_j \beta_j Z_{jt} + \sum_k \delta_k D_{kt} + \varepsilon_t$$

Here every wealth term enters as a standalone level with a *constant* MPC and *no* credit scaling; credit conditions appear only through the short-run term $\Delta_2 \ln CCI_{t-2}$ (`d2_logcci_lag2`). This is **not** the LIVES equation — it is a generic constant-MPC wealth ECM, and we label it the *conventional baseline* (Spec 6), not the preferred or headline specification. It is theoretically nested in the canonical form only in the degenerate sense that $\gamma_1 \cdot CCI \cdot (HA/4y)$ would reduce to a constant MPC if CCI were constant; on post-deregulation Australian data, where the credit liberalisation that identifies the channels largely predates the 1988Q3 start of ABS sectoral balance-sheet data (§3), that reduction is precisely what kills the housing-collateral channel.

The implications follow directly. The conventional baseline delivers a speed of adjustment of $\lambda = -0.239$ ($t = -2.55$, full sample, $n = 86$; `australia_all_results.csv`) — significant, but estimated on a sample less than two-thirds the faithful specification's, to which its short-run credit term binds it — and an insignificant standalone housing coefficient ($ha_y = +0.0022$, $t = 0.30$). Reading that insignificant ha_y as a *failed housing wealth effect* is the category error of §4.1: the LIVES form predicts that very coefficient is approximately zero absent the CCI

interaction. The faithful specification, by contrast, re-routes housing through $\gamma_1 \cdot \text{CCI} \cdot (\text{HA}/4y)$ alone, restores the autonomous-consumption loading $\zeta_c \cdot \text{CCI}$, and combines illiquid financial wealth ($\text{IFA} = \text{eq} + \text{super}$); the error-correction and core wealth structure then come alive (§7.0). We retain Spec 6 throughout as the conventional comparator against which the faithful form is the correctly-specified alternative — a matter of theory, not of fit.

1.5.6 4.6 Wealth definitions

Wealth enters as asset-to-annualised-income *ratios* — the end-of-quarter stock divided by four times current quarterly income, $x_t / (4 \cdot y_t)$. This is *contemporaneous* (closing-balance-sheet) timing: Williams' canonical equation dates the stock at $t-1$, and an earlier draft of this paper described our ratios that way, but the implementation uses the current quarter's closing stocks throughout. Because ABS 5232 stocks are closing values, the contemporaneous ratio embeds within-quarter revaluations, which weakens the predeterminedness defence of OLS for the wealth terms; the IV exercise of §8.1 is the corresponding robustness check. The three components are

- **Housing wealth (HA):** the value of the household-sector dwelling stock; it enters the faithful specification only through $\gamma_1 \cdot \text{CCI} \cdot (\text{HA}/4y)$.
- **Illiquid financial wealth (IFA):** equities plus superannuation; it enters as a plain MPC γ_2 (combined as $\text{ilfa}_y = \text{eq}_y + \text{super}_y$ in the faithful specification; entered separately as eq_y and super_y in the conventional baseline).
- **Net liquid wealth (NLA):** liquid assets minus total household debt, entered as a plain MPC γ_3 .

For NLA we follow the Italian implementation (De Bonis et al. 2020 §2; Williams 2010) in netting debt against liquid assets, and we test the underlying restriction $\gamma_{\text{LA}} + \gamma_{\text{LOANS}} = 0$ directly by refitting the disaggregated specifications with liquid assets and debt entered separately and conducting a Wald test of equal-and-opposite coefficients (`australia_nla_restriction_test.csv`). The restriction is not rejected in any specification or sample: in the conventional baseline the sum is 0.105 (SE 0.103, $t = 1.02$, $p = 0.31$) on the full sample and -0.020 ($t = -0.38$, $p = 0.71$) pre-COVID, and Spec 4 and Spec 5 likewise accept (all six rows `restriction_accepted = TRUE`). The honest reading is that the data cannot distinguish separate liquid-asset and debt MPCs — non-rejection by imprecision rather than confirmation of exact netting — which supports the NLA aggregation as a parsimonious choice rather than as a tested structural equality.

1.5.7 4.7 The de-meaning convention for the interactions

Following Williams' own convention, it is the *interacted economic variable*, not the CCI, that is de-meaned. The housing-collateral regressor is $\text{ha}_x_{\text{cci}} = (\text{HA}/4y - \text{mean}(\text{HA}/4y)) \cdot \text{CCI}$, with the mean taken over the 1980Q1-onwards estimation window on which the CCI is observed; the rate and permanent-income interactions are $r_x_{\text{cci}} = (r - \bar{r}) \cdot \text{CCI}$ and $\text{yp}_x_{\text{cci}} = (\ln(\hat{y}^p/y) - \bar{m}) \cdot \text{CCI}$, and the affordability term is $\text{hp}_x_{1\text{minus_cci}} = (\ln(\hat{p}^h/y) - \bar{m}) \cdot (1 - 1.2 \cdot \text{CCI})$. The CCI level `cci_williams` enters Spec 11 *raw* (peak-normalised, not de-meaned) as the autonomous-consumption loading. (An earlier draft described the convention backwards — de-meaned CCI times the raw variable — which is a materially different regression for the housing and rate channels, whose levels Spec 11 deliberately excludes; the description here matches the code.) Under this convention the interaction coefficients are the credit loadings evaluated at the sample-mean value of the interacted variable, and — because

the deployed CCI is zero before 2007Q3 (§5.1.1) — the un-interacted coefficients (γ_2, γ_3 , the intercept, ψ_0) are marginal propensities and gearings *at* $CCI = 0$, i.e. in the pre-2007 credit regime. Two consequences matter for interpretation. First, a pure re-centring of the credit regime is absorbed by the included CCI level, so the housing-collateral coefficient is invariant to the de-meaning constant; this is not true of specifications that omit the level. (The near-zero CCI counterfactual of §10 is run on the conventional baseline (Spec 6), where CCI enters only the short run and so has no long-run channel to integrate — confirming Spec 6 has no operative housing-collateral channel.) Second, de-meaning does not break the near-collinearity of the interactions — each remains approximately proportional to CCI and their absolute pairwise correlations are 0.66–0.97 on this sample (`australia_cci_interaction_corr.csv`; §5) — which is the structural reason the credit channels cannot be separately free-estimated off a single equation, and why Williams identifies them jointly through FIML.

1.5.8 4.8 Sign priors

Theoretical sign priors on the long-run coefficients are

- $\gamma_1 \geq 0$ (housing collateral, unlocked by CCI);
- $\gamma_2 \geq 0$ (illiquid financial wealth);
- $\gamma_3 \geq 0$ (net liquid wealth: buffer-stock plus inter-temporal substitution);
- $\alpha_4 \leq 0$ on $\ln(p^h/y)$ at credit-tight regimes (down-payment penalty), with the sign reversing toward positive at credit-loose regimes through the $(1 - \alpha \cdot CCI)$ term;
- $\alpha_1 \leq 0$ on $r \cdot CCI$ (inter-temporal substitution);
- ζ_c positive (autonomous-consumption loading on credit);
- $\psi_0, \psi_1 \geq 0$ (permanent-income gearing rising in credit);
- $\lambda < 0$ (stable error correction).

We use these as informal screens (§6.2) rather than imposing them as formal restrictions. Freely estimating the coefficients lets us report violations of the priors as substantive findings — for example the sign failures on r_x_cci and yp_x_cci in the faithful specification (§7.0), and the sign-prior GETS verdicts on the CCI knots (§5) — rather than burying them inside imposed constraints. This is the methodological discipline that distinguishes our single-equation estimates from Williams' FIML system: where his cross-equation restrictions deliver identification by imposition, our free estimates expose, rather than hide, the weak identification of the credit channels. The cost is power; the benefit is that the negative results (placebo failure, near-zero SUR efficiency gain, out-of-sample loss to a random walk, the calibration collapse, and the interaction collinearity) are visible as diagnostics pointing to why FIML and the pre-1988 back-extension are the routes forward (§8–§11).

1.6 5. Identification of credit conditions

This section is the methodological pivot of the paper. In the LIVES framework (Williams 2010; Muellbauer and Williams 2012) the credit conditions index, CCI, is not an observed series. It is a latent common factor that enters the consumption, house-price, mortgage-stock and home-equity-withdrawal equations simultaneously, and it is the object that interacts with the six channels inside the long-run bracket of equation (7). In Williams' setting it is identified by being shared across equations under cross-equation parameter restrictions estimated by full-information maximum likelihood (FIML). In a single-equation setting that joint identification is unavailable, and CCI can only be *proxied* — by an institutionally dated smoothed-step spline,

a state-space factor, a principal component, or a credit-to-income gap — or *calibrated* to Williams' published values (the calibration route is developed in §7.0.1 and shown to collapse the equilibrium).

The central new result of this section is an identification diagnosis, not a CCI success. We document directly why the credit channels cannot be separately recovered from a single equation: the five CCI-carrying regressors that implement the credit channels are between 0.66 and 0.97 mutually correlated in absolute value on this sample (`australia_cci_interaction_corr.csv`) because each is approximately proportional to CCI itself (§5.5). A single equation therefore cannot tell them apart. The placebo battery (§5.2) returns a split verdict on the spline itself: the literal Williams 4-knot construction, and the institutional knot choices under single-pass reduction, sit at or below the median of random smoothed-step draws — the detrending critique is vindicated for those constructions — while the *deployed* construction (the iterated reduction over the 15-knot basis) beats roughly four fifths of random draws, moderate support that remains protocol-dependent. This is the structural reason Williams identifies the credit channels through FIML, and it is why we treat four-equation FIML and pre-1988 back-extension, not further single-equation search, as the routes that could sharpen the credit channels. Throughout, the honest negatives — placebo failures for the literal construction, near-zero SUR residual correlation, selector divergence — are reported as substantive diagnostic findings, not embarrassments.

1.6.1 5.1 The Williams smoothed-step spline approach

Williams (2010) constructs CCI as a spline of SDMMMA smoothed-step dummies — a 5-quarter moving average of a 4-quarter moving average of a 0/1 step — at institutional turning points in the Australian financial-policy chronology. Each knot's coefficient is constrained by a sign prior derived from institutional history (deregulation episodes positive; retrenchment episodes negative), enforced by Hendry-Krolzig (2005) drop-on-violation general-to-specific reduction.

Williams' canonical paper uses four knots: 1979Q1 (Campbell Committee, removal of interest-rate ceilings), 1992Q1 (NBFI distress after the early-1990s recession), 1998Q1 (NBFI/securitisation expansion), and 2007Q1 (GFC retrenchment). The four-knot choice reflects the institutional information available over his 1977-2008 sample: STAMP- derived turning points and a deregulation calendar ending shortly after the GFC.

On our 1988Q3-onwards sample, only one of Williams' four canonical knots survives sign-prior reduction. A direct replication of the Williams 4-knot specification yields (`australia_knot_experiment_estimates.csv`):

Williams knot	Sign prior	Status on 1988+ sample
1979Q1	+	aliased (constant within window)
1992Q1	–	sign violator (data signal +ve)
1998Q1	+	sign violator (data signal –ve)
2007Q1	–	survives, coef ≈ -0.010

The 1979 deregulation knot is mechanically uninformative because the smoothed step reaches unity by 1980Q2, three years before our window opens. The 1992 and 1998 knots fail their

institutional sign priors: the post-1988 sample observes the recovery from the early-1990s banking distress (during which credit growth resumed and the OLS coefficient turns positive) and the late-1990s NBFI period without the contrast against the prior tight regime that would identify the loosening direction. This is a direct consequence of the data window. As noted in §3, ABS sectoral balance-sheet data begin only in 1988Q3, so the financial-liberalisation episode that most cleanly identifies the credit channels (Campbell, the early-1980s deregulation, the NBFI cycle) largely predates the sample on which the single-equation model can be estimated.

A direct 4-knot replication is therefore *not* identifying the four distinct credit-conditions episodes Williams' framework attributes to the spline. It is identifying one — the 2007 GFC tightening — plus a constant.

1.6.1.1 5.1.1 The maximal-GETS Australian CCI

Rather than impose Williams' published knot count on a sample that cannot identify three of his four knots, we adopt a **maximal-GETS approach**: starting from a richer 15-knot candidate set covering the documented Australian financial-policy chronology, we let drop-on-violation reduction prune knots that are aliased or violate their institutional sign prior. The candidate institutional events span Campbell '79, the housing-finance deregulation of '86, state-bank distress '90, banking distress '92/'93, the Wallis report and the establishment of APRA in '98, the GFC tightening of '07, the deposit-guarantee scheme '08, the FHB Boost '09, the APRA macroprudential rounds of '14 and '17, the Hayne Royal Commission of '19, the APRA cap removal and buffer reduction of '19Q3, the COVID/JobKeeper episode of '20, and the APRA buffer hike of '21.

On the 1988Q3-2024Q4 sample this candidate set reduces to **four surviving knots** under the iterated drop-on-violation reduction (`australia_williams_cci_knots.csv`):

Knot	Sign prior	Coef (OLS)	Reading
2007Q3	–	–0.0022	GFC tightening onset
2009Q1	+	+0.0123	First Home Buyer Boost (but see below)
2019Q1	–	–0.0338	Hayne Royal Commission lending crackdown
2020Q2	+	+0.0071	COVID/JobKeeper income support

(Nine candidate knots — 1990Q3, 1992Q1, 1993Q1, 1998Q3, 2008Q4, 2014Q4, 2017Q1, 2019Q3 and 2021Q4 — violate their institutional sign priors and are dropped; 1979Q1 and 1986Q1 are aliased, their smoothed step being constant within the estimation window.)

The `cci_williams` series used throughout the rest of the paper is constructed from these four surviving knots, peak-normalised to unity, and is committed in full (`australia_cci_williams_series.csv`; path figure `australia_cci_williams_path.png`). Its path should be read before any credit interpretation is placed on it: the index is **identically zero from 1976 until 2007Q3**, dips slightly negative over the GFC ramp (≈ -0.15 by late 2008), rises to its peak of 1 by 2010Q4, plateaus at 1 through 2018Q4, then falls steeply after 2019Q1 to a trough of -2.12 in 2020Q4 and settles at ≈ -1.63 from 2022 onwards (range -2.12 to $+1.00$). Four implications follow. First, every credit channel in Spec 8 and Spec 11 is identified off roughly **70 post-2007 quarters**: before 2007Q4 each CCI interaction is exactly zero, so

the pre-GFC half of the sample contributes nothing to the credit coefficients. Second, that all four surviving knots are post-2007 is itself part of the identification story (§5.3): the post-1988 sample carries usable sign-identifying variation only around the GFC, macroprudential and pandemic episodes. Third, the institutional reading of the surviving 2009Q1 "+" knot is contestable: the First Home Buyer Boost was a *fiscal* stimulus rather than a lending-standards easing, and the RBA's Financial Stability Reviews record lending standards *tightening* through 2009, so the "+" prior records a credit-demand event under a credit-supply label. Fourth, the candidate basis embeds a documented sign-prior conflict at 1993Q1: the maximal basis codes it -1 (the tail of the early-1990s bank retrenchment) while the sectional basis of §5.2.2 codes the same date +1 (mortgage-originator entry, Aussie Home Loans); both institutional readings are defensible for different events at the same date, and we disclose rather than silently resolve the conflict (`model_helpers.R`).

The reduction protocol also requires precise statement. At each pass the consumption ECM is re-fitted with the currently surviving candidate set and **all** knots whose coefficient violates its sign prior are dropped simultaneously; the loop iterates to a fixed point (`fit_consumption_with_williams_cci()`). This *differs* from Williams' one-at-a-time, strongest-violator-first reduction, and the survivor set is protocol-dependent: a single-pass reduction over the same 15-knot basis retains a *different* set of five knots — 1992Q1, 2007Q3, 2017Q1, 2019Q1 and 2020Q2, with 2009Q1 aliased rather than surviving — and a different λ (`australia_knot_experiment.csv`, `maximal_gets` row). We deploy the iterated protocol and placebo-test exactly that protocol in §5.2.

Finally, the construction is **two-step, with pre-test re-use of the dependent variable**. The knots are first selected as plain *additive* long-run regressors inside a constant-MPC Spec-4-style consumption equation estimated on the same $\Delta \ln c$ series; the surviving combination, peak-normalised, is then re-deployed *multiplicatively* (as `ha_x_cci`, `r_x_cci`, `yp_x_cci`, `hp_x_1_minus_cci` and the `cci_williams` level) in Spec 8 and Spec 11. Spec 11's fit statistics are therefore conditional on a CCI that was pre-fitted, under sign priors, to the same dependent variable — a pre-test problem that the placebo battery quantifies but does not remove.

The maximal-GETS construction remains defensible on two grounds: (i) the candidate set comes from documented Australian institutional history, not authorial choice of specific dates; and (ii) the surviving knots are those whose data signal aligns with their institutional sign prior, so the spline is *empirically* selected rather than imposed. We do **not**, however, claim that this delivers a structurally identified credit-conditions factor. The placebo battery (§5.2) gives the deployed protocol moderate — not strong — support against random smoothed-step constructions, and §5.5 documents the underlying obstacle: the credit channels the spline is meant to carry are near-collinear and cannot be separated off one equation. Williams' canonical 4-knot setup is retained as a robustness benchmark, and a sectional sign-prior alternative following Williams' (Aust paper §5.1) specification is also implemented and placebo-tested (§5.2.2).

1.6.2 5.2 The placebo battery

Whether the spline is identifying genuine credit-conditions turning points — rather than acting as flexible detrending of the consumption-equation residual — is empirically testable. We construct random-knot placebos: 200 draws of knot dates, compared like-for-like with the institutional construction under the *same* protocol. For the literal-Williams comparison

all four drawn knots are entered **unconditionally** — testing pure fit, with no sign-prior reduction — so the canonical 4-knot series is compared against random 4-knot series. For the maximal-GETS comparison each draw of 15 candidate knots is passed through a single sign-prior reduction pass. And — new in this draft — a **deployed-protocol placebo** replaces the 15 institutional knot dates with random dates carrying the same sign-prior pattern and runs exactly the iterated drop-on-violation reduction of §5.1.1, so the deployed `cci_williams` construction is placebo-tested as deployed, pre-test step and all. The institutional result's percentile rank in each placebo distribution measures whether the specific knot choice identifies something the data genuinely flags, versus whatever a flexible smoothed-step series could fit by chance.

1.6.2.1 5.2.1 Four placebo runs

Construction	Protocol	Sample	adj R ² %ile	$ \lambda $ %ile	Verdict
Literal Williams 4-knot	unconditional 4-knot	1988Q3+ (n=146)	45th	56th	Below R ² median — detrending critique vindicated
Literal Williams 4-knot	unconditional 4-knot	1976Q3+ (n=190)	36th	26th	Below median on both — critique persists
Maximal-GETS canonical	single-pass reduce	1976Q3+ (n=190)	48th	70th	Below R ² median — critique persists
Deployed cci_williams	iterated reduce	1988Q3+ (n=146)	84th	80th	Moderate support

Sources: `australia_williams_knot_placebo_verdict.csv` (literal, 1988Q3+; Williams adj-R² = 0.7252, $|\lambda|$ = 0.1890; 200 draws; committed verdict "DETRENDING CRITIQUE VINDICATED — Williams below median"); `australia_williams_knot_placebo_extended_summary.csv` (literal, 1976Q3+; Williams adj-R² = 0.6801 versus a placebo median of 0.6815, $|\lambda|$ = 0.2023 versus a placebo median of 0.2079); `australia_williams_knot_placebo_maximal_extended_summary.csv` (maximal-GETS, 1976Q3+; canonical adj-R² = 0.6836 versus a placebo median of 0.6846, $|\lambda|$ = 0.2563 versus a placebo median of 0.2329, canonical surviving knots 7 versus a placebo median of 8); `australia_williams_knot_placebo_deployed_verdict.csv` (deployed protocol, 1988Q3+; deployed adj-R² = 0.7540 at the 84th percentile, $|\lambda|$ = 0.2461 at the 80th, 4 surviving knots versus a placebo median of 5; committed verdict "MODERATE SUPPORT — deployed CCI beats most random draws"; full 200-draw table in `australia_williams_knot_placebo_deployed.csv`).

The verdict is split, and we report both halves. The **literal Williams 4-knot construction sits at or below the placebo median on both samples** (45th adjusted-R² percentile on the modern sample, 36th/26th on the extended sample): Williams' specific published knot dates, entered as published, do not outperform random dates, and the single-pass maximal-GETS reduction on the extended sample does no better (48th R² percentile, with the institutional

choice retaining *fewer* knots than the random median). For these constructions the detrending critique is vindicated, and the committed verdict labels say so. The **deployed construction fares better**: under the iterated reduction actually used to build `cci_williams`, the institutional knot dates beat 84 per cent of random draws on adjusted R^2 and 80 per cent on $|\lambda|$, while retaining fewer knots (4) than the placebo median (5) — i.e. the deployed CCI achieves more fit with less flexibility than typical random constructions. We read this as moderate — not strong — support: one in six random draws still matches the deployed fit, the percentile is specific to the iterated protocol (§5.1.1), and the construction re-uses the dependent variable. The standalone spline remains, at best, weakly distinguished from flexible detrending; it is not a structurally identified common factor.

1.6.2.2 5.2.2 Sectional sign priors (Williams Aust paper §5.1) tested

Williams (Aust paper §5.1) imposes sign priors over PERIODS rather than knot-by-knot:

Period	Sign prior	Rationale
1982–1990	non-negative	Financial deregulation
Early 1990s	non-positive	Banking sector distress
Mid-1990s–2006	non-negative	New entrants, securitisation
2007+	non-positive	GFC

We constructed a parallel CCI basis with one knot per period, extending Williams' four periods to cover post-2008 events (APRA 2014, APRA 2017, COVID 2020, APRA 2021). On the back-extended sample, with a random-period placebo (200 draws of 8 random knots and 8 random ± 1 priors), the sectional canonical sits at the **37th adjusted- R^2 percentile and 60th $|\lambda|$ percentile** (`sectional_placebo_summary.csv`; committed verdict "DETRENDING CRITIQUE PERSISTS — sectional below random median"). Williams' specific period dating does not outperform random period placements on the post-deregulation-extended window. In the committed side-by-side coefficient comparison (`sectional_cci_comparison.csv`) the sectional basis retains only two survivors against the maximal basis's four, with a weaker fit (adj- R^2 0.726 vs 0.754; $\lambda = -0.203$ vs -0.246) and a 0.69 correlation between the two indices — so the period-prior construction delivers a related but coarser credit signal than the maximal-GETS reduction. (An earlier draft reported the two fits as identical; that was a script bug — the sectional comparison had silently re-used the maximal basis — fixed in this revision.)

The takeaway across §5.2.1 and §5.2.2: neither the literal 4-knot construction, nor a single-pass maximal-GETS reduction, nor sectional sign priors delivers placebo support on the extended sample — the *institutionally dated* knots per se do not beat random dates. The one construction that does beat most random draws is the deployed iterated protocol on the modern sample, and even that support is moderate and conditional on the protocol. The CCI's standalone identification therefore remains closer to a single-equation OLS using flexible smoothed-step dummies than to a structurally identified common factor. The next two subsections set out *why* this is the expected outcome of single-equation estimation, and §5.5 supplies the direct mechanical reason.

1.6.3 5.3 Why the spline alone cannot identify the CCI as a common factor

The placebo evidence is consistent with the structural diagnosis in the LIVES literature itself. Williams (2010) and Duca, Muellbauer and Murphy (2013, ECB WP 1581) estimate the CCI inside a **multi-equation system** (consumption, house prices, mortgage stock, home equity withdrawal — Williams; consumption + refinancing rate — Duca, Muellbauer and Murphy) where the *same* CCI series enters all equations simultaneously. Williams imposes a normalisation ($\zeta_h = 1$ in the house-price equation) and estimates $\zeta_c, \zeta_m, \zeta_w$ as relative scalings; this cross-equation parameter restriction is what identifies the CCI as a common factor rather than as an equation-specific residual.

In a single-equation OLS estimation the spline is fit only to the consumption-equation residual. There is no constraint that the same knot loadings satisfy sign priors in the house-price, mortgage-stock or HEW equations simultaneously. The multi-equation scaffolding in our companion directory tests this directly.

1.6.3.1 5.3.1 Cross-equation joint sign-survival (LIVES phase 3)

We refit the Williams 15-knot maximal candidate set with the consumption equation, the house-price equation, the mortgage-stock equation and a home-equity-withdrawal proxy equation simultaneously, then require each knot to satisfy its institutional sign prior in every equation in which it is estimable (LIVES/outputs/lives_joint_cci_survival.csv).

Survival regime	Surviving knots	n
Consumption equation only (Spec 1 with <code>ln_networth_y_proxy</code> , extended sample)	1979, 1986, 1992, 2007Q3, 2017Q1, 2020Q2	6
Joint 3-equation (C $\cap$ H $\cap$ M)	1986, 2017Q1	2
Joint 4-equation (C $\cap$ H $\cap$ M $\cap$ HEW)	1986	1

Of the six knots that pass their sign prior when fitted to consumption alone (this is the single-pass reduction in `joint_cci_identification.R`, using the Spec-1 aggregate-proxy specification on the back-extended sample — distinct from the *iterated* reduction the canonical consumption pipeline applies in §5.1.1, which retains four knots, 2007Q3, 2009Q1, 2019Q1 and 2020Q2, on the 1988+ sample; the two reductions give different and only partially overlapping survivor sets, and 2009Q1 is aliased in the joint system), only **1986 (financial deregulation) and 2017Q1 (APRA macroprudential round II)** have signs consistent with their institutional priors across consumption, house prices and mortgage stock simultaneously, and only **1986** also survives the home-equity-withdrawal equation.

The consumption-only identification is therefore overstated: four of the six consumption-passing knots are consumption-equation-specific and do not survive a cross-equation common-factor restriction, and only one knot survives all four equations. This is the empirical content of the placebo results in §5.2: without imposing cross-equation sign consistency, the consumption-equation residual can be flexibly fit by smoothed-step dummies whose information content is consumption-specific.

1.6.3.2 5.3.2 What joint identification fixes

When we rebuild `cci_williams_joint` using the two knots that survive the three-equation test and re-estimate the house-price equation with the new CCI (`LIVES/outputs/lives_phase3_comparison.csv`):

HP equation, CCI loading	(a) cons-only CCI	(b) joint OLS	(c) joint SUR
Estimate	-0.016	+0.024	+0.024
Sign	× violator	✓	✓

The house-price equation's CCI loading flips from negative (under the consumption-only CCI) to positive (under joint identification) — Williams' cross-equation sign restriction working as intended. The mortgage-stock equation's CCI loading remains negative under this joint-sign-survival approximation, which weights surviving knots by consumption-equation coefficients; full FIML with parameter restrictions across all four equations would be needed to flip it. The exercise is therefore illustrative of the mechanism, not a substitute for the FIML estimate.

1.6.4 5.4 Two-equation SUR — joint estimation gives no efficiency gain

A complementary test of the multi-equation framework's value is whether SUR or FIML deliver efficiency gains over equation-by-equation OLS. A two-equation SUR (consumption + house prices, on the extended 1976Q3+ sample using Spec 1 with the aggregate net-worth proxy) finds an essentially negligible cross-equation residual correlation: the committed estimates are $\hat{\rho}(\varepsilon_C, \varepsilon_H) = -0.0109$ under equation-by-equation OLS and -0.0133 under SUR (`LIVES/outputs/lives_sur_2eq_resid_corr.csv`, committed verdict "NEGLIGIBLE cross-equation linkage — single-equation OLS approximately efficient"). The corresponding coefficient comparison is in `australia_joint_pi_robustness.csv`: moving from single-equation OLS to SUR_joint on the Spec-3 consumption equation shifts coefficients only within sampling noise — the net-worth loading moves from $+0.0011$ to $+0.0024$ (well under half a standard error, SE 0.0028; insignificant under both), the speed of adjustment from -0.191 to -0.204 (raw-units `ecm_lag`), and the permanent-income loading from $+0.196$ to $+0.187$. No coefficient changes sign in a way that alters the economic reading, and the standard errors are barely tightened.

The joint-estimation case for LIVES therefore does **not** rest on efficiency gains. It rests entirely on **cross-equation parameter restrictions** — Williams' FIML imposes that the same CCI loading enters all four equations with specific sign constraints. SUR alone imposes only residual-covariance flexibility, and at the quarterly frequency that covariance is negligible (§5.3.1). The two-equation SUR confirms that the cross-equation linkage between consumption and house prices is captured by shared regressors (CCI, the real rate, dummies for major events); it does not reside in unexplained residual covariance. We report the near-zero SUR residual correlation as another honest negative: it tells us that the value of the system is in its sign restrictions, not its error structure, and so single-equation SUR cannot be a shortcut to system identification.

1.6.5 5.5 Interaction collinearity — why the credit channels cannot be separated off one equation

The placebo failures (§5.2), the cross-equation sign-survival collapse (§5.3.1) and the absence of an SUR efficiency gain (§5.4) all point to the same underlying obstacle, which we now state

directly. In the faithful LIVES form, CCI does not enter once: it multiplies six channels jointly inside the long-run bracket of equation (7) — the autonomous-consumption loading ($\zeta_c \cdot \text{CCI}$), the rate channel ($\alpha_1 \cdot r \cdot \text{CCI}$), the housing-collateral channel ($\gamma_1 \cdot \text{CCI} \cdot (\text{HA}/4y)$), the down-payment/affordability composite ($\alpha_4 \cdot (1 - \omega \cdot \text{CCI}) \cdot \log(p^h/y)$), the permanent-income gearing ($\psi(\text{CCI}) \cdot \log(y^p/y)$ with $\psi(\text{CCI}) = \psi_0 + \psi_1 \cdot \text{CCI}$), and so the regressors that carry these channels are each approximately *proportional* to CCI over the sample.

The empirical consequence is that the five CCI-carrying regressors — `cci_williams`, `ha_x_cci`, `hp_x_1_minus_cci`, `r_x_cci` and `yp_x_cci` — have absolute pairwise correlations between **0.66 and 0.97** on this sample. The full correlation matrix is committed (`australia_cci_interaction_corr.csv`); the extremes are $\rho(\text{cci_williams}, \text{yp_x_cci}) = -0.967$ and $\rho(\text{cci_williams}, \text{ha_x_cci}) = +0.890$, with the weakest pair (`cci_williams`, `hp_x_1_minus_cci`) still at -0.681 . Five near-collinear regressors carrying distinct structural meanings cannot be separately free-estimated off a single equation: ordinary least squares will allocate a near-singular design among them more or less arbitrarily, producing wrong-signed and insignificant individual loadings even when the joint contribution of the credit block is real.

The committed, load-bearing evidence for this collinearity is threefold, and each piece is itself an honest negative:

1. **Sign failures and insignificance when the interactions are freed (Spec 8, §5.6 and §7).** When the credit interactions are estimated freely, the housing-collateral interaction is right-signed but insignificant, while three of the remaining interactions fail their institutional sign priors (`australia_spec8_sign_prior_verdicts.csv`). This is exactly the pattern a near-singular design produces.
2. **The identification-versus-detrending decomposition (`australia_cci_fit_decomposition.csv`).** Adding the Williams maximal-GETS CCI to the conventional baseline (Spec 6 $\rightarrow$ Spec 8) does *not* merely detrend: it shifts the permanent-income loading by +51 per cent (from +0.33, $t = 1.50$, to +0.49, $t = 3.42$) and the speed of adjustment by -92 per cent (from -0.239 to -0.458), with the net-liquid loading shifting +303 per cent and the superannuation loading +258 per cent. By the decomposition's own classification these are "IDENTIFICATION" (> 30 per cent) rather than "DETRENDING" (< 5 per cent) shifts. The credit interactions re-allocate identification across the income, liquid-wealth and speed-of-adjustment channels — which is precisely the symptom of collinearity (one near-singular block redistributing fit), not evidence of clean separate identification of each channel.
3. **The calibration collapse (Spec 10 and Spec 12, §7.0.1).** When Williams' calibrations are imposed instead of freed, the error-correction term collapses to $\lambda = -0.030$ ($t = -0.74$) in Spec 12, independently reproduced at $\lambda = -0.048$ ($t = -0.78$) by Spec 10. A block of regressors that can be jointly identified would not behave this way; a near-collinear block, whose joint mapping into the data is sharp but whose internal split is not, will.

Single-equation calibration of the credit channels is therefore empirically closed: the data pin down the *block* but not its *components*. This is the structural reason Williams uses FIML — the cross-equation sign restrictions break the within-equation collinearity by requiring the *same* loadings to be consistent across consumption, house prices, mortgage stock and HEW.

Two routes can sharpen the credit channels on Australian data: the four-equation FIML system (which the companion paper develops), and the pre-1988 back-extension, which adds the financial-liberalisation variation that most cleanly identifies CCI but which the post-1988 balance-sheet data exclude. Further single-equation search cannot.

1.6.6 5.6 Spec 8: the freely estimated CCI interactions (the collinearity demonstration)

When `cci_williams` is available, we estimate Spec 8, which incorporates the Williams interaction structure with the credit channels entered as free regressors (the autonomous-consumption CCI level is the one channel Spec 8 omits; Spec 11 restores it):

$$\dots + \gamma_{HA} \cdot ha_y + \gamma_{HA_cci} \cdot ha_y \cdot CCI + \gamma_{HP} \cdot \log(p^h/y) \cdot (1 - \omega \cdot CCI) + \alpha_r \cdot r \cdot CCI + \psi_1 \cdot \log(y^p/y) \cdot CCI + \dots$$

with ω calibrated to 1.2 following Williams (Aust paper §5.2 fn 9). We present Spec 8 not as a credit-conditions success but as the empirical *demonstration* of the collinearity diagnosed in §5.5. It is the free-estimation counterpart of the calibration-collapse specifications (Spec 10/12): freeing the block over-fits the within-equation split and re-allocates identification, whereas imposing the block collapses the equilibrium. Neither separately identifies the credit channels.

The freely estimated interaction loadings, with their institutional sign priors (australia_spec8_sign_prior_verdicts.csv, australia_williams_spec8_comparison.csv), are:

Interaction	Williams reference	Sign prior	OLS (Spec 8, full)	t	Verdict
ha_x_cci (housing CCI; γ_1)	+0.0488 (peak MPC)	+	+0.0043	0.87	right-signed, ns
r_x_cci (real rate $\times$ CCI; α_1)	-0.871 (at CCI=1)	-	+0.0028	2.05	sign FAIL
hp_x_1_minus_cci (down-payment; α_4)	-0.13 (at CCI=0)	-	+0.0299	1.38	sign FAIL
yp_x_cci (PI $\times$ CCI; ψ_1)	+0.93 (calibrated)	+	-0.5046	-2.43	sign FAIL

The pattern is diagnostic of the near-singular design, not of structure. The housing-collateral interaction is **right-signed** — consistent with Williams' prediction that the housing MPC rises with credit ease as collateral becomes spendable — but statistically insignificant ($t = 0.87$, $p = 0.38$). Of the remaining three interactions, all fail their institutional sign priors: the rate interaction is wrong-signed (and significant at 5 per cent, $p = 0.042$), the down-payment composite is wrong-signed and insignificant, and the permanent-income interaction is strongly wrong-signed (the free estimate is -0.50, $t = -2.43$, against Williams' calibrated +0.93). The one term

that survives its prior is the *non*-interacted permanent-income level ($\ln_{yp_over_y} = +0.49$, $t = 3.42$, PASS). In other words, when the credit block is freed the equation re-allocates almost all of its identification onto the income and speed-of-adjustment channels and away from the individual interactions — exactly the identification re-allocation §5.5 documents.

It is true that Spec 8 with the interactions included delivers a numerically larger speed of adjustment, $\lambda = -0.458$ ($t = -3.52$), exceeding Williams' calibrated -0.286 in magnitude, and a BIC of -952.8 — second only to the faithful Spec 11 (-954.8) among the $n = 146$ specifications. We do **not** read this as Spec 8 "closing the gap" with Williams or "exceeding" him in any substantive sense. The larger $|\lambda|$ and the low BIC reflect the re-allocation of fit that a near-collinear block produces, not separate identification of the credit channels: three of the four interactions are wrong-signed, and the one right-signed interaction is insignificant. The honest reading is that freeing the block buys fit at the cost of structurally meaningless component loadings. The faithful LIVES specification (Spec 11, §7.0), by contrast, enters housing *only* through its CCI interaction (ha_x_cci , de-meant, $+0.0025$, $t = 0.71$ — right-signed, insignificant) and restores the autonomous-consumption CCI loading, and it is the narrative-preferred headline on theoretical-form grounds; Spec 8 is retained as the collinearity demonstration and is not promoted as a CCI result.

1.6.7 5.7 CCI measure cross-checks

To confirm that the identification difficulty is a property of single-equation Australian data rather than of the smoothed-step construction in particular, we cross-check the Williams maximal-GETS spline against three alternative latent-CCI measures (*australia_cci_methods_summary.csv*): a Kalman state-space common factor estimated by ML on five credit indicators with a housing-loan-flow anchor ($n = 194$); the first principal component across the same five standardised indicators ($n = 146$); and a BIS-style credit-to-income HP-filter gap ($\lambda = 400,000$) on log household debt-to-income ($n = 146$). The pairwise correlations between the four measures (*australia_cci_method_4way.csv*) are low except between the two purely statistical factors:

Pair	ρ	n
Kalman vs Williams maximal-GETS	-0.004	194
PCA vs Williams maximal-GETS	-0.218	146
Credit/income gap vs Williams	0.335	146
Credit/income gap vs Kalman	0.304	146
Kalman vs PCA	0.764	146
Credit/income gap vs PCA	0.238	146

The institutional spline is essentially uncorrelated with the Kalman factor ($\rho = -0.004$) and weakly negatively correlated with the PCA factor ($\rho = -0.22$); only the two data-driven statistical factors agree strongly with one another (Kalman versus PCA, $\rho = 0.76$). The four candidate measures do not converge on a common Australian credit-conditions series. When the Kalman factor is used in place of the spline (Spec 9, §7), the liquid, illiquid-financial and permanent-income channels shift by 32–66 per cent relative to the no-CCI baseline (*australia_cci_fit_decomposition.csv*), again re-allocating identification rather

than sharpening it. The measure choice is therefore not innocuous, and no single-equation CCI proxy resolves the underlying weak identification — reinforcing that regime classification should rely on the joint system rather than on any one single-equation proxy (§10).

1.6.8 5.8 The wealth-coefficient profile on the back-extended sample

A natural follow-up question is whether *sample length*, rather than the single-equation framing itself, accounts for the residual gap between our estimates and Williams' Table 1. The cleanest test runs on the simpler disaggregated no-CCI specification (Spec 4), which is closest in form to Williams' long-run cointegrating regression. We refit Spec 4 on the back-extended 1976Q3+ sample using the disaggregated wealth proxies of §3:

LR coefficient	Baseline	1988+ (n=146)	Extended	1976+ (n=190)	% change	Williams Table 1
λ (ecm_lag)		-0.182		-0.203	+11.5	-0.286
nla_y		+0.0245		+0.0013	-94.6	+0.159
eq_y		-0.043		-0.041	-6.5	+0.022‡
super_y		+0.0137		-0.0052	-138	(incl.)‡
ha_y		+0.0193		+0.0177	-8.1	+0.0488
ln_yp_over_y		+1.11		+1.04	-6.5	+0.20

Source: spec46_extended_comparison.csv (structural long-run coefficients, $\gamma = \text{OLS}/|\lambda|$, per §4.2; the λ row is the OLS ecm_lag). ‡ Williams reports a single illiquid-financial-asset MPC ($\gamma_{\text{IFA}} = 0.022$), shown here against the combined eq_y + super_y. The Williams structural MPCs used throughout the paper ($\gamma_{\text{HA}} = 0.0488$, $\gamma_{\text{NLA}} = 0.159$, $\gamma_{\text{IFA}} = 0.022$) are the BIS-chapter values recorded in australia_williams_comparison.csv — the same authoritative source as §7.3 and §11.4.

On Spec 4 the speed of adjustment moves about a fifth of the remaining distance towards Williams' value ($-0.182 \rightarrow -0.203$, against -0.286), but the individual structural wealth coefficients become *smaller*, not larger; γ_{NLA} collapses to roughly zero ($+0.0245 \rightarrow +0.0013$), γ_{SUPER} flips sign, and γ_{EQ} retains a wrong sign on both samples. The long-run permanent-income coefficient remains far above Williams' calibrated +0.20 on both samples.

The reading is consistent with the rest of this section. First, the back-extension does not push the Spec 4 estimates *closer* to Williams' Table 1 in the cross-section of wealth loadings — the post-1988 sample window is not, in itself, what generates the residual gap; sample length is not the binding constraint. Second, triangulating with the placebo evidence (§5.2), the cross-equation sign-survival collapse (§5.3.1), the near-zero SUR residual correlation (§5.4) and the interaction collinearity (§5.5), we read the residual difference between any single-equation OLS estimate and Williams' system FIML as a consequence of the single-equation framing rather than of sample length, knot count, or sign-prior structure. The path to a tighter reconciliation runs through a full FIML build with cross-equation parameter restrictions and the pre-1988 financial-liberalisation variation — the two routes §5.5 identifies — which the companion paper develops.

1.7 6. Specifications and selection

We estimate fourteen specifications — a numbered ladder Spec 1 through Spec 12, plus two variants, the long-history Spec 6b and the measured-burden Spec 7b — that range from a single aggregate-net-worth error-correction model to the faithful Muellbauer–Williams LIVES form and its calibration-imposed counterpart. The ladder is then passed through a four-screen selection rubric. We report the automated selector outcome in full and honestly, but we state at the outset that the headline of this paper is chosen on theoretical-form grounds, not by the screen: the **faithful LIVES specification (Spec 11)** is the narrative lead, the **conventional constant-MPC disaggregated ECM (Spec 6)** is retained as the baseline against which it is contrasted, and the **Williams-calibration-imposed specification (Spec 12)** is the negative control that shows what fails to transfer. In the current vintage, BIC and LIVES theory *agree*: Spec 11 carries the best BIC of all fourteen specifications (−954.8). The automated rubric nevertheless returns a third, non-LIVES specification (Spec 3, aggregate net worth in levels), because no specification clears the cointegration screen and the conservative speed-of-adjustment ceiling ($|\lambda| < 0.30$) and the 2008Q3 Chow screen bind against Spec 11's COVID-inflated full-sample estimate. We treat that divergence not as a defect to be smoothed over but as a documented result (§6.3): it is diagnostic of how weakly a single equation can pin down the LIVES long run on post-deregulation Australian data, and of how much the conservative screens are doing.

1.7.1 6.1 The fourteen specifications

The ladder is organised so that each step adds one structural element. Specs 1–3 are aggregate-net-worth ECMs; Specs 4–7b disaggregate wealth and add cohort/burden terms; Specs 8–9 introduce the Williams CCI interactions (spline and Kalman extractions); and Specs 10–12 are the explicitly Williams-aligned forms — the calibrated-prior fixed point (Spec 10), the faithful free LIVES estimate (Spec 11, the headline) and the Williams-calibration-imposed LIVES (Spec 12). Long-run regressors follow the notation of §4; ha_x_cci is the de-meaned credit-scaled housing term $CCI \cdot (HA/4y)$, $ilfa_y$ is the combined illiquid-financial ratio (equities + superannuation), $cci_williams$ is the autonomous- consumption CCI loading ζ_c , and yp_x_cci is the credit-scaled permanent-income term.

Spec	Description	Long-run regressors / notes
1	Aggregate net worth	$\ln_networth_y$, $\ln_hp_over_y$, $real_rate$, $\ln_yp_over_y$, ecm_lag
2	Spec 1 + short-run CCI	adds $\Delta^2\log$ CCI lag 2 to short-run set
3	Net worth in levels	replaces $\ln_networth_y$ with $networth_y$
4	Disaggregated wealth	adds nla_y , eq_y , $super_y$, ha_y ; drops aggregates
5	Spec 4 + full short-run dynamics	adds $\Delta^2\log$ CCI, $\Delta\Delta_4income$, $\Delta^2\log$ unemp, $ \hat{\epsilon} $
6	Conventional constant-MPC disaggregated ECM (baseline)	Spec 5 + post-2008 PI break $\ln_yp_over_y_post2008$; plain $ha_y/eq_y/super_y/nla_y$, CCI short-run only
6b	Spec 6 with back-extension-compatible SR CCI	replaces $\Delta^2\log$ CCI with $\Delta^2\log$ RBA D02 credit; disaggregated wealth proxies; fits on $n = 180$ (1977Q3+ window less short-run lags)
7	Spec 6 + cohort terms + synthetic burden	adds $prime_age_share$, fhb_share
7b	Spec 7 with RBA E13 measured burden	post-2009 sample only
8	Williams CCI interactions (free)	Spec 4 with the plain $real_rate$ and $\ln_hp_over_y$ levels replaced by $r \times CCI$ and $\log(HP/y) \times (1 - 1.2 \cdot CCI)$, plus ha_x_cci and $\log(y^p/y) \times CCI$; plain ha_y retained
9	Spec 8 with Kalman state-space CCI	mirrors Spec 8's structure exactly (same de-meaned interaction set incl. $ha_x_cci_k$; plain ha_y retained) with cci_kalman in place of the spline, isolating the CCI series
10	Williams-prior calibrated	$\gamma_IFA = 0.022$, $\psi_0 = 0.20$, $\psi_1 = 0.93$, $\omega = 1.2$; iterative fixed-point OLS
11	Faithful LIVES (free, headline)	housing via ha_x_cci only; $\zeta_c \cdot CCI$ intercept $cci_williams$ restored; IFA combined ($ilfa_y$); nla_y , $hp_x_1_minus_cci$, r_x_cci , $\ln_yp_over_y$, yp_x_cci
12	Williams-calibration-imposed LIVES	Spec 11 form with $\psi_0 = 0.20$, $\psi_1 = 0.93$, $\gamma_IFA = 0.022$ imposed via iterative fixed-point

The crucial structural difference is between Spec 6 and Spec 11. Spec 6 carries the wealth components as plain, constant marginal propensities (ha_y , eq_y , $super_y$, nla_y) and lets the credit-conditions index enter only through a short-run term, $\Delta^2 \log CCI$ at lag 2. It is the form that the broader empirical literature, and an earlier draft of this work, treated as "the" LIVES equation. It is not. As §4 sets out, the LIVES theory predicts no classical housing-wealth effect: the housing marginal propensity is zero at $CCI = 0$ and is unlocked only as credit conditions ease, so housing must enter through the interaction term ha_x_cci and not as a standalone level. Spec 11 imposes exactly that form — housing enters only via ha_x_cci , the autonomous- consumption CCI loading ζ_c ($cci_williams$) is restored, and the two illiquid-financial components are combined into $ilfa_y$ as the theory specifies. The contrast between the two is therefore a matter of functional form dictated by theory, not of fit; §7 shows that the form is decisive for whether the error-correction and wealth structure come alive.

1.7.2 6.2 The four selection screens

Following the structural-econometrics tradition (Hendry and Krolzig 2005; Doornik 2009), we screen each estimable specification through four formal tests, with BIC as a tiebreak. The screens and their per- specification outcomes are recorded in `australia_spec_selection.csv`; the four-sample λ stability that feeds the stability screen is in `australia_lambda_robustness.csv`; the cointegration statistics are in `australia_cointegration.csv`.

1. **Sign screen** — every long-run coefficient carrying a non-ambiguous theoretical prior (§4.2) has the correct sign.
2. **Cointegration screen** — an Engle–Granger residual test: an ADF (with drift) on the residual of the static long-run regression rejects the no-cointegration null at 5 per cent, evaluated against MacKinnon (1991, 2010) critical values keyed to the number of regressors in the cointegrating regression, *not* the univariate Dickey–Fuller value. Phillips–Ouliaris results are reported alongside for Specs 1–3. A Johansen trace statistic is also reported, but it should be read for what it is: **one fixed trivariate subsystem per specification** — log consumption, log income, and either $\ln_networth_y$ (Specs 1–3) or ha_y (all other specs) — estimated with $K = 2$ lags and a restricted constant, testing only the $r = 0$ null. It is *not* a per-specification test of the specification's own long run, and its uniform rejection of $r = 0$ (`johansen_r1_pass = TRUE` for every estimable form in `australia_cointegration.csv`) says only that the small common subsystem cointegrates.
3. **Speed-of-adjustment screen** — λ has the correct (negative) sign and lies in the interval (0.02, 0.30).
4. **Stability screen** — Chow at 2008Q3 is not rejected at the 1 per cent level, *and* λ is sign-stable across at least 3 of the 4 sample variants (full, pre-COVID, COVID-dropped, COVID rich-dummies). The Chow statistic uses `strucchange::sctest` where computable; when the subsample design is singular (e.g. dummies that are all-zero on one side of the break) the pipeline falls back to a manual common-coefficient Chow F-test, and the method actually used per specification is recorded in the `chow_method` column of `australia_full_diagnostics.csv`. A Chow that is incomputable even after the fallback is treated as *neutral*, not as a failure — an earlier vintage treated incomputability as failure, which spuriously "failed" 12 of the 14 specifications.

Two screens warrant comment for the LIVES interaction specifications (Specs 8–12). The cointegration battery now covers Spec 8 and Spec 11 directly: both fail, like every other specification (Spec 11: EG ADF -3.13 against a MacKinnon 5 per cent critical value of -5.47 for nine regressors; Spec 8: -3.40 against -5.70 for ten). Three forms are skipped with an explanatory note in `australia_cointegration.csv`: Spec 9, because its de-measured Kalman interactions are constructed locally inside the estimation step and are not available to the static long-run regression, and Specs 10 and 12, because their long run is a calibrated offset rather than a freely estimated static regression, so an Engle–Granger residual test is not applicable. And the upper bound of the speed-of-adjustment screen ($|\lambda| < 0.30$) is binding for the strongest-adjusting forms: Specs 8 and 11 both adjust faster than the 0.30 ceiling on the full sample ($|\lambda| = 0.458$ and 0.448), and so are flagged as failing the $|\lambda|$ screen even though their λ is correctly signed, strongly significant, and sign-stable across all four sample variants. We return to this in §6.3 and §7: the full-sample λ for the faithful LIVES form is inflated by the COVID quarters, and the identified value is the pre-COVID estimate, $\lambda = -0.266$ ($t = -4.85$), which lies comfortably inside the screen interval and within roughly 7 per cent of Williams' -0.286 .

1.7.3 6.3 Selector outcome

Under the canonical `PI_METHOD = 'italy'` setting, **no specification passes all four screens** — the cointegration screen fails wherever it is computable — so the selector falls back to its documented most-passes rule (most screens passed, BIC tiebreak). Spec 1 and Spec 3 each pass three screens (signs, λ , stability), and the BIC tiebreak returns **Spec 3** (aggregate net worth in levels, -919.4 against Spec 1's -919.3) as the automated pick — a conservative, non-LIVES form. The full screen card, taken directly from `australia_spec_selection.csv`, is:

Spec	Signs	Coint	λ	Stability	BIC
1	✓	×	✓	✓	-919.3
2	×	×	✓	✓	-501.7
3 (selector-preferred)	✓	×	✓	✓	-919.4
4	×	×	✓	✓	-905.3
5	×	×	✓	✓	-494.6
6 (conventional baseline)	×	×	✓	✓	-492.5
6b	✓	×	✓	×	-1114.0
7	×	×	×	✓	-500.6
7b	×	×	×	✓	-364.5
8	×	×	×	✓	-952.8
9	×	NA	✓	×	-890.6
10	✓	NA	✓	×	-493.2
11 (faithful LIVES, headline)	✓	×	×	×	-954.8
12	✓	NA	✓	×	-893.8

Source: `australia_spec_selection.csv` (the BIC column is the Schwarz criterion recorded there and in `australia_full_diagnostics.csv`). Spec 6b carries the lowest (best) raw BIC only because it is fitted on the longest, $n = 190$ back-extended sample and so is not directly comparable to the $n = 146$ or $n = 86$ forms.

Several patterns emerge, and we read them as a coherent statement about the limits of single-equation identification rather than as a clean verdict.

No specification clears the cointegration screen. Evaluated against MacKinnon critical values keyed to the regressor count (`coint_adf_5pct_cv` ranging from -4.42 to -6.13 across the estimable forms), no single-equation specification rejects the no-cointegration null. The richer forms come closest — Specs 4/5 reach ADF -3.12 against -5.23 , Spec 6 -3.12 against -5.47 , the faithful Spec 11 -3.13 against -5.47 , Spec 8 -3.40 against -5.70 , and the back-extended Spec 6b -3.47 against -5.47 — but none crosses its critical value. The aggregated Specs 1–3 fall far short (ADF -1.75 to -1.95 against -4.42). Phillips–Ouliaris likewise fails for Specs 1–3. The Johansen column rejects $r = 0$ for every estimable form, but — as §6.2 cautions — it tests one fixed trivariate subsystem (consumption, income, and one wealth ratio; $K = 2$, $r = 0$ only), not each specification's own long run, so it cannot rescue the screen. A static single-equation long run between consumption and its wealth/income determinants is therefore not formally established on this sample. This is one of the paper's recurring themes (§5, §7.3, §9): the long-run identification Williams obtains comes from his cross-equation FIML system, not from any single equation. Because the screen never passes, selection falls to the remaining three screens with the BIC tiebreak.

The sign screen passes for the faithful LIVES form — and now fails for Spec 8. Spec 11 passes the sign screen: its long-run coefficients with unambiguous priors are all correctly signed, with $nla_y = +0.0269$ ($t = 3.75$), $ilfa_y = +0.0155$ ($t = 3.09$) and $ln_yp_over_y = +0.459$ ($t = 4.04$) significant and right-signed on the full sample, and $ha_x_cci = +0.0025$ ($t = 0.71$) right-signed though insignificant. Spec 8, the free-interaction form, *fails* the sign screen in the current vintage because its separately entered equities component is wrong-signed ($eq_y = -0.0014$, $t = -0.11$). The disaggregated Specs 4–6 fail for the same reason — small negative coefficients on eq_y (Spec 6 full: -0.016 , $t = -0.30$) tip the screen. We read this as an identification artefact of the constant-MPC disaggregation, not as a substantive reversal of the illiquid-financial channel, and it is precisely the artefact that the faithful LIVES form removes by combining the illiquid components into $ilfa_y$ (which is then correctly signed and significant).

The faithful LIVES form fails the $|\lambda|$ ceiling and the 2008Q3 Chow on the full sample. Spec 11's full-sample $\lambda = -0.448$ ($t = -3.57$) exceeds the 0.30 ceiling, as does Spec 8's $\lambda = -0.458$ ($t = -3.52$). Both are correctly signed, strongly significant and sign-stable across all four sample variants; the $|\lambda|$ failure is not a sign or significance failure but reflects the full-sample λ being inflated by the COVID quarters. Spec 11 additionally fails the stability screen through its 2008Q3 Chow ($p = 0.0066$ under the manual common-coefficient fallback, rejected at the 1 per cent threshold) — an honest flag that the faithful form's coefficients shift across the GFC boundary on the full COVID-inclusive sample. On the pre-COVID sample Spec 11's $\lambda = -0.266$ ($t = -4.85$) lies inside the screen interval and within roughly 7 per cent of Williams' -0.286 — which is why §7 treats the pre-COVID estimate as the identified value.

The calibration-imposed forms pass the $|\lambda|$ screen for the wrong reason. Specs 10 and 12 pass the speed-of-adjustment screen on the full sample, but only because their λ has *collapsed* toward zero (Spec 10: -0.048 , $t = -0.78$; Spec 12: -0.030 , $t = -0.74$), placing it just inside the lower edge of the $(0.02, 0.30)$ interval. Both fail the stability screen — their λ is not sign-stable across samples (`lambda_sign_stable_across_samples = FALSE` in `australia_lambda_robustness.csv`; Spec 12's pre-COVID λ flips to $+0.041$, Spec 10's COVID-dropped λ flips to $+0.009$). A screen that is satisfied by a near-zero, sign-unstable adjustment speed is exactly the case the stability screen exists to catch. The economic content of this collapse — that imposing Williams' Australian calibrations wrecks the equilibrium — is developed in §7.0.1 and §9.

BIC and theory now agree on the headline; only the conservative screens dissent. In the current vintage the faithful Spec 11 carries the best BIC of all fourteen specifications (-954.8), ahead of Spec 8 (-952.8), so the fit criterion and the LIVES theoretical form point to the *same* specification. What stands between Spec 11 and the automated pick is the conservative screen pair: the $|\lambda| < 0.30$ ceiling, which the COVID-inflated full-sample estimate (-0.448) breaches, and the 2008Q3 Chow. The selector therefore returns Spec 3 — correctly, by its own documented rules — while the narrative leads with Spec 11. This is a much narrower divergence than in earlier vintages (where selector, BIC and theory pointed to three different specifications), but we still treat it **as a documented limitation, not a result to be argued away**: the long run does not cointegrate in any single equation, the five CCI-carrying regressors are 0.66 – 0.97 mutually collinear (§5), and a mechanical rule built around a 0.30 adjustment ceiling cannot endorse a form whose full-sample speed is inflated by the pandemic quarters. We do *not* promote Spec 8's fast adjustment ($\lambda = -0.458$) as a CCI success; its interaction coefficients are wrong-signed or insignificant (§7) and it re-allocates identification across the interaction block rather than sharpening any single channel.

Accordingly, the body of this paper leads with the **faithful LIVES specification (Spec 11)** on theoretical-form grounds — it is the form Williams (2010) and the LIVES tradition adopt, it passes the sign screen, it carries the best BIC, and (on the identified pre-COVID sample) it recovers Williams' error-correction speed. We retain **Spec 6 as the conventional constant-MPC baseline** — not the headline or the preferred specification — because it is the form prior work treated as LIVES and because it permits the $\gamma_{LA} + \gamma_{LOANS} = 0$ net-liquid restriction test (§8). We carry **Spec 12 (and its independent reproduction Spec 10) as the negative control** that shows the structure transfers but the calibrations do not, and **Spec 3 as the automated selector-best** comparator. The full per-specification coefficient vectors, with Newey–West HAC standard errors and now including the Spec 11 and Spec 12 columns, are in Appendix B.

1.8 7. Results

1.8.1 7.0 The LIVES headline specification (Spec 11)

The specification we lead with, Spec 11, is the faithful Muellbauer–Williams Eq (7) form. It differs from the conventional disaggregated ECM (Spec 6, §7.1 below) in three structural respects, each dictated by the theory rather than by fit: (i) housing wealth enters **only** through its credit-conditions interaction $CCI \cdot (HA/4y)$ — there is no standalone ha_y level, because the Williams housing m.p.c. is zero at $CCI = 0$; (ii) the autonomous-consumption intercept $\zeta_c \cdot CCI$

is restored; and (iii) illiquid financial assets are combined into a single ratio (equities + super-annuation) rather than split, which removes a collinear, wrong-signed equities coefficient. The CCI interactions de-mean the interacted variable over the 1980Q1+ window, following Williams (§4.7). Spec 11 omits Williams' demographic $\Delta_4\text{DEMFTB} / \Delta_4\text{WAPOP}$ terms, his DSRISK downside-risk index and the $\Delta_4\ln c$ habit term (none constructible on our data, §3.8), and enters the unemployment uncertainty term un-interacted — so it is faithful to Eq (7)'s *credit architecture*, not a term-for-term replication.

Full-sample (1988Q3–2024Q4, $n = 146$, $\text{adj-}R^2 = 0.82$) long-run coefficients (australia_all_results.csv, Spec 11 rows; the full coefficient vector under all four sample treatments is in australia_spec11_variants.csv):

Term	OLS coef	t-stat	Implied γ (= OLS/ $ \lambda $)	Williams
ha_x_cci (γ_1 , housing $\times$ CCI)	+0.0025	+0.71	0.005	0.049
nla_y (γ_{NLA})	+0.0269	+3.75	0.060	0.159
ilfa_y (γ_{IFA})	+0.0155	+3.09	0.035	0.022
cci_williams (ζ_c)	+0.0001	+0.01	0.000	0.190
hp_x_1_minus_cci (α_4)	+0.0279	+3.08	0.062	−0.130
r_x_cci (α_1)	+0.0028	+4.55	0.006	−0.871
ln_yp_over_y (ψ_0)	+0.459	+4.04	1.024	0.20
yp_x_cci (ψ_1)	−0.510	−1.51	−1.138	0.93
ecm_lag (λ)	−0.448	−3.57	(= 1)	−0.286

And the headline speed of adjustment and credit-invariant wealth coefficients across the four sample treatments (australia_spec11_variants.csv):

Variant	n	λ (t)	nla_y (t)	ilfa_y (t)	ha_x_cci ln_yp_over_y (t)	(t)
Full sample	146	−0.448 (−3.6)	+0.0269 (3.8)	+0.0155 (3.1)	+0.0025 (0.7)	+0.459 (4.0)
Pre-COVID (to 2019Q4)	126	−0.266 (−4.8)	+0.0159 (1.8)	+0.0093 (1.7)	+0.0019 (0.9)	+0.298 (5.8)
COVID quarters dropped	138	−0.248 (−6.7)	+0.0170 (2.1)	+0.0098 (2.0)	+0.0023 (1.1)	+0.281 (8.7)
Quarterly COVID dummies	146	−0.242 (−6.2)	+0.0141 (3.1)	+0.0084 (3.3)	+0.0016 (0.8)	+0.278 (8.2)

The contrast with Spec 6 (§7.1) is stark, and — unlike in an earlier draft — it does not rest on the COVID quarters. **The error-correction and core wealth structure come alive.** The identified speed of adjustment is $\lambda \approx -0.25$, nearly invariant across the pre-COVID, COVID-dropped and quarterly-dummy treatments (−0.266 / −0.248 / −0.242, t-ratios −4.8 to −6.7) and

close to Williams' -0.286 ; the full-sample -0.448 is inflated by the COVID quarters (the three pulse dummies are demonstrably insufficient — replacing them with quarterly dummies halves $|\lambda|$) and fails the upper-bound $|\lambda|$ screen, so we do not headline it. The net-liquid and illiquid-financial m.p.c.s are correctly signed in every variant and significant at 5 per cent in the full-sample, COVID-dropped and quarterly-dummy treatments (pre-COVID they retain the same magnitudes with $t \approx 1.7$ – 1.8); the delta-method intervals on the structural values are $\gamma_{\text{NLA}} = 0.060$ $[0.022, 0.098]$ and $\gamma_{\text{IFA}} = 0.035$ $[0.012, 0.057]$ (`australia_gamma_inference.csv`). The permanent-income response is strong in every variant ($t = 4.0$ – 8.7). Because the CCI-spline interactions replace Spec 6's 2002Q3-binding `cci_ratio` short-run term, the model estimates on $n = 146$ rather than $n = 86$ — though the *credit channels* themselves have identifying variation only on the ~ 70 post-2007 quarters where the deployed CCI moves (§5.1.1). Reading a standalone, insignificant `ha_y` in Spec 6 as a failed housing wealth effect was therefore a category error: the theory predicts that coefficient to be ≈ 0 , and the housing effect lives in the credit interaction.

Three honest qualifiers temper the headline. First, **the credit-conditions interactions themselves remain weakly identified**. The housing-collateral term `ha_x_cci` (γ_1) is correctly signed but insignificant in every variant (full-sample $t = 0.71$; implied peak m.p.c. 0.0055 $[-0.010, 0.021]$ against Williams' 0.049 , which the interval excludes); the ζ_c intercept is essentially zero on the full sample though right-signed and significant in the COVID-controlled variants ($+0.0199$, $t = 2.77$ pre-COVID; $+0.0148$, $t = 4.32$ COVID-dropped); the real-rate interaction is wrong-signed and significant in every variant (α_1 should be negative); and the affordability interaction is wrong-signed (positive) and significant on the full sample. The permanent-income credit slope `yp_x_cci` is wrong-signed full-sample (-0.51 , $t = -1.5$) but flips to the correct sign in the pre-COVID ($+0.19$, $t = 1.8$) and COVID-dropped ($+0.10$, $t = 1.7$) variants. This is the signature of two compounding problems documented in §5: the CCI-scaled regressors are 0.66 – 0.97 mutually correlated in absolute value (each $\approx$ proportional to the latent index), and the deployed CCI has no variation before 2007Q3, so the liberalisation episode that identifies the credit channels in Williams' 1978–2008 sample is absent from ours.

Second, **the structural permanent-income gearing violates the theory bound**. By the §4.2 recovery rule, $\hat{\psi} = \text{OLS}/|\lambda|$ is 1.02 on the full sample and 1.12 – 1.13 in the COVID-controlled variants — above the admissibility bound $\psi \leq 1 - \eta \approx 0.95$ implied by the discounting that defines y^p . The breach is not the GFC ogive (removing it gives $\lambda = -0.574$ and $\hat{\psi} \approx 1.05$; `australia_spec11_ogive_robustness.csv`), and the delta-method interval $[0.86, 1.18]$ does not exclude 0.95 , so the violation is not itself statistically decisive. Candidate explanations are the unit-income restriction forcing $\ln(y^p/y)$ to absorb low-frequency consumption-to-income drift, and the measure's non-causal, post-2014 tail-extrapolated construction (§4.4); under the causal real-time variant the coefficient reverses sign entirely (§7.4), so the strong positive gearing is a property of the full-sample *measurement*, not an operational forecasting relationship.

Third, the wealth-magnitude comparison with Williams now *rejects* as well as *accepts*: γ_{NLA} 's interval $[0.022, 0.098]$ excludes his 0.159 , while γ_{IFA} 's interval comfortably includes his 0.022 (§7.3.1). The agreement is on form and on the illiquid-financial channel; the net-liquid magnitude is genuinely smaller in Australia post-1988 than in his 1978–2008 sample.

1.8.2 7.0.1 The calibration route does not transfer (Spec 12, Spec 10)

Because the interactions cannot be freely identified, the natural single-equation response is Williams' own: calibrate the credit channels and estimate only what the data can support. Spec 12 imposes Williams' scale-robust calibrations ($\gamma_{\text{IFA}} = 0.022$, $\psi_0 = 0.20$, $\psi_1 = 0.93$) via an iterative fixed-point offset and frees only the housing-collateral m.p.c., the net-liquid m.p.c. and λ . (His real-rate, affordability and intercept loadings cannot be imposed at their published magnitudes: the repo's percent-scaled real rate against a unit-normalised CCI makes $\alpha_1 = -0.871$ roughly thirty times too large, diverging the fixed point. The CCI-support problem cuts deeper than scale: on the deployed index, which ranges over $[-2.1, 1]$ rather than Williams' $[0, 0.8]$, the imposed gearing $\psi(\text{CCI}) = 0.20 + 0.93 \cdot \text{CCI}$ is *negative* in the post-2019 regime and breaches Williams' own $\psi \leq 0.95$ cap at the peak — part of the collapse below is therefore the calibration being evaluated on a support it was never defined for, a caveat that applies to any structural reading of the imposed-calibration specifications.)

The result is decisive and negative: **imposing Williams' permanent-income calibration collapses the error-correction to $\lambda \approx 0$** (Spec 12: $\lambda = -0.030$, $t = -0.74$, flipping sign pre-COVID; independently reproduced by the pre-existing Spec 10, $\lambda = -0.048$, $t = -0.78$, which keeps the rate and affordability channels free). The mechanism is that the Australian data freely estimate a structural permanent-income gearing of order one (§7.0) — several times Williams' 0.20 — so forcing his value injects a large, mis-signed contribution that destroys the equilibrium. The LIVES *structure* transfers; Williams' Australian *calibrations* do not. This sits with, and explains, the companion paper's Wald result that the joint calibration is *not rejected* ($\chi^2(6) = 7.55$, $p = 0.27$; LIVES/outputs/williams_calibration_wald.csv): the freely estimated Spec 6 coefficients are too imprecise to reject Williams' values jointly, but imposing them still ruins the fit — low power is not the same as good fit. Sharpening the credit channels therefore requires either the four-equation FIML system (whose cross-equation restrictions supply the identifying variation the single equation lacks) or a pre-1988 back-extension that recovers the financial-liberalisation episode — not a single-equation calibration.

1.8.3 7.1 Conventional disaggregated specification (Spec 6) — baseline

Spec 6 is the constant-marginal-propensity disaggregated ECM. We retain it here not as the preferred or headline result — that role belongs to the faithful LIVES form of §7.0 — but as the **conventional baseline** against which Spec 11 is the faithful alternative. It is the specification an earlier draft, and much of the Australian aggregate-wealth-effect literature, would treat as "the LIVES equation": it carries plain ha_y , eq_y , $super_y$ and nla_y with no credit scaling, and CCI enters only as a short-run term ($d2_logcci_lag2$). It is therefore the right object against which to make the form-is-decisive point concrete, not the equation we recommend.

Over the full 1988Q3–2024Q4 sample Spec 6 fits on $n = 86$ after lag truncation. The binding constraint is cci_ratio from ABS Cat 5601.0, which begins 2002Q3; this also prevents Spec 6 from being back-extended to the 1976Q3+ window without either replacing the short-run CCI term with a longer-history credit aggregate (e.g. $\Delta^2 \log$ of $credit_total_d02$) or setting it to zero pre-2002. We retain the 2002Q3+ binding constraint here and report the back-extension exercise on the simpler Spec 4 in §7.3 and §8.15.

The long-run coefficients of Spec 6 under canonical Italy LP are (australia_full_results.csv, Spec 6 rows):

Term	OLS coef	NW SE	t-stat	Implied γ (= OLS/ $ \lambda $)	Sign OK
ha_y	+0.0022	0.0076	+0.30	+0.009	✓
nla_y	+0.0083	0.0351	+0.24	+0.035	✓
eq_y	-0.0156	0.0517	-0.30	-0.065	×
super_y	+0.0060	0.0091	+0.66	+0.025	✓
ln_hp_over_y	+0.0102	0.0444	+0.23	+0.043	n/a
real_rate	-0.00053	0.0011	-0.46	-0.0022	✓
ln_yp_over_y	+0.3253	0.2173	+1.50	+1.363	n/a
ln_yp_over_y_post2008	+0.1704	0.1965	+0.87	+0.714	n/a
ecm_lag (λ)	-0.2386	0.0935	-2.55	(=1)	✓

(Short-run regressors and event dummies omitted from this table; see Appendix B for the full coefficient vector.)

The contrast with Spec 11 is now sharper than a simple significant-versus-insignificant split. In summary:

- **Speed of adjustment.** $\lambda = -0.239$ (NW SE 0.093), $t = -2.55$, $p = 0.013$ — significant at the 5 per cent level and about 83 per cent of Williams' published -0.286 . But the significance leans on the COVID quarters: the pre-COVID estimate collapses to -0.087 ($t = -0.79$, *australia_precovid_results.csv*), where Spec 11's pre-COVID λ is -0.266 with $t = -4.85$ (§7.0). The error-correction is identified in this form only when the COVID episode supplies the variation.
- **Housing wealth.** OLS coefficient $+0.0022$, $t = 0.30$ — statistically indistinguishable from zero as a *standalone* level. This is the coefficient the theory predicts to be ≈ 0 absent the CCI interaction; reading it as a failed housing wealth effect is the category error §7.0 identifies. The implied structural $\gamma_{HA} = 0.009$ is now far below Williams' 0.0488 (the numerical coincidence an earlier draft reported did not survive the deflator fix to *ln_hp_over_y* and the pipeline re-run; §7.3).
- **Net liquid assets.** OLS $+0.0083$ ($t = 0.24$, insignificant), implied $\gamma_{NLA} = 0.035$ — about a fifth of Williams' 0.159, though correctly signed. The $\gamma_{LA} + \gamma_{LOANS} = 0$ cross-equation restriction is accepted at the 5 per cent level (§8.5).
- **Illiquid financial wealth.** Decomposed into equities ($\gamma = -0.065$, wrong-signed but $t = -0.30$, statistically indistinguishable from zero) and superannuation ($\gamma = +0.025$, $t = 0.66$); the combined $\gamma_{IFA} = -0.040$ is wrong-signed, dragged below zero by the equities component. The negative point estimate on equities is a small-sample identification artefact of the disaggregated split: combining the two illiquid components into a single *ilfa_y* ratio (as Spec 11 does, §7.0) restores a positive, significant coefficient ($+0.0155$, $t = 3.09$).
- **House-price affordability.** OLS $+0.0102$ ($t = 0.23$), implied $\gamma = +0.043$; Spec 6 does not include the affordability $\times (1 - \omega \cdot \text{CCI})$ interaction that Williams' framework uses to identify this channel, so the level coefficient is not a like-for-like comparison with his -0.13 .
- **Real mortgage rate.** OLS -0.00053 , insignificant in the level; the credit-conditions-contingent rate effect Williams models enters only through the $r \times \text{CCI}$ interaction, which Spec 6 omits.

- **Permanent income.** Base coefficient +0.325 (SE 0.217, $t = 1.50$), plus a post-2008 break of +0.170 (SE 0.196, $t = 0.87$) — neither individually significant. The implied structural gearing at $CCI = 0$ is 1.36, well above Williams' calibrated 0.20 (§7.3.1).
- **Diagnostics.** $\text{adj-}R^2 = 0.80$, Durbin–Watson 2.15, $AR(1)$ $p = 0.34$, $AR(4)$ $p = 0.30$ (no serial correlation), RESET $p = 0.0003$ (functional-form misspecification remains), heteroskedasticity structural (NW HAC SEs throughout; `australia_full_diagnostics.csv`). λ is sign-stable across all four sample variants (full -0.239 , pre-COVID -0.087 , COVID-dropped -0.162 , COVID-rich -0.177 ; `australia_lambda_robustness.csv`).

The key reading is comparative, not absolute: Spec 6 now delivers a significant full-sample error-correction, but its wealth channels are individually unidentified (no wealth t -ratio exceeds 0.7), one of the four is wrong-signed, and the λ identification evaporates pre-COVID. Spec 11 recovers the same theory on the same data with significant, correctly signed core wealth channels and a λ that survives every COVID treatment — the difference is the functional form, not the sample, the data vintage, or the estimator.

1.8.4 7.2 Diagnostics summary

Diagnostic results for all fourteen specifications (full sample) are summarised below; full per-spec output is in `australia_full_diagnostics.csv` (pre-COVID counterparts in `australia_precovid_diagnostics.csv`).

	Spec	n	adj R ²	DW	AR(1) p	AR(4) p	RESET p	Het p (diagnosis)	BIC
	1 (LogNet-Worth)	146	0.731	2.32	0.026	0.093	<0.001	<0.001 (struct)	-919.3
	2 (LogNetWorth_CCI)	86	0.772	2.44	0.020	0.165	0.003	<0.001 (struct)	-501.7
	3 (Level-NetWorth)	146	0.731	2.32	0.025	0.088	<0.001	<0.001 (struct)	-919.4
	4 (Disagg_NoCCI)	146	0.726	2.32	0.018	0.067	<0.001	<0.001 (struct)	-905.3
	5 (FullDisagg)	86	0.802	2.25	0.107	0.087	<0.001	<0.001 (struct)	-494.6
	6 (conventional baseline)	86	0.804	2.15	0.339	0.301	<0.001	<0.001 (struct)	-492.5
	6b (LongHist-SRCCI)	180	0.703	2.12	0.275	0.134	<0.001	<0.001 (struct)	-1114.0
	7 (Cohort-Burden)	86	0.834	2.21	0.203	0.050	0.018	0.002 (struct)	-500.6
	7b (RBABurden)	64	0.872	2.16	0.327	0.004	0.109	0.003 (struct)	-364.5
	8 (CCI_Interactions)	146	0.827	1.92	0.494	0.057	0.001	<0.001 (struct)	-952.8
	9 (KalmanCCI)	146	0.735	2.18	0.140	0.018	<0.001	<0.001 (struct)	-890.6
	10 (WilliamsPrior)	86	0.778	2.10	0.510	0.230	0.085	<0.001 (struct)	-493.2
	11 (LIVES_Headline)	146	0.824	1.90	0.448	0.055	0.001	<0.001 (struct)	-954.8
	12 (LIVES_Calibrated)	146	0.686	2.08	0.555	0.009	0.003	<0.001 (struct)	-892.3

(Table regenerated from `australia_full_diagnostics.csv`; AR(1)/AR(4) are Breusch–Godfrey p-values, "Het p" is the Breusch–Pagan-type LM p-value and the parenthetical is the `het_diagnosis` field. The diagnostics file also records a `chow_method` column for the 2008Q3 stability statistic: where `strucchange::sctest` is singular — because event dummies are all-zero in one subsample — the p-value comes from a manual split-sample Chow F-test on the coefficients estimable in both subsamples, labelled `manual_common_coef`; previously those specifications "failed" the stability screen by incomputability rather than by evidence.)

Four diagnostic patterns are worth noting. First, **heteroskedasticity is structural in every full-sample specification** — the LM rejection survives dropping the event quarters in every case — so the Newey–West HAC standard errors used throughout are necessary rather than a precaution. Second, **RESET rejects in every specification bar the Williams-prior Spec 10** ($p = 0.085$) **and the short-sample Spec 7b** ($p = 0.109$), indicating functional-form misspecification that the single-equation framing does not fully resolve (consistent with §5.3/§7.3/§9); the faithful Spec 11 is not exempt (RESET $p = 0.001$). Third, **low-order serial correlation is concentrated in the aggregated and disaggregated-no-CCI forms**: the Breusch–Godfrey AR(1) test rejects for Specs 1–4 but for none of the richer forms, so the disaggregated and credit-spline dynamics absorb the serial dependence the aggregated specifications leave in the residual; at the fourth order the rejections sit instead in the burden, Kalman and calibrated forms (Specs 7, 7b, 9, 12), with Spec 8 ($p = 0.057$) and Spec 11 ($p = 0.055$) borderline. Fourth, **the headline LIVES form (Spec 11) now carries the best BIC of any $n = 146$ specification, including the over-parameterised Spec 8** (-954.8 vs -952.8): the form correction is not bought at the cost of fit. The automated selector nonetheless prefers Spec 3, because Spec 11 fails the λ -bound and stability screens (§6.3) — the selector divergence is about admissibility screens, not fit.

1.8.5 7.3 Comparison with Williams (2010, 2012): the conventional baseline

The faithful comparison with Williams — where the LIVES structure recovers his error-correction speed and core wealth m.p.c.s — is in §7.0 and is taken up systematically in §9. Here we present the comparison for the *conventional baseline* Spec 6, because that comparison is what an earlier draft (and much of the literature) would have offered as the headline.

We compare Spec 6 to Williams' published Table 1 estimates from the BIS chapter (Muellbauer and Williams 2012). Williams reports **structural** long-run coefficients γ ; our OLS coefficients relate to those γ by the ECM identity $\text{OLS_coef} = \lambda \times \gamma$, so a difference in the implied γ can come from either the OLS coefficient or λ . Reporting both forms separates the two channels (australia_williams_comparison.csv):

Term	Williams γ	Williams implied OLS	Our OLS	Our γ	OLS gap	γ gap
λ	-0.2860	(same)	-0.2386	(same)	-17 %	(same)
Housing wealth ha_y	0.0488	0.0140	0.0022	0.0094	-84 %	-81 %
Illiquid eq_y + super_y	0.0220	0.0063	-0.0096	-0.0403	wrong sign	wrong sign
Net liquid nla_y	0.1590	0.0455	0.0083	0.0350	-82 %	-78 %
log(HP/y)	-0.1300	-0.0372	+0.0102	+0.0429	wrong sign	wrong sign
ψ at CCI = 0	0.2000	0.0572	0.3253	1.3632	—	—

(The combined illiquid row sums the eq_y and super_y OLS coefficients and their structural values from `australia_full_results.csv`; the negative total is driven entirely by the wrong-signed equities split, §7.1.)

Mechanically, the picture has inverted relative to an earlier draft of this paper (which was estimated on a deflator-contaminated `ln_hp_over_y` regressor; Appendix D). The speed of adjustment is now the coefficient that *agrees* with Williams — our -0.239 sits 17 per cent below his -0.286 , well inside one standard error — while the disaggregated wealth OLS coefficients sit roughly 80 per cent below his implied OLS values. Because the λ gap is now small, the ECM identity $\gamma = \text{OLS}/|\lambda|$ no longer rescales the wealth deficits away: γ_{HA} comes out at 0.0094 against Williams' 0.0488, γ_{NLA} at 0.035 against 0.159, and the combined illiquid γ is wrong-signed. There is no offsetting-deficits arithmetic to report; the conventional baseline matches Williams on speed and misses him on every wealth channel.

Two readings discipline this. First, none of the wealth gaps is sharply estimated: §7.3.1 shows the Spec 6 delta-method intervals are wide enough to contain Williams' value *and* zero for every wealth coefficient, so the 80 per cent shortfalls are imprecise, not rejections. Second, and more fundamentally, Spec 6 is not the LIVES equation: its housing channel is a standalone level that the theory predicts to be ≈ 0 , so neither a match nor a miss against Williams' credit-unlocked m.p.c. carries structural content. The faithful Spec 11 makes the housing channel explicit ($\gamma_1 = 0.0055$, $t = 0.71$) — right-signed, weakly identified, and below Williams numerically — which is the honest read on what the single-equation Australian data support. The §9 comparison is therefore built on Spec 11, not on this baseline.

A natural follow-up question is whether the simpler disaggregated no-CCI specification (Spec 4) — closer in form to Williams' Table 1 long-run cointegrating regression than Spec 6 is — would align with Williams' Table 1 on a longer sample. Refitting Spec 4 on the back-extended 1976Q3+ sample using the disaggregated wealth proxies (§3.13; `spec46_extended_comparison.csv`; structural long-run coefficients $\gamma = \text{OLS}/|\lambda|$):

LR coefficient (γ)	1988+ baseline (n=146)	1976+ extended (n=190)	Williams Table 1
λ (ecm_lag)	-0.182	-0.203	-0.286
nla_y	+0.025	+0.001	+0.159
eq_y	-0.043	-0.041	+0.022‡
super_y	+0.014	-0.005	(incl.)‡
ha_y	+0.019	+0.018	+0.0488

‡ Williams reports a single illiquid-financial-asset MPC ($\gamma_{\text{IFA}} = 0.022$), shown here against the combined eq_y + super_y. Williams' structural MPCs ($\gamma_{\text{HA}} = 0.0488$, $\gamma_{\text{NLA}} = 0.159$, $\gamma_{\text{IFA}} = 0.022$) are the BIS-chapter values in `australia_williams_comparison.csv`, the same source used in §7.3's headline comparison and §11.4.

On Spec 4 the speed of adjustment moves 11.5 per cent toward Williams ($-0.182 \rightarrow -0.203$, still 29 per cent short of -0.286), but the individual wealth coefficients become *smaller*, not larger: γ_{NLA} collapses by 95 per cent toward zero, γ_{SUPER} flips sign, and γ_{EQ} retains a wrong sign. The back-extension therefore does not push the Spec 4 estimates closer to Williams'

Table 1; the post-1988 sample window is not, in itself, what generates the divergence between Spec 4 and Williams' values. **Sample length is not the binding constraint.**

Reading the two exercises together points to the same conclusion as §7.0.1: the wedge between any single-equation OLS estimate and Williams' system FIML is the single-equation framing itself, not sample length, knot count, or sign-prior structure. The placebo evidence (§5.2) and the two-equation SUR result (§5.4) — in which joint estimation delivers a negligible cross-equation residual correlation and no efficiency gain — corroborate this reading.

We do *not* present Spec 8 (the free six-interaction form, $\lambda = -0.458$) as a CCI success here. Adding the full Williams CCI interaction set re-allocates the long-run identification across wealth components ($\gamma_{\text{HA}} = 0.007$, $\gamma_{\text{IFA}} = 0.044$, $\gamma_{\text{NLA}} = 0.073$) without moving systematically toward Williams' Table 1, and three of its four sign-priced interaction coefficients carry the wrong sign against their priors (§5.5, §8.4). The magnitude of its λ reflects the collinear interaction set absorbing variation, not a sharpening of the credit channel.

1.8.6 7.3.1 How precisely is the structural profile identified?

The implied structural coefficients $\gamma_i = \beta_i/|\lambda|$ are ratios of two imprecisely estimated quantities, so they inherit sampling uncertainty that the point-estimate comparison above conceals. We compute, for both the conventional baseline (Spec 6, $n = 86$) and the faithful LIVES headline (Spec 11, $n = 146$), delta-method standard errors from the Newey–West covariance of (β_i, λ) — which carries the correlation between each numerator and the speed of adjustment — and, as a cross-check, a seeded moving-block residual bootstrap (block length 8, $B = 1000$). Both blocks are committed in `australia_gamma_inference.csv`.

For **Spec 6** the intervals are wide enough to be almost uninformative:

Term (Spec 6)	Implied γ	95% CI (delta)	Williams Table 1	Williams in CI?
Housing ha_y	0.009	[−0.056, 0.075]	0.0488	✓
Net liquid nla_y	0.035	[−0.262, 0.332]	0.159	✓
Equities eq_y	−0.065	[−0.468, 0.337]	0.011	✓
Super $super_y$	0.025	[−0.040, 0.091]	0.011	✓
$\log(\text{HP}/y)$	0.043	[−0.308, 0.393]	−0.130	✓
Permanent income	1.363	[0.285, 2.442]	0.200	×
Wealth aggregate (Σ)	0.004	[−0.659, 0.668]	0.230	✓

Williams' Table 1 value lies inside the 95 per cent interval for every Spec 6 coefficient except the permanent-income gearing (whose interval [0.29, 2.44] excludes his calibrated 0.20 from above) — but the same intervals also contain zero for every wealth channel and values far from Williams. The bootstrap agrees (median $\gamma_{\text{HA}} = 0.008$, 95 per cent interval [−0.049, 0.107]). On the conventional baseline, then, the data cannot distinguish Williams' wealth profile from a broad range of alternatives, including no wealth effect at all: **non-rejection driven by imprecision, not confirmation.**

For **Spec 11** the intervals are several times tighter, and they become informative in *both* directions:

Term (Spec 11)	Implied γ	95% CI (delta)	Williams Table 1	Williams in CI?
Housing $\times$ CCI ha_x_cci (γ_1)	0.0055	$[-0.010, 0.021]$	0.0488	$\times$
Net liquid nla_y	0.060	$[0.022, 0.098]$	0.159	$\times$
Illiquid ilfa_y	0.035	$[0.012, 0.057]$	0.022	$\checkmark$
Affordability hp_x_1_minus_cci	0.062	$[0.027, 0.098]$	-0.130	$\times$
Permanent income (ψ_0)	1.024	$[0.865, 1.184]$	0.200	$\times$
PI $\times$ CCI yp_x_cci (ψ_1)	-1.138	$[-2.099, -0.177]$	0.930	$\times$
Wealth aggregate (Σ)	0.100	$[0.036, 0.164]$	0.230	$\times$

The bootstrap intervals are again close to the delta-method ones (e.g. γ_{NLA} [0.013, 0.111], γ_{IFA} [0.005, 0.065]). The Spec 11 read is therefore no longer a blanket non-rejection. The data **accept** Williams' illiquid financial m.p.c. (0.022 sits comfortably inside [0.012, 0.057]) but **reject** his net-liquid magnitude (0.159 against an upper bound of 0.098), his credit-unlocked housing m.p.c. (0.0488 against $[-0.010, 0.021]$), the sign of his affordability loading, the magnitude of his permanent-income gearing, and the sign of his permanent-income credit slope. The agreement with Williams is on *form* — the error-correction speed and the significance pattern of the wealth channels — not on the credit-channel calibrations, which is the same verdict §7.0 and §9 reach from the point estimates.

For use in a calibrated model (MARTIN, §10.3), the policy-relevant summary is Spec 11's aggregate long-run wealth coefficient $\gamma_W = 0.100$ (95% CI [0.036, 0.164]): positive and bounded away from zero, but below both Williams' 0.230 and MARTIN's calibrated net-wealth elasticity of ≈ 0.17 (the latter sits just outside the upper bound). The Spec 6 aggregate (0.004, CI $[-0.66, 0.67]$) is too imprecise to discipline anything.

(Caveat: both the delta method and the residual bootstrap hold the right-hand side fixed, so they propagate sampling uncertainty in the ECM coefficients but **not** the first-stage uncertainty in the generated permanent-income and CCI regressors — the permanent-income projection and the CCI are held at their full-sample values in both methods; the true intervals are therefore at least this wide. The real-time permanent-income sensitivity of §7.4 and §8.9 gives a partial read on the first-stage component.)

1.8.7 7.4 The Italy / AR comparison and the real-time check

The permanent-income measure matters materially for two coefficients — the speed of adjustment and the long-run permanent-income coefficient. Refitting Spec 6 ($n = 86$) under each measure on a common data flow (australia_pi_realtime_robustness.csv):

Term	AR (real-time)	Italy full-sample (headline measure)	Italy real-time	Williams
ecm_lag (λ)	-0.095 (t = -1.68)	-0.239 (t = -2.55)	-0.159 (t = -2.39)	-0.286
ln_yp_over_y	-0.158 (t = -1.68)	+0.325 (t = 1.50)	-0.145 (t = -2.23)	(calib. 0.20)
ha_y (OLS)	+0.0135 (t = 1.76)	+0.0022 (t = 0.30)	+0.0024 (t = 0.30)	(implied 0.014)

(The refit now reproduces the canonical pipeline's Spec 6 estimates exactly under the full-sample Italy measure — $\lambda = -0.2386$, matching `australia_full_results.csv` — since the pipeline-isolation fix described in §8.2. The `ha_y` row reports the OLS coefficient, which equals the implied γ only after dividing by $|\lambda|$, consistent with the CSV.)

Two readings follow. First, the full-sample Italy measure multiplies $|\lambda|$ by roughly two and a half relative to AR ($-0.095 \rightarrow -0.239$) and flips the long-run permanent-income coefficient from negative (-0.158 , $t = -1.68$ — the "Australian permanent-income puzzle") to positive ($+0.325$, in sign agreement with theory and Williams' calibrated value, though insignificant at $t = 1.50$). Second — the operational caveat — **only part of either move survives a causal real-time projection**. The real-time Italy variant keeps roughly two-thirds of the full-sample $|\lambda|$ (-0.159 against -0.239) and is now itself significant ($t = -2.39$), so about a third of the headline $|\lambda|$ is attributable to look-ahead rather than to the Italy projection per se. The permanent-income coefficient, by contrast, returns to significantly negative under the causal measure (-0.145 , $t = -2.23$). The positive-PI sign is therefore a property of the full-sample, two-sided measure, not of the real-time forecaster a model like MARTIN would use.

We retain the structural reasons the AR and Italy measures diverge — the rolling AR(8) forecaster lacks the labour-force-share predictor that captures Australia's slow-moving demographic effects on trend income, compounds short-run AR misspecification across 40 horizons, and over-estimates persistence after large income shocks, all of which the one-step direct projection avoids — but we read the puzzle's reversal as a feature of full-sample permanent-income *measurement* rather than a clean real-time resolution. We carry the full-sample measure as the headline and the real-time and AR variants as robustness columns (§8.9), and we flag throughout that the strong permanent-income response in the faithful Spec 11 ($t = 4.0$, §7.0) inherits this look-ahead caveat: it is conditioned on the full-sample permanent-income forecaster, and the real-time read is the operational lower bound.

1.9 8. Robustness

We run the Italian-style robustness suite of De Bonis, Liberati, Muellbauer and Rondinelli (2020) on the automated-selector specification (Spec 3, §6.3), on the conventional constant-MPC baseline (Spec 6) where the battery requires the disaggregated wealth split, and — in parallel, with a `_spec11` suffix on the committed outputs — on the faithful LIVES specification (Spec 11). The suite is deliberately weighted toward exposing rather than concealing weakness: it spans estimator choice (OLS, IV, SUR), break testing, the credit-conditions construction, the permanent-income measure, the affordability adjustment, sample length, and

out-of-sample forecast accuracy. Several of the batteries return honest negatives — the multi-step out-of-sample loss to a random walk, the near-zero cross-equation residual correlation, the below-median placebo percentiles for the literal Williams construction, and the partial reversal under a causal permanent-income measure. We report these prominently. They are substantive findings: a freely-estimated single-equation framework permits them to surface, whereas the imposed restrictions of a four-equation FIML system would hide them, and each points to why FIML and pre-1988 back-extension are the routes forward (§5, §9).

1.9.1 8.1 OLS vs IV on current income (Hall 1978 endogeneity)

Because current income enters both the error-correction term ($\text{ecm_lag} = \ln c_{\{t-1\}} - \ln y_t$) and the permanent-income gap ($\ln_{yp_over_y}$), the exercise now instruments **both** regressors — an upgrade on an earlier draft, which treated ecm_lag as exogenous and instrumented only the PI gap. The instrument set is income lagged one, two and four quarters, unemployment lagged one and two quarters, and the mortgage rate lagged one quarter (six instruments); both the OLS and IV columns carry Newey–West HAC standard errors (`australia_iv_robustness.csv`, Spec 3, $n = 146$).

Instrumenting moves the speed of adjustment *away* from zero: $\text{ecm_lag} = -0.191$ (OLS) $\rightarrow -0.258$ (IV, HAC SE 0.110), +35 per cent, and the permanent-income coefficient from +0.196 to +0.280 (+43 per cent); the small net-worth coefficient flips sign (+0.0011 $\rightarrow$ -0.0009) but is statistically indistinguishable from zero either way. The IV diagnostics (`australia_iv_diagnostics.csv`) say the exercise is well-posed but the endogeneity it corrects is marginal: the weak-instrument F-statistics are 73.6 ($\ln_{yp_over_y}$) and 28.9 (ecm_lag), both far above conventional thresholds; the Wu–Hausman test does not reject OLS exogeneity at 5 per cent ($p = 0.095$); and the Sargan over-identification test does not reject instrument validity at 5 per cent ($p = 0.074$), though both are close enough to the boundary to keep honest.

Run on the faithful LIVES specification (`australia_iv_robustness_spec11.csv`, `australia_iv_diagnostics_spec11.csv`, $n = 146$), the exercise is more demanding and correspondingly less conclusive, and we read it diagnostically rather than confirmatorily. Correcting an error in an earlier run — which had let three of the CCI interactions instrument themselves — the endogenous set is now the five income- and wealth-carrying terms (ecm_lag , $\ln_{yp_over_y}$, yp_x_cci , ha_x_cci and $hp_x_1_minus_cci$) against the same six instruments, which leaves only a single overidentifying restriction. The Wu–Hausman test still rejects exogeneity ($p = 0.002$), so current-income endogeneity is a live concern in the LIVES form; but the first stages are uneven — the weak-instrument F-statistics are 126.0 ($\ln_{yp_over_y}$), 55.5 (ha_x_cci), 32.5 (ecm_lag) and 15.2 ($hp_x_1_minus_cci$), yet only 6.5 for yp_x_cci , below the conventional weak-instrument threshold. λ does move away from zero ($-0.448 \rightarrow -0.503$, +12 per cent), in the direction of a *faster* error-correction; but the wealth-channel point estimates cannot bear weight. nla_y attenuates to statistical insignificance ($0.0269 \rightarrow 0.0040$, HAC $t \approx 0.3$), and the two housing interactions flip sign relative to OLS ($ha_x_cci +0.0025 \rightarrow -0.013$; $hp_x_1_minus_cci +0.028 \rightarrow -0.015$). With only one overidentifying restriction (Sargan $p = 0.251$, a near-vacuous test at $df = 1$) and a weak first stage on yp_x_cci , these movements are not reliable evidence for or against the wealth channel in either direction — the IV design is underpowered for Spec 11. What the exercise establishes is narrower than the earlier draft claimed: that endogeneity matters here (Wu–Hausman), not that the LIVES wealth m.p.c.s survive instrumentation.

1.9.2 8.2 Joint permanent-income + consumption SUR

Estimating the consumption equation jointly with the permanent-income equation by SUR (`australia_joint_pi_robustness.csv`, Spec 3, $n = 146$) leaves every coefficient within sampling noise of the single-equation OLS values. The largest level shift among the long-run terms is in the speed of adjustment (`ecm_lag` $-0.191 \rightarrow -0.204$, +7.0 per cent in relative terms but barely a hundredth in level), with the permanent-income coefficient moving -4.5 per cent ($0.196 \rightarrow 0.187$). On the faithful LIVES specification (`australia_joint_pi_robustness_spec11.csv`, $n = 146$) the stability is even tighter: λ $-0.448 \rightarrow -0.457$ (+1.9 per cent), `nla_y` -0.2 per cent, `ilfa_y` +2.6 per cent, `ln_yp_over_y` -1.4 per cent. Single-equation OLS is therefore an acceptable estimator for the consumption block, and — consistent with the two-equation SUR result in §8.18 — the case for the multi-equation framework rests on cross-equation parameter restrictions, not on residual covariance or efficiency gain.

Two honest disclosures attach to this battery. First, the second equation is a *one-step-ahead* income-growth forecast, not the $k = 40$ -quarter discounted projection that defines the permanent-income regressor, and the SUR does not propagate the first-stage uncertainty of the generated permanent-income series into the consumption-equation standard errors — it tests residual covariance, not generated-regressor bias (the §7.3.1 caveat applies here too). Second, the IV and SUR tables in a previous draft had been produced on data contaminated by a pipeline-ordering bug: the diagnostic that builds the AR permanent-income series had overwritten the canonical Italy-method series in the shared model dataset before the robustness batteries ran. The pipeline now isolates each sub-script's data flow, and the committed tables here and in §8.1 are estimated on the canonical data (which is why the §7.4 refit now matches `australia_full_results.csv` exactly).

1.9.3 8.3 Chow battery and multiple-break tests

Chow tests on the selector specification (`australia_chow_battery.csv`, Spec 3, $n = 146$) do not reject parameter stability at 1995Q1 (stat 0.380, $p = 0.978$; $n_{\text{pre}} = 27$, $n_{\text{post}} = 119$), 2000Q1 (0.740, $p = 0.730$) or 2008Q3 (0.529, $p = 0.912$), while the 2020Q1 break is strongly rejected (Chow stat 11.12, $p = 8.9 \times 10^{-16}$; $n_{\text{pre}} = 127$, $n_{\text{post}} = 19$) — the COVID structural break that the event dummies absorb. On the faithful LIVES specification (`australia_chow_battery_spec11.csv`) 1995Q1 ($p = 0.844$) and 2000Q1 ($p = 0.152$) are stable, but the 2008Q3 break is rejected at 5 per cent (stat 1.914, $p = 0.017$) — consistent with the GFC being exactly where Spec 11's credit-interaction structure starts to bind (the deployed CCI's first knot is 2007Q3, §5.1.1) — and the 2020Q1 statistic is not computable (`sctest` singular on the COVID-dummied subsample; the CSV records the failure). The Bai–Perron / CUSUM battery (`australia_breaks.csv`, Spec 3) returns a single dominant break: $\text{supF} = 169.96$ ($p = 0$) dated 2019Q1 — the trimming convention places the date at the segment boundary adjacent to the COVID episode — with a CUSUM p -value of 0.971 (recursive residuals stable away from that episode). For the selector specification the break structure is therefore concentrated at COVID, not at the GFC; for the LIVES form a 5 per cent GFC break is the honest additional finding.

1.9.4 8.4 Williams CCI interactions (Spec 8) — reallocation, not identification

Spec 8 enters all six Williams CCI interactions freely on the 1988Q3+ sample with the reduced-form `cci_williams`. Following the §4.7 convention (Williams' own), it is the inter-

acted economic *variable* that is de-means — over the 1980Q1-onwards window on which the CCI is observed — while the CCI itself enters the interaction raw, so that each interaction term has a clean conditional interpretation rather than absorbing an implicit linear CCI level shift. The sign-prior verdicts (`australia_spec8_sign_prior_verdicts.csv`; full vector in `australia_full_results.csv`, $n = 146$) are:

Williams interaction	Sign prior	OLS coef	t	p	Verdict
$r \times \text{CCI}$	–	+0.00283	+2.05	0.042	wrong sign, significant – FAIL
$\log(\text{HP}/y) \times (1 - 1.2 \cdot \text{CCI})$	–	+0.0299	+1.38	0.171	wrong sign on composite, insignificant – FAIL
$\log(y^p/y) \times \text{CCI}$	+	–0.5046	–2.43	0.016	wrong sign, significant – FAIL
$\log(y^p/y)$	+	+0.4921	+3.42	0.001	right sign – PASS
$\text{HA} \times \text{CCI} (\gamma_1)$	+	+0.00426	+0.87	0.384	right sign, insignificant

Three of the four sign-priced interactions fail. Of the free interactions only the permanent-income level term ($\ln_{yp_over_y}$, +0.4921, $t = 3.42$) passes its prior cleanly; the housing-collateral interaction ha_x_cci carries its theoretically correct positive sign but is far from significant (+0.00426, $t = 0.87$). Against Williams' Table 1 the raw-coefficient gaps remain structural (`australia_williams_spec8_comparison.csv`): his α_{c1} ($r \times \text{CCI}$) = –0.871 against our structural +0.0062; his α_{c4} ($\text{HP}/y \times (1 - 1.2 \cdot \text{CCI})$) = –0.13 against our structural +0.065; his calibrated $\psi_1 = +0.93$ against our freely-estimated structural –1.101.

The substantive effect of Spec 8 is to *re-allocate* the long-run identification, not to recover Williams' channels. Standalone non-housing wealth strengthens (nla_y +0.0337***, struct +0.073; $super_y$ +0.0215**, struct +0.047; ha_y stays small at +0.0030, $t = 0.48$) and the speed of adjustment shifts from –0.239 (Spec 6) to –0.458 ($t = -3.52$) on the full sample — past Williams' value in magnitude. Spec 8 also attains the highest adjusted R^2 among the $n = 146$ specifications (0.827) and the second-best BIC (–952.8, behind Spec 11's –954.8). We do **not** read any of this as a credit-conditions success. The pre-COVID estimates are revealing (`australia_precovid_results.csv`): there the standalone housing level is significant (ha_y +0.0154***, $t = 2.87$) but the *interactions* turn negative — ha_x_cci at –0.0069 ($t = -1.42$) and $hp_x_1_minus_cci$ at –0.0292 ($t = -1.54$), the former the wrong sign for the LIVES collateral channel — while λ halves to –0.219 ($t = -3.95$). Read together with the calibration collapse (§9, Spec 10/12) and the identification-vs-detrending decomposition (§5), the honest reading is that the six CCI-interacted regressors in a single equation act as flexible parameter time-variation rather than as the structurally identified common-factor channel that Williams' four-equation system delivers. Their mutual collinearity (each is approximately proportional to CCI) is the structural reason FIML is required; we discuss it as a first-class identification result in §5.

1.9.5 8.5 Net-liquid-assets restriction $\gamma_{LA} + \gamma_{LOANS} = 0$

We refit each disaggregated specification with $deposits/y$ and $debt/y$ entered separately and conduct a Wald test of $H_0 : \gamma_{LA} + \gamma_{LOANS} = 0$ using `car::linearHypothesis` with the

Newey–West variance estimator. The restriction is **accepted at the 5 per cent level in every specification × sample combination** (`australia_nla_restriction_test.csv`):

Spec	Sample	$\gamma_{LA} + \gamma_{LOANS}$	NW SE	t	p	Restriction
4	full	+0.0325	0.0445	0.731	0.465	accepted
5	full	+0.0422	0.0999	0.422	0.673	accepted
6	full	+0.0202	0.0987	0.204	0.838	accepted
4	pre-COVID	+0.0282	0.0285	0.989	0.322	accepted
5	pre-COVID	-0.0222	0.0499	-0.445	0.656	accepted
6	pre-COVID	-0.0026	0.0509	-0.051	0.959	accepted

The data cannot distinguish separate liquid-asset and debt propensities. We read this as non-rejection-by-imprecision rather than as positive confirmation of exact netting, but it validates the Italian convention of netting deposits against debt and supports the use of the constructed `nla_y` series — the net-liquid channel that carries a significant, correctly-signed marginal propensity in the faithful LIVES specification (Spec 11, `nla_y` +0.0269***, struct 0.060; §7.0).

1.9.6 8.6 Drehmann amortising-mortgage adjusted real rate

De Bonis et al. (2020) apply the BIS Drehmann, Juselius and Korinek (2017) amortisation-adjusted rate $\text{adjR} = R / (1 - (1+R)^{-N})$. For Australia we set $N = 25$ years (100 quarters), consistent with the longer Australian average mortgage maturity (`australia_drehmann_robustness.csv`, Spec 3, $n = 146$). The substitution is now essentially inert, matching the Italian invariance result: `ecm_lag` -0.1906 (base) $\rightarrow$ -0.1902 (Drehmann), the net-worth coefficient 0.00108 $\rightarrow$ 0.00129, the permanent-income coefficient 0.1963 $\rightarrow$ 0.1959, and the real-rate coefficient itself -0.00020 $\rightarrow$ -0.00015 (insignificant either way). (The large Drehmann sensitivity an earlier draft reported was an artefact of the contaminated data flow disclosed in §8.2.) The exercise is undefined for the faithful LIVES specification: Spec 11 carries no plain `real_rate` term — its rate enters only through the `r × CCI` interaction — so the Drehmann substitution cannot be applied, and `australia_drehmann_robustness_spec11.csv` records the attempt as FAILED for that reason.

1.9.7 8.7 Scaled-income robustness

De Bonis et al. (2020) average disposable income with labour-plus-transfer income to down-weight property-income mismeasurement. Re-running this construction (`australia_scaled_income_robustness.csv`, Spec 3, $n = 146$) shifts the speed of adjustment modestly from -0.1906 to -0.2111 and the permanent-income coefficient from 0.196 to 0.163, with the net-worth coefficient roughly halving (0.00108 $\rightarrow$ 0.00061) from an already negligible base. On the faithful LIVES specification (`australia_scaled_income_robustness_spec11.csv`) the income-measure choice likewise moves λ from -0.448 to -0.525 while *strengthening* the wealth m.p.c.s (`nla_y` 0.0269 $\rightarrow$ 0.0344, `ilfa_y` 0.0155 $\rightarrow$ 0.0175) and trimming the permanent-income coefficient (0.459 $\rightarrow$ 0.376). The income-measure choice moves $|\lambda|$ by 0.02–0.08 in level — not negligible — but does not change the substantive ranking or signs of the wealth coefficients in either frame.

1.9.8 8.8 Williams non-property income (NPY) robustness

Replacing the disposable-income series with `npv_real_pc` constructed per Williams (2009) §4.2.1 (`australia_williams_income_robustness.csv`, Spec 3, $n = 146$) provides the closest methodological match to Williams' income concept (property income stripped, but not symmetrically averaged with labour-plus-transfer income). The substitution is now close to inert on the speed of adjustment — `ecm_lag` $-0.1906 \rightarrow -0.1857$, a 2.6 per cent change — with the permanent-income coefficient easing from 0.196 to 0.164 and the net-worth coefficient staying negligible ($0.00108 \rightarrow 0.00044$). (An earlier draft reported the NPY substitution roughly halving $|\lambda|$; that result did not survive the §8.2 data fix.) On the faithful LIVES specification (`australia_williams_income_robustness_spec11.csv`) the NPY measure trims λ from -0.448 to -0.395 while *raising* the wealth m.p.c.s (`nla_y` $0.0269 \rightarrow 0.0413$, `ilfa_y` $0.0155 \rightarrow 0.0203$) and easing the permanent-income coefficient ($0.459 \rightarrow 0.352$). The income concept is therefore not what separates our estimates from Williams': under his own income measure the conventional λ is essentially unchanged and the LIVES wealth channels strengthen.

1.9.9 8.9 Permanent-income method comparison (AR, full-sample Italy, real-time Italy)

§7.4 reports the headline three-way comparison in the Spec 6 frame; the committed batteries span two specification frames, and we label each row accordingly. The method comparison (`australia_pi_method_comparison.csv`) runs on the selector specification (Spec 3, $n = 146$ — its Italy-column estimates match the Spec 3 rows of `australia_full_results.csv` exactly), while the real-time battery (`australia_pi_realtime_robustness.csv`) runs on the conventional baseline (Spec 6, $n = 86$):

PI measure	Frame	λ (<code>ecm_lag</code>)	t	$\log(y^p/y)$	t	adj-R ²
AR (expanding-window)	Spec 3, $n = 146$	-0.0479	-1.12	-0.0036	-3.96	0.696
Italy LP (full-sample)	Spec 3, $n = 146$	-0.1906	-2.90	$+0.1963$	$+3.41$	0.731
AR (expanding-window)	Spec 6, $n = 86$	-0.0948	-1.68	-0.1583	-1.68	—
Italy LP (full-sample)	Spec 6, $n = 86$	-0.2386	-2.55	$+0.3253$	$+1.50$	—
Italy LP (real-time)	Spec 6, $n = 86$	-0.1594	-2.39	-0.1451	-2.23	—

(Adjusted R² from `australia_pi_method_meta.csv`; the Italy LP forecaster fits better in the Spec 3 frame, 0.731 vs 0.696. The AR forecaster is *always* expanding-window — it refits the AR(8) on data through each forecast origin — so there is no separate "full-sample AR" variant to report.)

The full-sample Italy LP measure is the headline measure (framed as a *measurement*, §4.3), and the AR forecaster delivers the negative "Australian permanent-income puzzle" coefficient in both frames (significantly so in the Spec 3 frame, $t = -3.96$). The real-time Italy LP variant is the operationally honest benchmark: it is causal, re-fitting the projection at each t on data whose full k -quarter horizon is realised by t , so it is usable at forecast time. It shows that

roughly two-thirds of the full-sample Italy $|\lambda|$ is genuine (-0.159 against -0.239 , and now itself significant at $t = -2.39$) — the look-ahead accounts for about a third — but that the positive permanent-income sign is **not**: it reverses to -0.145 ($t = -2.23$) under the causal measure. We carry the full-sample measure as the headline and disclose explicitly that its positive-PI sign and part of its λ magnitude are full-sample, non-causal properties. The same look-ahead caveat attaches to the strong permanent-income coefficient in the faithful LIVES specification (Spec 11, $\ln_yp_over_y +0.459$, $t = 4.0$); the real-time column is the operational robustness check.

1.9.10 8.10 Permanent-income filter sensitivity

A grid over discount factor $\delta \in \{0.90, 0.95, 0.97\}$ and horizon $k \in \{20, 40, 60\}$ quarters, with the GFC learning-weight ogive on and off (`australia_permanent_income_sensitivity.csv`), shows the speed of adjustment to be extremely stable within the AR method on the selector specification (Spec 3 — the grid's λ values match the AR column of §8.9's Spec 3 frame): λ ranges only from -0.0470 to -0.0482 across the eighteen $\delta \times k \times$ ogive cells (baseline $\delta = 0.95$, $k = 40$, ogive on: $\lambda = -0.0479$, structural PI weight -0.076 , structural net-worth weight $+0.122$). The GFC ogive toggle is now a real switch — an earlier version of the grid carried it as a no-op, so its on/off rows were identical; the fixed toggle moves λ by less than 0.0003 and the structural PI weight by at most 0.03 within each $\delta \times k$ cell, so the original "no effect" conclusion survives the fix, now for a demonstrated rather than vacuous reason. The PI weight is more sensitive to the horizon (-0.21 at $k = 20$ down to -0.04 at $k = 60$) but never changes sign within the AR method. Switching to an HP-filter permanent income ($\lambda = 1600$) moves λ to -0.0820 and flips the structural PI weight to $+1.238$. The within-AR-method PI tuning is therefore not what drives the $|\lambda|$ gap with Williams; the dominant factor is the AR-versus-Italy-direct-forecast method choice itself (§8.9).

For the headline LIVES specification the corresponding sensitivities are the GFC-ogive toggle on the Italy measure and the real-time variant. Removing the ogive from the headline measure (`australia_spec11_ogive_robustness.csv`) moves Spec 11's λ from -0.448 to -0.574 ($t = -4.63$) and the permanent-income coefficient from $+0.459$ to $+0.604$, leaving the structural gearing essentially unchanged ($\psi \approx 1.05$ against the headline 1.02) — so the §7.0 bound violation is not an ogive artefact. The real-time variant is §7.4/§8.9.

1.9.11 8.11 COVID-period robustness

All fourteen specifications now carry all **four** sample variants (full, pre-COVID, COVID-dropped, COVID-rich) in `australia_lambda_robustness.csv` — the COVID-rich variant was previously missing for Specs 7–12, so this is the first complete battery. The speed of adjustment is sign-stable across all four variants for every specification except the two Williams-calibration-imposed ones. For the conventional baseline (Spec 6) λ runs -0.239 (full) / -0.087 (pre-COVID) / -0.162 (COVID-dropped) / -0.177 (COVID-rich) — all correctly signed, though the pre-COVID estimate is small and insignificant (§7.1). The faithful LIVES specification (Spec 11) gives -0.448 (full) / -0.266 (pre-COVID) / -0.248 (COVID-dropped) / -0.242 (COVID-rich) — sign-stable, with the full-sample value inflated by the COVID quarters and the tightly clustered -0.24 to -0.27 of the three COVID-controlled variants treated as the identified value (close to Williams' -0.286 ; full coefficient vectors for all four variants in `australia_spec11_variants.csv`). Only Spec 10 (Williams-prior calibrated; sign-flips to $+0.009$ COVID-dropped and $+0.051$ COVID-rich) and Spec 12 (Williams calibrations imposed;

sign-flips to +0.041 pre-COVID and +0.015 COVID-rich) are not sign-stable across samples — an artefact of the calibration collapse documented in §9, not of the COVID episode itself.

1.9.12 8.12 Rolling-window estimation

A 60-quarter rolling estimation of the selector specification (Spec 3; 87 windows ending 2003Q2–2024Q4, `australia_rolling_coefs.csv`) shows λ holding in a band of roughly -0.13 to -0.27 for the windows ending before 2020, the net-worth coefficient trending down from about +0.012 in the earliest windows to near zero through the macroprudential era (consistent with that era flattening the wealth-consumption transmission) and mildly negative in the latest windows (-0.011 at 2024Q4, within one standard error of zero). The COVID quarters destabilise the short windows: λ briefly flips to +0.15 in the windows ending around 2021Q3 before settling at -0.44 (SE 0.19) in the final window, where the COVID quarters dominate a 60-quarter span. We do not interpret this as model instability but as a symptom of the limited identifying variation in the post-deregulation portion of the sample (§5): the financial-liberalisation episode that would identify the credit channels largely predates the 1988Q3 start of ABS sectoral balance-sheet data.

1.9.13 8.13 Out-of-sample forecast validation

We run a rolling out-of-sample validation (`australia_oos_rmse.csv`) on six structural specifications (Spec 4 disagg-no-CCI, Spec 6 conventional baseline, Spec 7 cohort-burden, Spec 8 Williams-CCI interactions, Spec 9 Kalman-CCI, and now the faithful LIVES Spec 11) over 36 expanding-window cuts at horizons $h \in \{1, 4, 8\}$ quarters ($n = 36$ at $h = 1, 4$; $n = 32$ at $h = 8$), against random-walk-with-drift and AR(1) benchmarks:

Specification	$h = 1$ RMSE	$h = 4$ RMSE	$h = 8$ RMSE
Benchmark RW drift	0.03094	0.03094	0.03282
Benchmark AR(1)	0.03703	0.03102	0.03283
Spec 4 (disagg, no CCI)	0.03175	0.03182	0.03896
Spec 6 (conventional baseline)	0.03231	0.03293	0.04180
Spec 7 (cohort-burden)	0.03247	0.03164	0.03540
Spec 8 (Williams CCI)	0.02901	0.03323	0.04038
Spec 9 (Kalman CCI)	0.03206	0.03299	0.03929
Spec 11 (LIVES headline)	0.02919	0.03517	0.06402

At $h = 1$ the two CCI-interaction forms now **beat** the random-walk benchmark — Spec 8 (0.0290) and Spec 11 (0.0292) against RW-drift's 0.0309 — with the remaining structural specifications close behind; this is a better one-step showing than an earlier draft reported. At $h = 4$ and $h = 8$ the random walk with drift still dominates **every** structural specification (best structural at $h = 4$ is Spec 7 at 0.0316 vs 0.0309; at $h = 8$, Spec 7 at 0.0354 vs 0.0328), and Spec 11 is the *worst* performer at $h = 8$ (0.0640) — plausibly the full-sample permanent-income measure's extrapolated tail compounding over long horizons. This is the standard macro-forecasting pattern: the LIVES framework's identification advantage is in interpreting historical co-movement, not in beating naive benchmarks at multi-step prediction. We record it honestly rather than overstating forecast performance.

A construction caveat applies to the whole exercise, and we correct an earlier draft's claim here. As committed, the validator's permanent-income input is the **full-sample Italy-method measure**, not a real-time forecaster (the earlier statement that "the AR permanent-income input is real-time" was wrong for the committed numbers), and the credit-conditions series and its de-mean constants are likewise full-sample objects. Both generated regressors therefore embed information from beyond each expanding-window cut, so these RMSEs measure **fit stability under re-estimation, not genuine real-time forecast accuracy** — they are an upper bound on what a fully real-time forecaster would deliver, and the $h = 1$ wins over the random walk should be read in that light.

1.9.14 8.14 Back-extension robustness — Spec 1 on the 1976Q3+ sample

Refitting Spec 1 (aggregate net worth) on the back-extended sample using `ln_networth_y_proxy` (`spec1_extended_comparison.csv`; structural long-run coefficients $\gamma = \text{OLS}/|\lambda|$):

LR coefficient (γ)	1988+ baseline (n = 146)	1976+ extended (n = 190)	% change
λ (<code>ecm_lag</code>)	-0.1934	-0.2090	+8.1
<code>ln_networth_y</code>	+0.0195	+0.0467	+140
<code>ln_hp_over_y</code>	+0.0443	+0.0375	-15.5
<code>real_rate</code>	-0.00115	+0.00005	sign flip
<code>ln_yp_over_y</code>	+1.0379	+0.9913	-4.5

The speed of adjustment and the permanent-income elasticity are essentially stable across samples (λ -0.193 $\rightarrow$ -0.209; γ_{yp} 1.04 $\rightarrow$ 0.99), so doubling the sample length and adding the deregulation-era regime does not disturb the equilibrium structure. The aggregate wealth elasticity, however, more than doubles (0.019 $\rightarrow$ 0.047) — both values small and imprecisely estimated (baseline $t = 0.17$ on the OLS coefficient, `australia_full_results.csv`), so we read this as the pre-deregulation regime adding identifying variation to a coefficient the modern sample barely pins down, not as parameter instability that can be dated. The house-price-to-income coefficient eases toward zero on the longer sample — consistent with lower `hp_over_y` variation in the pre-deregulation 1970s — and the real-rate coefficient sign-flips, though both estimates are economically negligible. The aggregate net-worth proxy correctly includes the household M3 liquid component (the earlier \$bn/\$m unit defect that had made the M3 term numerically inert is fixed and propagated through a reproducible cold rebuild; see Appendix D).

1.9.15 8.15 Spec 4 on the back-extended sample

The disaggregated-wealth proxies of §3 allow Spec 4 to fit on the back-extended sample (`spec46_extended_comparison.csv`; structural γ): λ moves 11.5 per cent toward Williams (-0.1824 $\rightarrow$ -0.2034, still 29 per cent short of -0.286), but the individual wealth coefficients become *smaller* rather than larger — γ_{NLA} collapses by 95 per cent (+0.0245 $\rightarrow$ +0.0013), γ_{SUPER} flips sign (+0.014 $\rightarrow$ -0.005), γ_{HA} eases (+0.019 $\rightarrow$ +0.018) and γ_{EQ} retains its wrong sign (-0.043 $\rightarrow$ -0.041). The exercise establishes that sample length is **not** the binding constraint on whether the disaggregated single-equation form reproduces Williams' Table 1:

the longer sample sharpens the speed of adjustment but blunts, rather than sharpens, the individual wealth channels.

1.9.16 8.15.1 Spec 6b — conventional baseline on the back-extended sample

Spec 6 binds at 2002Q3+ on the baseline sample because `cci_ratio` (ABS Cat 5601.0 housing-loan flow) begins there. Spec 6b retains the Spec 6 long-run and short-run structure but replaces the short-run CCI regressor with the second difference of log RBA D02 total credit (available from 1976Q3) and switches the wealth components to their back-extended proxies. This lets the conventional baseline fit on the full back-extended sample ($n = 190$ in the cointegration screen; $n = 180$ full / 160 pre-COVID in the diagnostics) (`australia_full_results.csv`; `australia_full_diagnostics.csv`; structural γ):

LR coefficient (γ)	Spec 6 ($n = 86$)	Spec 6b ($n = 180$)	Williams Table 1
λ (<code>ecm_lag</code>)	-0.239 ($t = -2.55$)	-0.248 ($t = -3.96$)	-0.286
<code>ha_y</code> / <code>ha_y_proxy</code> γ	0.009	0.012	0.049
<code>nla_y</code> / <code>nla_y_proxy</code> γ	0.035	0.015	0.159
<code>eq_y</code> / <code>eq_y_proxy</code> γ	-0.065	-0.008	(calibrated 0.011)
<code>super_y</code> / <code>super_y_proxy</code> γ	0.025	-0.001	(calibrated 0.011)
<code>ln_hp_over_y</code> γ	+0.043	+0.024	-0.130
<code>ln_yp_over_y</code> (CCI = 0) γ	+1.363	+1.113	+0.20 (calibrated)
BIC	-492.5	-1 114.0	n/a

Two patterns recur. First, the speed of adjustment moves modestly closer to Williams' published value — $\lambda = -0.248$ on the back-extended sample (87 per cent of Williams' -0.286 , vs 83 per cent on Spec 6) — and is far more sharply estimated, the t -statistic improving from -2.55 to -3.96 , with the pre-COVID estimate nearly identical at -0.240 ($t = -4.61$, `australia_lambda_robustness.csv`): unlike Spec 6, whose λ collapses without the COVID quarters, the back-extended conventional baseline is identified in every sample variant (COVID-dropped -0.240 , COVID-rich -0.234). Second, the wealth γ profile shifts toward still-smaller individual elasticities — γ_{NLA} falls from 0.035 to 0.015 and γ_{SUPER} flips sign, mirroring the Spec 4 back-extension finding — while γ_{HA} stays positive but at a quarter of Williams' 0.049. This is consistent with the substantive reading throughout the paper: on the back-extended sample the disaggregated wealth proxies do not separately identify with Williams-like precision, even when the canonical short-run dynamics, the post-2008 permanent-income break, and a long-history credit proxy are all available. Sample length sharpens $|\lambda|$ but not the individual wealth γ profile — the residual gap is structural to the single-equation framing, not a sample-length artefact.

(The proxy caveats of §3.14 apply with full force: in particular `eq_y_proxy` is held constant at its 1988Q3 value pre-1988, so the equities coefficient is identified only off the modern subsample, and the near-zero back-extended γ_{EQ} and γ_{SUPER} should be read as proxy-limited rather than as estimates of a true pre-1988 propensity.)

1.9.17 8.16 Maximal-GETS placebo on the back-extended sample

We place the canonical knot constructions against 200 random draws under the same sign-survival protocols on the back-extended 1976Q3+ sample. The committed summary files report two extended-sample placebo variants (`australia_williams_knot_placebo_extended_summary.csv`, `australia_williams_knot_placebo_maximal_extended_summary.csv`):

Placebo variant (1976Q3+)	Canonical adj-R ²	adj-R ² percentile	$ \lambda $ percentile	Verdict
Extended (Williams literal 4-knot)	0.6801	36th	26th	"detrending critique persists — below median"
Maximal-GETS (canonical 15- knot reduction)	0.6836	48th	70th	"detrending critique persists — below median"

On the back-extended sample the literal Williams 4-knot construction sits below the placebo median on both metrics (adj-R² 0.6801 versus a placebo median of 0.6815; $|\lambda|$ 0.2023 versus 0.2079). The maximal-GETS reduction does better on mean reversion — its $|\lambda|$ of 0.2563 beats 70 per cent of random draws against a placebo median of 0.2329 — but still sits just below the random median on fit (48th percentile, 0.6836 versus 0.6846), while retaining *fewer* knots (7) than the placebo median (8), i.e. achieving near-median fit with less flexibility. Neither extended-sample construction is the deployed CCI: the deployed-protocol placebo — random knot dates run through exactly the iterated drop-on-violation reduction used to build `cci_williams`, on the 1988Q3+ estimation sample — is the §5.2 result, where the deployed construction reaches the **84th adjusted-R² and 80th $|\lambda|$ percentiles** (`australia_williams_knot_placebo_deployed_verdict.csv`). The honest reading is unchanged from §5.2: on the back-extended sample, where the institutional knot dates predate the data's identifying variation, the canonical constructions do not beat random flexibility, and most of what lift there is comes from the adaptiveness of the reduction protocol rather than from Williams' specific knot or prior choice. We carry the below-median extended-sample percentiles as a core negative result, not as an embarrassment: they are the empirical expression of why single-equation CCI is weakly identified and why the joint system is needed for regime classification (§10).

1.9.18 8.17 Sectional sign-prior CCI

Williams (Aust paper §5.1) imposes sign priors over periods rather than knot by knot. We construct a sectional CCI basis with one knot per period (1982 / 1990 / 1993 / 2007, plus 2014 / 2017 / 2020 / 2021 extensions) and re-run the placebo on the back-extended sample (`LIVES/outputs/sectional_placebo_summary.csv`). The sectional canonical (adj-R² 0.6805, $|\lambda|$ 0.2244) sits at the 37th adjusted-R² percentile and 60th $|\lambda|$ percentile — below the random median on fit, like every extended-sample construction in §8.16, and between the literal 4-knot and maximal-GETS constructions on mean reversion. Williams' specific period dating therefore does not outperform random period placements on the back-extended sample. On the modern 1988Q3+ sample the sectional reduction retains two survivors against the deployed protocol's four, fitting somewhat worse (adj-R² 0.726 vs 0.754; $\lambda = -0.203$ vs -0.246) with a 0.69

correlation between the two indices (LIVES/outputs/sectional_cci_comparison.csv) — a related but coarser credit signal, consistent with the period priors being a constrained subset of the maximal candidate set.

1.9.19 8.18 Two-equation SUR (consumption + house prices)

Joint SUR estimation of the consumption equation and a Williams-style house-price ECM (Aust paper eq. 11) on the back-extended 1976Q3+ sample yields a residual correlation $\hat{\rho}(\varepsilon_C, \varepsilon_H) = -0.0109$ under equation-by-equation OLS and -0.0133 under SUR (LIVES/outputs/lives_sur_2eq_resid_corr.csv; committed verdict "NEGLIGIBLE cross-equation linkage — single-equation OLS approximately efficient"). The SUR coefficients move by under 5 per cent of themselves relative to OLS for every substantive term (10 per cent on the near-zero intercept) — the consumption equation's λ shifts from -0.2101 to -0.2118 (+0.8 per cent) and its SUR standard errors are within 0.01 per cent of the OLS ones (LIVES/outputs/lives_sur_2eq_compare.csv). Joint estimation gives no efficiency gain at the quarterly frequency. The case for the multi-equation framework therefore rests on cross-equation parameter restrictions, not on residual covariance — the same conclusion as the consumption + PI SUR of §8.2.

1.9.20 8.19 Three-equation joint cross-equation CCI identification

We extend the maximal-GETS protocol to require sign-prior survival across **three** equations simultaneously (consumption + house prices

- mortgage stock), and then across **four** (adding the wealth equation). Of the fifteen candidate knots, seven pass consumption-only fitting; only **two** pass the three-equation joint test and only **one** survives all four equations (LIVES/outputs/lives_joint_cci_survival.csv):

Survival regime	Surviving knots
Consumption only	1979Q1, 1986Q1, 1992Q1, 2007Q3, 2009Q1, 2017Q1, 2020Q2
Joint 3-eq (C $\cap$ H $\cap$ M)	1986Q1, 2017Q1
Joint 4-eq (C $\cap$ H $\cap$ M $\cap$ W)	1986Q1

The joint-identified CCI flips the house-price equation's CCI loading from significantly negative (-0.0156 under the consumption-only CCI) to positive ($+0.0240$ under the 3-equation joint CCI; $+0.0236$ under the 4-equation variant; LIVES/outputs/lives_phase3_comparison.csv), consistent with Williams' cross-equation sign restrictions working as intended. The mortgage-stock equation's loading moves the other way — positive under consumption-only CCI ($+0.0026$), negative under the joint variants (-0.0060 / -0.0069) — a reminder that joint sign-survival is a sign restriction, not a parameter-equality restriction; full FIML would be required to discipline all loadings simultaneously. The consumption equation's error-correction is only modestly affected — λ moves from -0.271 (consumption-only) to -0.213 (3-eq) / -0.227 (4-eq) — while its housing/net-worth long-run coefficient sign-corrects under joint identification (-0.031 consumption-only $\rightarrow +0.037$ / $+0.017$ joint), and the SUR residual correlations remain modest ($\hat{\rho}(\varepsilon_C, \varepsilon_H) = -0.119$, $\hat{\rho}(\varepsilon_C, \varepsilon_M) = +0.102$, $\hat{\rho}(\varepsilon_H, \varepsilon_M) = -0.244$). Joint sign-survival therefore does real, if limited, identification work — it re-signs the house-price loading and the consumption wealth term — without closing the magnitude gap against Williams' Table 1, confirming that the residual gap is structural to the single-equation framing, not a CCI-

construction artefact, and that the route to sharpening the credit channels runs through the four-equation FIML system, not through further single-equation tuning.

1.10 9. Comparison with Williams (2010, 2012)

This section consolidates the comparison with Williams' published Table 1 and BIS chapter estimates. The organising thesis is the one that runs through the paper: **the LIVES structure transfers to Australia, but Williams' Australian calibrations do not**. The faithful single-equation LIVES form (Spec 11) recovers Williams' error-correction speed and the correctly signed wealth-m.p.c. structure; imposing his permanent-income gearing (Spec 12, Spec 10) collapses the equilibrium; and the six CCI interactions through which his framework identifies the credit channels are too mutually collinear to be separated off a single equation. We deliberately retire the earlier draft's "the preferred specification reproduces Williams' profile almost exactly" framing: that reading rested on the conventional constant-m.p.c. baseline (Spec 6), on an offsetting-deficit coincidence, and on confidence intervals so wide that they contain both Williams' values and zero. The honest ceiling is consistency, not confirmation.

1.10.1 9.1 Where the structure transfers — the faithful LIVES form recovers Williams' speed and wealth structure

The decisive agreement is on the error-correction speed and the *sign and significance* of the core wealth channels, and it appears only once the equation is written in the faithful LIVES form.

On the COVID-controlled samples the faithful LIVES specification (Spec 11) estimates a speed of adjustment tightly clustered at $\lambda \approx -0.25$ — pre-COVID -0.266 ($t = -4.85$), COVID quarters dropped -0.248 ($t = -6.66$), quarterly COVID dummies -0.242 ($t = -6.25$) — against Williams' FIML estimate of **-0.286 (SE 0.083 , $t = -3.45$)**: about 13 per cent below his value, and tightly identified (`australia_spec11_variants.csv`; `australia_lambda_robustness.csv`). The full-sample estimate is $\lambda = -0.448$ ($t = -3.57$); it is inflated by the COVID quarters and fails the $|\lambda|$ upper-bound screen, so we read the COVID-controlled cluster as the identified speed (§7.0). Either way the contrast with the conventional baseline is sharp: Spec 6 returns $\lambda = -0.239$ ($t = -2.55$) on its $n = 86$ sample — about 17 per cent below Williams — and collapses to an insignificant -0.087 once the COVID quarters are excluded.

Within the faithful form the wealth structure is correctly signed and, for two of the three components, individually significant on the full sample (`australia_all_results.csv`; `australia_gamma_inference.csv`):

- net liquid assets (γ_{NLA}): OLS $+0.0269$ ($t = 3.75$, ***), implied structural m.p.c. **0.060** [0.022 , 0.098], against Williams' calibrated 0.159 — same sign, but genuinely smaller: the interval *excludes* his value (§9.4);
- illiquid financial assets (γ_{IFA} , equities + superannuation combined): OLS $+0.0155$ ($t = 3.09$, ***), implied structural m.p.c. **0.035** [0.012 , 0.057], against Williams' calibrated **0.022** — same sign, and the interval comfortably includes his value;
- housing-collateral (γ_i , the $\text{CCI} \cdot (\text{HA}/4y)$ interaction): OLS $+0.0025$ ($t = 0.71$), implied structural m.p.c. **0.0055** [-0.010 , 0.021], against Williams' peak housing m.p.c. of 0.0488 — correctly signed but insignificant, with an interval that excludes his peak value while containing zero.

The permanent-income response is strong and correctly signed ($\ln_{yp_over_y}$ OLS +0.459, $t = 4.04$) — though its structural gearing breaches the admissibility bound (§7.0, §9.8). And critically, because the CCI-spline interactions replace Spec 6's 2002Q3-binding cci_ratio short-run term, the faithful form estimates on $n = 146$ rather than $n = 86$ — a near-doubling of the estimation window relative to the baseline against which the previous draft compared Williams (though the credit channels themselves are identified only off the ~70 post-2007 quarters where the deployed CCI moves; §5.1.1).

The single most important interpretive point for the comparison is that **the housing channel only appears when the form is faithful**. In the LIVES theory there is no classical housing wealth effect: the housing m.p.c. is zero at CCI = 0 and is unlocked as credit conditions loosen. Reading Spec 6's insignificant standalone ha_y (OLS +0.0022, $t = 0.30$) as a *failed* housing wealth effect was therefore a category error — the theory predicts that coefficient to be ≈ 0 , and the housing effect lives in the credit interaction ha_x_cci , where it is correctly signed (if statistically unproven).

1.10.2 9.2 Where the calibrations do not transfer — imposing Williams' gearing collapses the equilibrium

The opposite result holds for Williams' Australian *calibrations*. The natural single-equation response to weakly identified interactions is Williams' own — calibrate the credit channels and estimate only what the data support — but the data reject that route decisively.

Spec 12 imposes Williams' scale-robust calibrations ($\gamma_{IFA} = 0.022$, $\psi_0 = 0.20$, $\psi_1 = 0.93$) via an iterative fixed-point offset and frees only the housing-collateral m.p.c., the net-liquid m.p.c. and λ . The result is a near-total collapse of the error-correction mechanism: $\lambda = -0.030$ ($t = -0.74$) on the full sample, and it flips to the wrong sign (+0.041, $t = 2.03$, significant at 5 per cent) pre-COVID, so it is not even sign-stable across samples (*australia_lambda_robustness.csv*). This is independently reproduced by the pre-existing Williams-prior specification (Spec 10), which keeps the rate and affordability channels free and still returns $\lambda = -0.048$ ($t = -0.78$) full-sample and a still-insignificant -0.025 ($t = -0.57$) pre-COVID, flipping to the wrong sign once the COVID quarters are dropped or dummied (*australia_lambda_robustness.csv*). Two separate calibration routes therefore reach the same conclusion.

The mechanism is straightforward: the Australian data freely estimate a structural permanent-income gearing of order one — by the §4.2 recovery rule, $\hat{\psi} = OLS/|\lambda|$ is 1.02 on the full sample and 1.12–1.13 in the COVID-controlled variants, roughly five times Williams' 0.20 — so forcing his much lower value injects a large, mis-signed contribution that destroys the long-run equilibrium. (The comparison must be made on a consistent scale: structural ≈ 1.0 –1.1 against his $\psi_0 = 0.20$, or OLS +0.459 against his implied $\phi \cdot \psi_0 \approx 0.057$ — not an OLS-to-structural mix. The freely estimated gearing itself breaches the $\psi \leq 0.95$ admissibility bound, which we flag openly in §7.0; the ψ_0/ψ_1 split is in any case not sharply estimable on a single equation, so we report the contrast as order-of-magnitude rather than as a precise ratio.)

Williams' rate, affordability and autonomous-consumption loadings cannot be imposed at all at their published magnitudes. His raw real-rate loading $\alpha_r = -0.871$ enters his FIML system on differently scaled regressors; on this repository's *percent*-scaled real rate against a *unit*-normalised CCI it is roughly thirty times too large, and the iterative fixed point diverges. This

is itself a finding: the LIVES structure is portable, but its numerical calibrations are specific to Williams' scaling conventions and sample, and cannot be transplanted mechanically.

1.10.3 9.3 Reconciling the companion paper's Wald non-rejection

The companion analysis reports that a joint Wald test on the freely estimated Spec 6 coefficients does **not reject** Williams' calibration: $\chi^2(6) = 7.55$, $p = 0.27$ for all six coefficients jointly, $\chi^2(4) = 1.83$, $p = 0.77$ for the four wealth coefficients, and no individual coefficient rejects either (LIVES/outputs/williams_calibration_wald.csv). Read alongside §9.2 this looks paradoxical: how can a calibration that *collapses* the equilibrium when imposed also be *not rejected* by a test? The resolution is power. The freely estimated single-equation coefficients are so imprecise that they cannot statistically reject Williams' values — but the same imprecision means the data also cannot reject zero, or any number of other points. **Low power is not the same as good fit.** A calibration can be non-rejected by an underpowered test and still wreck the model when imposed; both facts hold here simultaneously, and the gamma-inference confidence intervals (§9.4) make the imprecision explicit: in the Spec 6 frame every Williams *wealth* value lies inside our 95 per cent CI, and every such CI also contains zero. The one place the free estimate is strong enough to disagree is the permanent-income gearing: the Wald test on the OLS scale does not reject ($p = 0.20$), but the delta-method interval on the structural ratio $\gamma = \text{OLS}/|\lambda|$, $[0.28, 2.44]$, excludes Williams' 0.20 — and that is exactly the channel whose imposition produces the §9.2 collapse.

1.10.4 9.4 The structural-gamma comparison: the inference now cuts both ways

The earlier draft's headline — that the implied structural γ profile "matches Williams almost exactly" — was a property of the conventional baseline (Spec 6), and it does not survive the corrected data. Under the ECM identity $\text{OLS} = \lambda \times \gamma$, Spec 6's $|\lambda|$ now runs about 17 per cent below Williams' while its OLS wealth coefficients run roughly 80 per cent below his implied OLS values (e.g. $\text{ha_y} + 0.0022$ against the 0.0140 his $\gamma = 0.0488$ implies at $\lambda = -0.286$; $\text{nla_y} + 0.0083$ against 0.0455), so the implied γ profile sits far *below* Williams' rather than coincidentally close to it ($\gamma_{\text{HA}} 0.0094$ vs 0.0488; $\gamma_{\text{NLA}} 0.035$ vs 0.159). The earlier "offsetting deficits leave γ near Williams" reading is therefore retired on the corrected data, not merely hedged. What keeps Spec 6 *consistent* with Williams is imprecision: every wealth-component interval contains both his value and zero (australia_gamma_inference.csv, Spec 6 rows):

Term (Spec 6, n = 86)	OLS	Implied γ	95% CI	Bootstrap 95% CI	Williams	In CI?
ha_y (housing)	+0.0022	0.009	[-0.056, 0.075]	[-0.049, 0.107]	0.0488	✓
nla_y (net liquid)	+0.0083	0.035	[-0.262, 0.332]	[-0.183, 0.361]	0.159	✓
eq_y (equities)	-0.0156	-0.066	[-0.468, 0.337]	[-0.303, 0.208]	0.011	✓
super_y (super)	+0.0060	0.025	[-0.040, 0.091]	[-0.064, 0.141]	0.011	✓
ln_hp_over_y	+0.0102	0.043	[-0.308, 0.393]	[-0.400, 0.325]	-0.130	✓
ln_yp_over_y	+0.3253	1.363	[0.285, 2.442]	[0.605, 2.192]	0.20	×
WEALTH_AGG	+0.0010	0.004	[-0.659, 0.668]	—	0.230	✓

In the faithful frame the picture sharpens decisively, and in both directions (australia_gamma_inference.csv, Spec 11 rows):

Term (Spec 11, n = 146)	OLS	Implied γ	95% CI	Williams	In CI?
ha_x_cci (γ_1 , housing $\times$ CCI)	+0.0025	0.0055	[-0.010, 0.021]	0.0488	×
nla_y (γ_{NLA})	+0.0269	0.060	[0.022, 0.098]	0.159	×
ilfa_y (γ_{IFA})	+0.0155	0.035	[0.012, 0.057]	0.022	✓
hp_x_1_minus_cci (α_4)	+0.0279	0.062	[0.027, 0.098]	-0.130	×
ln_yp_over_y (ψ_0)	+0.4591	1.024	[0.865, 1.184]	0.20	×
yp_x_cci (ψ_1)	-0.5101	-1.138	[-2.099, -0.177]	0.93	×
WEALTH_AGG	+0.0448	0.100	[0.036, 0.164]	0.230	×

The honest summary is therefore no longer "everything is inside a wide interval". The gamma inference now **cuts both ways**: Spec 11's intervals are tight enough to *reject* parts of Williams' profile — his net-liquid m.p.c. (0.159) lies outside [0.022, 0.098], his peak housing m.p.c. (0.0488) outside [-0.010, 0.021], and his affordability loading has the wrong sign — while his illiquid-financial m.p.c. (0.022) is comfortably *inside* [0.012, 0.057]. The Spec 6 frame remains consistency-by-imprecision on a small $n = 86$ sample; the Spec 11 frame upgrades the comparison to agreement on form and on the illiquid-financial channel, disagreement on the net-liquid magnitude, and an underpowered housing-collateral channel whose interval contains zero.

1.10.5 9.5 Adding the CCI interactions does not close the gap (Spec 8)

Estimating the CCI interactions freely alongside the standalone wealth ratios (Spec 8) raises the speed of adjustment to $\lambda = -0.458$ ($t = -3.52$) full-sample on $n = 146$, in magnitude above Williams' -0.286 , with the second-best BIC in the ladder (-952.8 , behind Spec 11's -954.8). We do not present this as a credit-channel success. The wealth coefficients shift relative to the baseline ($\gamma_{NLA} \rightarrow 0.073$, $\gamma_{super} \rightarrow 0.047$, standalone housing $\rightarrow 0.007$) without moving systematically toward Williams' Table 1, and three of the four interaction sign priors fail outright (`australia_spec8_sign_prior_verdicts.csv`):

- `r_x_cci` (Williams $\alpha_{c1} = -0.871$ at CCI = 1): prior negative, OLS $+0.0028$, $p = 0.042 \rightarrow$ **FAIL** (wrong sign, significant);
- `hp_x_1_minus_cci` (Williams $\alpha_{c4} = -0.13$ at CCI = 0): prior negative, OLS $+0.0299$, $p = 0.171 \rightarrow$ **FAIL**;
- `yp_x_cci` (Williams $\psi_1 = +0.93$): prior positive, OLS -0.505 , $p = 0.016 \rightarrow$ **FAIL** (wrong sign, significant);
- `ln_yp_over_y` (Williams $\psi_0 \approx 0.20$): prior small positive, OLS $+0.492$, $p = 0.0008 \rightarrow$ **PASS**.

The raw-coefficient comparison confirms that the interactions do not land where Williams' framework predicts (`australia_williams_spec8_comparison.csv`): his $\alpha_{c1} = -0.871$ against our structural $+0.0062$ ($t = 2.05$), his $\alpha_{c4} = -0.13$ against our $+0.065$ ($t = 1.38$), his calibrated $\psi_1 = +0.93$ against our freely estimated -1.10 ($t = -2.43$). Adding the interactions therefore **re-allocates** the long-run identification across components — and raises λ — but does not close the residual gap with the joint FIML estimates. The earlier draft's framing that Spec 8 "exceeds Williams" is withdrawn.

1.10.6 9.6 The interactions are collinear: the structural reason Williams uses FIML

The reason single-equation estimation cannot deliver Williams' result is, at root, an identification problem. The CCI-interacted regressors are 0.66–0.97 mutually correlated in absolute value on this sample (`australia_cci_interaction_corr.csv`; §5.5) because each is approximately proportional to the latent CCI; they cannot be separately identified off one equation. The committed evidence runs in the same direction from three sides: the correlation matrix itself; the wrong-signed Spec 8 interaction coefficients of §9.5; and the Spec 10/Spec 12 collapse of §9.2.

This is precisely the structural reason Williams (2010) uses a four-equation FIML system rather than single-equation OLS. The same CCI enters all four of his equations with sign constraints; the affordability multiplier ω in the wealth $\times (1 - \omega \text{CCI})$ interaction is shared across equations; and $\zeta_h = 1$ normalises the house-price equation. Those cross-equation restrictions supply the identifying variation that lets the credit channels be separated — variation that no single-equation specification has. The joint-survival result of §5 corroborates this from the other direction: of the fifteen candidate knots, seven pass the consumption-equation sign prior, but only two of those survive the three-equation requirement (1986 and 2017Q1) and only one survives all four equations (1986) (`LIVES/outputs/lives_joint_cci_survival.csv`), so the single-equation identification was largely equation-specific. This aligns with both Williams' framework (CCI as a common factor under parameter restrictions) and the Duca, Muellbauer and Murphy (2013) state-space implementation in which the latent factor is identified jointly across equations.

1.10.7 9.7 Sample length is not the binding constraint

A natural conjecture is that the divergence from Williams reflects sample period — his 1978Q1–2008Q2 window spans the financial-liberalisation episode that identifies the credit channels, whereas our balance-sheet data begin only in 1988Q3, largely after deregulation. The back-extension exercise tests this directly and finds it is not the binding constraint.

Refitting the simpler disaggregated no-CCI specification (Spec 4) on the back-extended 1976Q3+ sample moves λ about 12 per cent toward Williams ($-0.182 \rightarrow -0.203$, still 29 per cent short of -0.286 ; `spec46_extended_comparison.csv`) — and the aggregate Spec 1 moves similarly ($-0.193 \rightarrow -0.209$; `spec1_extended_comparison.csv`) — but the individual wealth coefficients *shrink* rather than grow on the longer window: γ_{NLA} collapses by 95 per cent ($+0.025 \rightarrow +0.001$), γ_{super} flips sign, and γ_{EQ} retains a wrong sign (§8). Lengthening the sample slightly sharpens the speed of adjustment but not the wealth structure, so the post-1988 window is not, in itself, what generates the divergence between any single-equation OLS estimate and Williams' jointly identified Table 1. The back-extension is retained as a contribution (it builds a 1976Q3-anchored master dataset and permits the direct sample-length test), but it is not the route to closing the gap with Williams.

1.10.8 9.8 The permanent-income measure and the Australian PI puzzle

A final point of comparison concerns the permanent-income channel and the measure on which it depends. Under the AR-method forecaster we replicate the often-noted "Australian permanent-income puzzle": the long-run coefficient on $\log(y^p/y)$ is negative — near-zero on the full-sample (look-ahead) AR variant (-0.004) and negative though insignificant once the AR forecaster is run in real time (-0.158 , $t = -1.68$) (`australia_pi_method_comparison.csv`, `australia_pi_realtime_robustness.csv`). Under the full-sample Italian direct-forecast *measure* (De Bonis et al. 2020, Appendix A.2; §2.5) it is strongly positive ($+0.196$ on the net-worth forecaster regression; $+0.325$ in the Spec 6 frame; $+0.459$ in the faithful Spec 11). The structural reasons the measures diverge are real — the rolling-AR forecaster lacks the labour-force-share predictor that captures slow demographic effects, compounds short-run AR misspecification across 40 horizons, and over-estimates persistence after large income shocks, all of which the one-step direct forecast avoids. But the positive sign is a property of the full-sample, non-causal construction of the measure: under a causal real-time direct forecast the coefficient flips to significantly negative (-0.145 , $t = -2.23$), even though the error-correction term itself survives the real-time treatment ($\lambda = -0.159$, $t = -2.39$, against the AR real-time -0.095). The strong positive permanent-income response that the faithful LIVES form recovers therefore rests on the full-sample (look-ahead) permanent-income measure; we flag this explicitly and direct readers to the real-time robustness column rather than treating the puzzle as resolved.

1.10.9 9.9 Summary

The comparison with Williams resolves into a clean two-part statement, with one honest refinement. The LIVES *structure* transfers: the faithful single-equation form recovers his error-correction speed (COVID-controlled $\lambda \approx -0.25$ against -0.286 , a gap of about 13 per cent) and the correctly signed wealth-m.p.c. structure (net-liquid and illiquid-financial m.p.c.s individually significant; housing-collateral correctly signed). His Australian *calibrations* do not transfer: imposing his permanent-income gearing collapses λ to ≈ 0 (Spec 12, Spec 10), because Australia freely estimates a structural gearing of order one — roughly five times his 0.20 on

the consistent structural scale — and his rate loading cannot even be imposed at its published scale. The refinement is that the magnitude comparison now cuts both ways: the gamma inference rejects his net-liquid m.p.c. (0.159 outside [0.022, 0.098]) and his peak housing m.p.c. (0.0488 outside [-0.010, 0.021]) while matching his illiquid-financial m.p.c. (0.022 inside [0.012, 0.057]), so the agreement is on form and on the IFA channel, not on every magnitude. The companion Wald non-rejection ($\chi^2(6) = 7.55, p = 0.27$) is reconciled as low-power non-rejection in the imprecise Spec 6 frame. The CCI interactions are 0.66–0.97 collinear in absolute value and so weakly identified off a single equation, which is exactly why Williams' identification comes from cross-equation FIML restrictions; the back-extension shows sample length is not the binding constraint. The path to a tighter reconciliation with Williams' published values therefore runs through a full four-equation FIML build, not through any single-equation OLS refinement.

1.11 10. Decomposition and policy implications

This section reads the estimated long run and the policy counterfactuals through the lens of the faithful LIVES specification (Spec 11) of §7.0, while being explicit about which committed decompositions run on which long-run parameterisation. Two notes frame everything below. First, the long-run contributions decomposition is now committed in two parameterisations: the Spec 11 LIVES bracket (`australia_longrun_contributions_spec11.csv`), which is the lead decomposition of §10.1, and the selector-preferred net-worth Spec 3 (`australia_longrun_contributions.csv`), which we retain as a cross-check. Second, the counterfactuals (`australia_counterfactuals_summary.csv`) were generated on the conventional baseline (Spec 6 dummies) and the interaction spec (Spec 8); we report them as committed but read the CCI counterfactual through the de-meaning convention rather than as a structural housing-collateral experiment. The policy reading in §10.3 is anchored on the Spec 11 channels.

1.11.1 10.1 Long-run contributions decomposition

The long-run decomposition (an Australian counterpart to Williams (2010) Charts 2–8) splits fitted de-meaned $\log(c/y)$ into the partial contribution of each long-run regressor over the 146 quarters from 1988Q3 to 2024Q4. The lead decomposition is on the Spec 11 LIVES bracket: net liquid assets (`nla_y`), illiquid financial assets (`ilfa_y`), the credit-scaled housing-collateral term (`ha_x_cci`), the credit-scaled affordability term (`hp_x_1_minus_cci`), the rate and permanent-income interactions (`r_x_cci`, `yp_x_cci`), the CCI intercept (`cci_williams`) and the permanent-income ratio (`ln_yp_over_y`); each term is de-meaned and so sums to approximately zero by construction over the window, and the actual de-meaned path and the residual are reported alongside (`australia_longrun_contributions_spec11.csv`).

At the last observation (2024Q4) the dominant wedge is now the credit-scaled affordability term at +0.098 — its largest value in the sample, reflecting house prices elevated relative to income with the CCI in its post-2022 tight regime ($CCI \approx -1.6$, so the affordability multiplier $1 - 1.2 \cdot CCI$ is large and positive) — offset by the two permanent-income terms (–0.057 level, –0.088 interaction), with the financial-wealth channels smaller (`ilfa_y` +0.039, `ha_x_cci` –0.017, `nla_y` –0.004) against an actual de-meaned $\log(c/y)$ of –0.020 and a residual of only +0.013. The headline of the corrected decomposition is that the fitted long-run sum now tracks the actual path closely: the correlation between fitted and actual de-meaned $\log(c/y)$ is +0.945 and

the residual carries only 11 per cent of the variance of the actual path, so the estimated drivers – not an unexplained residual – account for most of the movement in the consumption-to-income ratio.

Three features of the path stand out. First, because the deployed CCI is identically zero before 2007Q3, the rate, permanent-income and housing-collateral interactions (r_x_cci , yp_x_cci , ha_x_cci) are held flat over 1988–2007 and only the affordability term moves (and only with house prices, since $1 - 1.2 \cdot CCI = 1$ there); the pre-GFC swings are therefore carried by permanent income and the financial-wealth ratios. The early-2000s credit boom is a permanent-income episode in this arithmetic – at 2003Q4 an actual de-meaned $\log(c/y)$ of +0.062 is almost entirely the +0.063 contribution of $\ln_yp_over_y$ – and the 2009Q1 GFC dip (actual –0.027) is carried by net liquid assets (–0.016) with permanent income turning below trend (–0.009). Second, once the CCI is live the credit terms do most of their work in the tightening episodes: through the 2010s the permanent-income level term runs persistently negative (PI below realised income, e.g. –0.046 at 2015Q4) largely offset by rising illiquid financial wealth (+0.020) and the permanent-income interaction (+0.046); and in the post-2022 tightening regime the rate interaction swings to +0.049 in 2022Q3 while the affordability wedge widens steadily to its end-of-sample extreme of +0.098. Third – the substantive reversal from the pre-fix decomposition – the residual is now small and no longer carries the story. Across 2002–2007 it sits within ± 0.02 (against the +0.09 to +0.16 level the sign-inverted decomposition had reported), and even the COVID collapse is largely captured by the long-run bracket rather than left to the dynamic dummies: at the trough (2020Q2, actual –0.213) the drivers still leave a –0.107 residual, but by 2020Q3 the permanent-income interaction (–0.158) and its level term (–0.077) absorb nearly all of the –0.170 dip, leaving a residual of only +0.017. The COVID quarters are where the drivers explain the most, not the least.

The honest qualifier is that the credit-scaled contributions inherit the weak identification of their coefficients (§5, §7.0): γ_1 is statistically indistinguishable from zero ($t = 0.71$), and the affordability and rate interactions are wrong-signed against Williams' priors, so the Spec 11 attribution should be read as the model's fitted arithmetic, not as well-identified structural channels. As a cross-check, the selector-preferred net-worth decomposition (Spec 3; `australia_longrun_contributions.csv`) tells a consistent end-of-sample story with coarser channels: net worth +0.018, house-price affordability +0.015, real rate ≈ 0.000 and permanent income –0.057 at 2024Q4, with a residual of +0.004. Spec 3 tracks the actual path less tightly than Spec 11 (fitted–actual correlation +0.81, residual variance 36 per cent of the actual) – lacking the CCI interactions it cannot capture the COVID dip, leaving a –0.103 residual at 2020Q3 – but both parameterisations agree that permanent income is the single largest source of variation in the fitted equilibrium, with the affordability/house-price wedge the next-largest moving part. On the wealth side the two decompositions also agree in the corrected sign: net worth (Spec 3, +0.018) and illiquid financial assets (Spec 11, +0.039) both sit modestly above their sample means at end-2024, a small positive support to the ratio rather than the drag the sign-inverted decomposition had shown.

Source:

`australia_longrun_contributions_spec11.csv`, `australia_longrun_decomposition_spec11.png`,
`australia_longrun_contributions.csv`, `australia_longrun_decomposition.png`.

1.11.2 10.2 Counterfactuals

We report three policy counterfactuals, holding all non-counterfactual regressors at their observed values and integrating the implied $\Delta \log c$ paths back to log-consumption levels (`australia_counterfactuals_summary.csv`). The APRA and COVID scenarios are computed on the conventional baseline event dummies; the CCI scenario is computed on the interaction spec. We retain them as committed evidence, with the hedges noted below.

Scenario	Event date	Basis	h = 4 q gap	h = 8 q gap	End-of-sample gap
No 2014/2017 APRA macro-prudential	2014Q4	Spec 6 dummies	-1.0 %	-2.3 %	+2.0 %
No COVID income support	2020Q1	Spec 6 dummies	-8.3 %	-8.3 %	-8.3 %
CCI at peak vs CCI = 0	1988Q4	Spec 8 interactions	n/a	n/a	≈ 0

(Gaps are cumulative deviations in $\log(c)$ from the baseline path, expressed in per cent; h-quarter values are measured from the relevant event date.)

Counterfactual 1 — no 2014/2017 APRA macroprudential. On the re-estimated Spec 6 the two APRA ogive dummies are small and statistically weak, and they now disagree in sign ($d_apra_{2014} +0.0050$, $t = 0.57$; $d_apra_{2017} -0.0069$, $t = -1.43$; `australia_full_results.csv`) — note that the Spec 11 headline estimates the opposite sign pattern (§10.3), so the profile is not robust across specifications. Zeroing both dummies implies consumption about 1.0 per cent *lower* four quarters after the 2014 round (-0.0103 in log points), 2.3 per cent lower after eight quarters (-0.0232), and 2.0 per cent *higher* by end of sample ($+0.0197$), as the negative 2017 dummy's removal eventually dominates. The honest reading is that the corrected data do not support a large, persistent APRA consumption drag through these event dummies: the counterfactual gaps are of the order of 1–2 per cent and not robustly signed, and an earlier draft's decade-horizon figure of $\approx +28$ per cent is withdrawn with the old estimates that produced it.

Counterfactual 2 — no COVID income support. Zeroing the `d_jobkeeper_2020`, `d2020_covid` and `d2020_rebound` dummies implies that consumption would have been about 8.3 per cent *lower* in the COVID period (-0.0835 in log points). The cumulative gap is constant across horizons because the COVID event dummies are bounded in time (zero before and after the 2020–21 window), so unlike the persistent APRA ogive the deviation does not continue to compound past the event window; the scenario is modelled as a one-off.

Counterfactual 3 — CCI at peak vs zero. Evaluating the interaction spec with the CCI-interacted regressors at CCI = 1 (historical peak) versus CCI = 0 (no liberalisation) implies an essentially zero cumulative consumption gap across the sample (-2.7×10^{-15} , i.e. zero to numerical precision). This is a mechanical consequence of the de-meaning convention: under de-meant interactions the average contribution of CCI variation to fitted $\Delta \log c$ is zero by construction, so the *cumulative* effect of a permanent CCI shift integrates to zero across the sample. The CCI interactions therefore operate as pure timing/distribution effects — they

reallocate where consumption growth lands across the cycle without shifting its unconditional level. This is the LIVES-theoretic reading: credit conditions matter for *when* households extract housing equity and respond to permanent-income news, not for the long-run level of the consumption-to-income ratio. It is not a structural housing-collateral experiment, and it should not be read as evidence on the magnitude of γ_1 (for which see §7.0, where the credit-scaled collateral MPC is right-signed but insignificant).

Source: australia_counterfactuals.csv, australia_counterfactuals_summary.csv,
australia_counterfactual_paths.png.

1.11.3 10.3 Policy implications

We read the policy implications off the faithful LIVES specification (Spec 11), distinguishing the channels that come through cleanly from those that remain weakly identified.

Wealth channel of monetary policy. Under Spec 11 the two plain-MPC wealth channels are significant and correctly signed: the net-liquid-asset MPC is $\gamma_3 = 0.060$ [0.022, 0.098] (OLS coefficient +0.027, $t = 3.75$) and the illiquid-financial-asset MPC is $\gamma_2 = 0.035$ [0.012, 0.057] (OLS +0.015, $t = 3.09$), both as structural propensities recovered as $OLS/|\lambda|$ on the full sample ($|\lambda| = 0.448$), both significant at 5 per cent in the full-sample and COVID-dummy variants, weakening to 10 per cent in the pre-COVID subsample. Housing wealth, by contrast, enters *only* through the credit-scaled collateral channel: the implied structural MPC is $\gamma_1 = 0.0055$ — right-signed but statistically insignificant (OLS +0.0025, $t = 0.71$), against Williams' calibrated peak of 0.0488, which the interval excludes. The policy reading is therefore asymmetric across asset classes: liquid and illiquid financial wealth transmit to consumption with well-identified marginal propensities, whereas the housing-collateral channel that LIVES theory makes conditional on credit conditions is, on Australian single-equation post-deregulation data, of the predicted sign but unproven. Movements in mortgage rates that change housing values propagate to consumption with a speed of adjustment we identify on the COVID-controlled samples at $\lambda \approx -0.25$ (pre-COVID -0.266 , $t = -4.85$) — close to Williams' -0.286 — implying about a 25 per cent closing of any equilibrium gap in the first quarter and, since $(1 - 0.25)^8 \approx 0.10$, roughly 90 per cent of the adjustment completed within two years. The full-sample λ of -0.448 is inflated by the COVID quarters and fails the $|\lambda|$ upper-bound screen (§6); the COVID-controlled value is the policy-relevant speed.

Housing wealth is not a free-standing channel. The central form-is-decisive result of the paper carries a direct policy corollary: reading an insignificant standalone housing-wealth coefficient as evidence of "no housing wealth effect" is a category error. LIVES predicts that coefficient to be approximately zero absent the credit-conditions interaction; the housing MPC is unlocked only as credit conditions ease. For macroprudential and financial-stability analysis this means the consumption response to house prices is regime-dependent — larger when credit is loose, muted when it is tight — rather than a fixed elasticity. The qualitative structure is the policy-relevant object even where the point estimate of γ_1 is imprecise.

Macroprudential effects. The 2014 and 2017 APRA episodes enter as smoothed-step ogive dummies with small $\Delta \log c$ coefficients. On the Spec 11 headline only the 2014 round is materially negative and marginally significant ($d_{apra_2014} = -0.0108$, $t = -1.92$); the 2017 dummy is positive and insignificant ($d_{apra_2017} = +0.0099$, $t = 1.61$), so any macroprudential drag is carried by the first round. The Spec 6-based counterfactual in §10.2 estimates the opposite sign pattern on its shorter sample and implies cumulative gaps of only 1–2 per cent

that are not robustly signed across horizons. The honest policy statement is that these event dummies detect, at most, a small and specification-dependent consumption effect of the APRA rounds — not the large persistent drag an earlier draft reported.

Permanent-income transmission. Permanent income is the strongest channel in the faithful form: the Spec 11 coefficient on $\log(y^p/y)$ is +0.46 ($t = 4.04$) on the full sample and +0.30 ($t = 5.81$) pre-COVID, both highly significant — though the implied structural gearing $\psi = \text{OLS}/|\lambda|$ of 1.02–1.13 sits *above* the theoretical admissibility bound $\psi \leq 0.95$, a breach we disclose as an open puzzle rather than re-scale away (§7.0). For fiscal-multiplier work this implies Australian households respond meaningfully and durably to credible permanent-income shocks. Two caveats apply. The headline estimate uses the full-sample (look-ahead) Italian direct-forecast permanent-income measure; under the real-time, no-look-ahead variant the speed of adjustment shrinks (Italy real-time $\lambda = -0.159$, $t = -2.39$, versus full-sample -0.239 in the Spec 6 frame) and the permanent-income coefficient flips sign (+0.325 $\rightarrow$ -0.145 ; §7.4, §8), so the real-time column is the operationally-relevant version for any forward-looking application. And the freely-estimated structural gearing of order one is roughly five times Williams' calibrated $\psi_0 = 0.20$; imposing his value collapses the equilibrium (§7.0.1, §9), so the Australian permanent-income gearing is a domestic estimate, not a transferred calibration.

Credit-conditions identification caveat. Section 5 documents that the CCI's identification in a single-equation OLS is weak, and that neither the back-extended sample, the sectional sign priors, nor the time-varying housing-wealth interaction changes this. The deployed index itself is degenerate over much of the sample — identically zero before 2007Q3, so the credit channels are identified off roughly seventy quarters — and the CCI-interacted regressors are very highly mutually correlated (pairwise $|\rho|$ of 0.66–0.97, each approximately proportional to CCI; `australia_cci_interaction_corr.csv`), so they cannot be separately identified off one equation — the structural reason Williams uses four-equation FIML. Policymakers using a single-equation CCI series for regime classification (tightening versus easing diagnoses) should treat the spline coefficients as consumption-equation residual identification rather than a structurally identified common credit-conditions factor. For policy use, regime classification should rely on the joint system, with the Kalman state-space CCI (Spec 9; available on the back-extended sample) as a less-imposed cross-check that does not require institutional knot choices.

1.11.4 10.4 Nesting in MARTIN

A natural use of a freely-estimated equation is to discipline the calibrated consumption block of the RBA's MARTIN model (Ballantyne et al. 2019), whose long run is homogeneous of degree one in real income and real net wealth — $\log c = w \cdot \log y + (1 - w) \cdot \log NW + (\text{rate term})$ — with a calibrated net-wealth elasticity $(1 - w) \approx 0.17$. We test that structure directly (`australia_martin_nesting.csv`) by fitting the static long run $\log c = \beta_0 + \beta_y \cdot \log y + \beta_{NW} \cdot \log(NW/y) + \beta_{rr}$. Because $\log(NW_real_pc) = \log(NW/y) + \log y + \log 4$, the coefficient on log income equals the sum of MARTIN's income and wealth weights (which homogeneity sets to one) and the coefficient on the net-worth ratio is the net-wealth elasticity:

Quantity	Estimate	MARTIN
Net-wealth elasticity (β_{NW})	0.1155	0.17
Income + wealth weight (β_y)	0.7198	1.00 (imposed)
Real-rate semi-elasticity (β_r)	-0.0086	small, calibrated

Three findings bear on integration. First, the unrestricted net-wealth elasticity, 0.1155, is below MARTIN's calibrated 0.17 but of the same order. Second, **MARTIN's long-run homogeneity restriction is rejected** on the Australian data: the income-plus-wealth weight is 0.7198, and the test $\beta_y = 1$ gives $\chi^2(1) = 16.41$ ($p = 0.0001$). Imposing homogeneity by force does not help — the restricted net-wealth elasticity flips to -0.0832 (restricted real-rate -0.0038), so the data resist the balanced-growth parameterisation in this simple form. Third, the static long run does not cointegrate under Engle–Granger MacKinnon critical values (ADF -2.3828 vs 5 per cent CV for four variables -4.10), echoing the cointegration-screen result of §6 — and this caveat bites on the second finding too: the homogeneity Wald test sits inside a non-cointegrated static levels regression, where its distribution is non-standard, so the rejection is indicative rather than decisive.

The reading for MARTIN is that the freely-estimated single-equation long run does **not** nest MARTIN's calibrated balanced-growth block: homogeneity would have to be imposed rather than tested-and-accepted (with the non-cointegration caveat above), and the comparable unrestricted wealth elasticity (0.12) sits below MARTIN's 0.17 without a committed standard error to adjudicate the gap. The disaggregated evidence is now more informative than in an earlier draft — Spec 11's aggregate-wealth structural MPC is 0.100 with a 95 per cent CI of $[0.036, 0.164]$ (`australia_gamma_inference.csv`), significantly positive rather than zero-spanning — but an MPC on the wealth-to-income ratio is not the same object as MARTIN's log-wealth elasticity, so it corroborates a positive wealth channel of the right order rather than recalibrating MARTIN's parameter. The honest position remains that this equation is *not yet MARTIN-ready as a source of point estimates*. The MARTIN-operational version of the permanent-income measure is the real-time Italian direct forecaster (§7.4), not the full-sample look-ahead measure used for the headline; any integration must use the real-time variant, under which the speed of adjustment is materially weaker. A productive integration path is therefore to import the *qualitative* structure — disaggregated wealth channels and a credit-conditions interaction, with the signs and rough magnitudes of the channels — while retaining MARTIN's calibrated long-run weights, using this paper as a freely-estimated benchmark the calibration is consistent with rather than as a source of point estimates precise enough to replace it.

Source: `australia_martin_nesting.csv`, `australia_gamma_inference.csv`.

1.12 11. Conclusion

This paper re-estimates the Williams (2010, 2012) Australian LIVES consumption equation on a sample extended to 2024Q4, and in doing so revisits what an earlier draft and much of the prior Australian literature had treated as a *weak* single-equation result. The central finding is methodological: the functional **form** of the LIVES equation is what identifies it. When housing wealth is entered faithfully through its credit-conditions interaction alone, the

autonomous-consumption CCI intercept is restored, and illiquid financial assets are combined, the error-correction mechanism and the core wealth structure come alive in a single equation – and they stay alive under every COVID treatment. When housing instead enters as a plain, constant-MPC level term – as in the conventional disaggregated error-correction model that prior work implicitly tested – the equilibrium is weaker and sample-fragile and the stand-alone housing coefficient is insignificant. Reading that insignificant standalone coefficient as a failed housing-wealth effect is a category error: LIVES theory predicts that coefficient is approximately zero absent the credit-conditions interaction, because there is no classical housing-wealth channel in the model.

A second, equally central finding is that the LIVES *structure* transfers to Australia while Williams' Australian *calibrations* do not – and the corrected data now grade the transfer channel by channel. The faithful specification recovers Williams' speed of adjustment to within about 13 per cent and matches his illiquid-financial m.p.c., but its intervals are tight enough to reject his net-liquid magnitude; imposing his calibrated permanent-income gearing collapses the equilibrium. We document why – the credit-conditions interactions are mutually collinear in the post-deregulation sample and the deployed index has no variation before 2007Q3 – and we are honest throughout about the limitations of single-equation LIVES estimation: a split placebo verdict on the credit-conditions index (the literal Williams construction sits at the placebo median, the deployed protocol earns only moderate support), a structural permanent-income gearing above its theoretical admissibility bound, no efficiency gain from joint estimation at the quarterly frequency, an out-of-sample loss to a random walk beyond one quarter, and a look-ahead permanent-income measure whose coefficient reverses under a real-time construction.

1.12.1 11.1 Form is decisive: the faithful LIVES specification

The headline result is the faithful LIVES specification (Spec 11), in which housing wealth enters *only* through the de-meaned credit-conditions interaction $ha_x_cci = CCI \times (HA/4y)$, the autonomous-consumption loading $\zeta_c\text{-CCI}(cci_williams)$ is restored, and illiquid financial assets are combined into a single ratio ($ilfa_y = equities + superannuation$). On this form the error-correction and core wealth structure are recovered (`australia_spec11_variants.csv`; `australia_lambda_robustness.csv`):

Quantity	Full sample (n = 146)	Pre-COVID (n = 126)	COVID dropped (n = 138)	Williams (2010)
λ (ecm_lag)	-0.448 (t = -3.57)	-0.266 (t = -4.85)	-0.248 (t = -6.66)	-0.286
NLA m.p.c. (nla_y)	+0.027 (t = 3.75)***	+0.016 (t = 1.81)*	+0.017 (t = 2.11)**	0.159
IFA m.p.c. (ilfa_y)	+0.015 (t = 3.09)***	+0.009 (t = 1.74)*	+0.010 (t = 2.00)**	0.022
housing-collateral (ha_x_cci, γ_1)	+0.0025 (t = 0.71)	+0.0019 (t = 0.86)	+0.0023 (t = 1.05)	0.0488
permanent income (ln_yp_over_y)	+0.459 (t = 4.04)***	+0.298 (t = 5.81)***	+0.281 (t = 8.68)***	0.20–0.95

The COVID-controlled speed of adjustment is tightly clustered at $\lambda \approx -0.25$ ($-0.266 / -0.248 / -0.242$ across the pre-COVID, COVID-dropped and quarterly-dummy treatments), about 13 per cent below Williams' (2010) -0.286 , and is the value we treat as identified: the full-sample $\lambda = -0.448$ is inflated by the COVID quarters and fails the upper-bound speed screen. The net-liquid and illiquid-financial marginal propensities are correctly signed in every variant and significant at 5 per cent in the full-sample, COVID-dropped and quarterly-dummy treatments, and permanent income enters strongly everywhere ($t = 4.0$ – 8.7). The housing-collateral term γ_1 is right-signed but insignificant in every variant ($ha_x_cci + 0.0025$, $t = 0.71$ full sample). Scaled by $|\lambda| = 0.448$, the implied structural marginal propensities are NLA 0.060 [$0.022, 0.098$], IFA 0.035 [$0.012, 0.057$], and housing-collateral γ_1 0.0055 (right-signed, insignificant, versus Williams' 0.0488 , which the interval excludes). The one theory casualty is the permanent-income gearing: $\hat{\psi} = OLS/|\lambda|$ is 1.02 – 1.13 , above the admissibility bound $\psi \leq 0.95$, a breach that survives removal of the GFC ogive and that we disclose as an open puzzle (§7.0).

By contrast the conventional constant-MPC disaggregated ECM (Spec 6), which carries plain $ha_y/eq_y/super_y/nla_y$ with no credit scaling and a 2002Q3-binding short-run credit term, delivers a sample-fragile equilibrium on $n = 86$ ($\lambda = -0.239$, $t = -2.55$, collapsing to -0.087 pre-COVID) and an insignificant standalone housing coefficient ($ha_y + 0.0022$, $t = 0.30$). This is the conventional baseline, not the preferred specification. The contrast between Spec 11 and Spec 6 is a matter of theory rather than of fit: the two forms test different equations, and only Spec 11 is the LIVES equation. Much of the apparent weakness of single-equation Australian LIVES estimates is, on this reading, a specification artefact.

1.12.2 11.2 Structure transfers; Williams' calibrations do not

The faithful form recovers Williams' structure when freely estimated. It does *not* survive imposition of his Australian calibrations. Specification 12 imposes Williams' permanent-income gearing ($\psi_0 = 0.20$, $\psi_1 = 0.93$) and his illiquid-financial m.p.c. ($\gamma_IFA = 0.022$) via an iterative fixed point; the error-correction mechanism collapses to $\lambda = -0.030$ ($t = -0.74$) and flips sign pre-COVID ($+0.041$, $t = 2.03$), with the net-liquid coefficient turning wrong-signed

(australia_all_results.csv; australia_lambda_robustness.csv). Specification 10 (Williams-prior calibrated) independently reproduces the collapse ($\lambda = -0.048$, $t = -0.78$). The mechanism is that Australia freely estimates a structural permanent-income gearing of order one ($\hat{\psi} = 1.02\text{--}1.13$) — roughly five times Williams' calibrated $\psi_0 = 0.20$ on the consistent structural scale; forcing his much lower value removes the equilibrium. Williams' rate, affordability and autonomous-consumption loadings cannot even be imposed at their published magnitudes: his raw $\alpha_r = -0.871$ is some thirty times too large on the repository's percent real-rate $\times$ unit-normalised CCI scaling, and diverges the fixed point.

This reconciles the companion Wald non-rejection of the joint Williams calibration ($\chi^2(6) = 7.55$, $p = 0.27$; LIVES/outputs/williams_calibration_wald.csv). The non-rejection is driven by imprecision, not by good fit: in the Spec 6 frame every Williams wealth value lies inside a wide structural confidence interval that also contains zero (australia_gamma_inference.csv). The free Spec 6 estimates are too imprecise to reject Williams' values, yet imposing those values wrecks the fit. Low power is not the same as good fit. In the faithful Spec 11 frame, by contrast, the inference cuts both ways: the intervals reject his net-liquid m.p.c. (0.159 outside [0.022, 0.098]) and his peak housing m.p.c. (0.0488 outside [-0.010, 0.021]) while matching his illiquid-financial m.p.c. (0.022 inside [0.012, 0.057]) — agreement on form and on the IFA channel, genuine disagreement on the net-liquid magnitude.

1.12.3 11.3 Why the credit channels are weakly identified in one equation

The reason the credit-conditions calibrations cannot be sharpened off a single equation is identification, not sample length. The deployed CCI is degenerate over much of the sample — only four of fifteen candidate knots survive, all post-2007, so the index is identically zero from 1988 to mid-2007 and the credit channels are identified off roughly seventy quarters — and the CCI-interacted regressors are each approximately proportional to the index and therefore 0.66–0.97 mutually correlated in absolute value (australia_cci_interaction_corr.csv). They cannot be separately free-estimated from one equation — the structural reason Williams uses four-equation FIML. The evidence converges from three directions. First, when the interactions are entered freely (Spec 8), the individual interaction coefficients are wrong-signed or insignificant against Williams' priors even though the fit is strong (BIC -952.8, second only to Spec 11's -954.8), and the credit channels do not move toward Williams' Table 1. Specification 8 reallocates identification; it does not close the gap, and we do not present its $\lambda = -0.458$ as a credit-conditions success. Second, the calibration collapse of Specs 10 and 12 (§11.2) shows the channels are not jointly recoverable by imposition either. Third, the credit-conditions placebo battery returns a split verdict rather than a clean failure: the literal Williams four-knot specification sits at the placebo median (45th adjusted- R^2 percentile) and below it on the extended samples (36th literal, 48th maximal, 37th sectional), while the deployed maximal-GETS construction, evaluated under its own iterated reduction protocol, beats 84 per cent of random draws — moderate support, short of strong identification (australia_williams_knot_placebo_verdict.csv; australia_williams_knot_placebo_deployed_verdict.csv; extended and maximal summaries). These are honest mixed-to-negative results, and they are diagnostic: single-equation calibration of the credit channels is empirically closed, and a four-equation FIML build is the route to sharpen them.

1.12.4 11.4 The back-extension: sample length is not the binding constraint

A natural alternative hypothesis is that the residual gap between the freely estimated Australian channels and Williams' Table 1 reflects the post-1988 sample window, which begins only after the financial-liberalisation episode that identifies the credit channels. To test this we assembled a back-extended master dataset to 1976Q3 ($n = 190\text{--}194$ quarters) — using a Treasury TRYM long-run house-price series, the RBA D03 M3 monetary aggregate, the RBA D02 total-credit splice, a historical labour-force compilation, and aggregate and disaggregated wealth proxies anchored at 1988Q3 — and refit the simpler disaggregated no-CCI specification (Spec 4) on the longer window (spec46_extended_comparison.csv):

Structural coefficient	1988+ baseline	1976+ extended	Williams (2010)
λ	−0.182	−0.203	−0.286
NLA (γ_{NLA})	+0.025	+0.001	+0.159
IFA equities (γ_{EQ})	−0.043	−0.041	—
IFA super (γ_{super})	+0.014	−0.005	—
IFA combined (γ_{IFA})	−0.030	−0.046	+0.022
housing (γ_{HA})	+0.019	+0.018	+0.049

The Williams (2010) reference values are his housing wealth m.p.c. (0.0488 at the CCI peak), net-liquid m.p.c. (0.159) and calibrated illiquid-financial m.p.c. (0.022, not split between equities and superannuation); the back-extension figures are the Spec 4 disaggregated structural coefficients (spec46_extended_comparison.csv). The speed of adjustment improves by about 12 per cent ($-0.182 \rightarrow -0.203$, still 29 per cent short of -0.286 ; the aggregate Spec 1 moves similarly, $-0.193 \rightarrow -0.209$, spec1_extended_comparison.csv), but the individual wealth coefficients shrink rather than converging on Williams' values: the net-liquid m.p.c. collapses by 95 per cent toward zero, the equities coefficient retains a wrong sign, and superannuation flips sign. Sample length is therefore not, in itself, what generates the divergence; the diagnosis remains the single-equation framing and the collinearity of the credit channels. The back-extension stands as a contribution in its own right — it makes this test possible for the first time — and supports the conclusion that the route forward is joint estimation rather than a longer single equation.

1.12.5 11.5 Honest negative results

The paper reports its negative results prominently, because a freely estimated single-equation framework permits diagnostics that imposed restrictions would hide. Each points to why FIML and back-extension are the natural next steps rather than further single-equation tuning.

- **Credit-conditions placebo: split verdict.** The literal Williams four-knot construction sits at the placebo median (45th adjusted- R^2 percentile) and below it on the extended variants (36th–48th; sectional 37th), sustaining the detrending critique; the deployed maximal-GETS protocol beats 84 per cent of random draws — moderate support, not strong identification (australia_williams_knot_placebo_verdict.csv; australia_williams_knot_placebo_deployed_verdict.csv).
- **No system-efficiency gain.** A two-equation SUR of consumption and house prices yields negligible cross-equation residual correlation ($\hat{\rho} = -0.0133$ against -0.0109 OLS; LIVES/out-

puts/lives_sur_2eq_resid_corr.csv), and the SUR coefficient estimates are within sampling noise of single-equation OLS (australia_joint_pi_robustness.csv): joint estimation buys no efficiency at the quarterly frequency in this sample.

- **Out-of-sample: a one-quarter win, multi-step losses.** At $h = 1$ Spec 8 (RMSE 0.0290) and Spec 11 (0.0292) beat the random walk with drift (0.0309), but at $h = 4$ and $h = 8$ the random walk beats every structural specification, and Spec 11 is the worst performer at $h = 8$ (0.0640) (australia_oos_rmse.csv). The exercise uses full-sample permanent-income and CCI inputs, so it measures fit stability under re-estimation, not real-time forecast accuracy (§8.13). The framework's value is structural interpretation, not point forecasting.
- **Selector divergence, now narrower.** The automated screen falls back to a most-passes rule (no specification clears the Engle–Granger cointegration screen at the correct MacKinnon critical values; australia_cointegration.csv) and returns the net-worth Spec 3, while BIC and LIVES theory now *agree* on Spec 11 (BIC –954.8; australia_spec_selection.csv). We headline Spec 11 on joint BIC-and-theory grounds and report the rubric fallback's divergence — and the universal cointegration failure — as documented limitations.
- **Look-ahead permanent income.** The headline permanent-income result uses the full-sample (non-causal) Italian direct forecaster. Under a causal real-time construction the speed of adjustment shrinks but survives ($\lambda = -0.159$, $t = -2.39$, against -0.239 full-sample) while the permanent-income coefficient flips sign ($+0.325 \rightarrow -0.145$; australia_pi_realtime_robustness.csv): the positive permanent-income loading is reported as a property of the full-sample measure, not a real-time resolution of the Australian permanent-income puzzle.
- **The ψ admissibility bound is breached.** The freely estimated structural gearing $\hat{\psi} = 1.02\text{--}1.13$ exceeds the theoretical $\psi \leq 0.95$ bound, and the breach is not an artefact of the GFC ogive (australia_spec11_ogive_robustness.csv). We disclose it as an open puzzle (§7.0) rather than re-scaling it away.

1.12.6 11.6 MARTIN nesting and the wealth elasticity

Nesting the long run against the RBA's MARTIN consumption block, the unrestricted net-wealth elasticity is 0.1155, against MARTIN's calibrated 0.17 (australia_martin_nesting.csv). Long-run homogeneity of the combined income-and-wealth term is rejected ($\chi^2(1) = 16.41$, $p = 0.0001$) — though the test sits inside a static levels regression that does not cointegrate (EG ADF -2.38 against a -4.10 critical value), so the rejection is indicative rather than decisive — and the wealth elasticity is not robustly identified: imposing homogeneity drives the restricted net-wealth elasticity to -0.083 , so the estimate is too sensitive to the restriction to discipline MARTIN's calibration. The appropriate reading is qualitative: the LIVES structure can inform MARTIN's functional form, and Spec 11's aggregate-wealth structural MPC of 0.100 [0.036, 0.164] corroborates a positive wealth channel of the right order, but the single-equation point estimates are not precise enough — nor on the right scale — to recalibrate MARTIN's parameters.

1.12.7 11.7 Contributions

The paper makes five contributions to the Australian household consumption literature.

1. **The faithful single-equation LIVES estimate (Spec 11) with the form correction as the lead methodological result.** Entering housing only through its credit-conditions interaction, restoring the autonomous-consumption CCI intercept, and combining illiquid

financial assets recovers a Williams-consistent, COVID-robust error-correction speed ($\lambda \approx -0.25$ across the COVID-controlled treatments, about 13 per cent below Williams' -0.286) and right-signed wealth structure (NLA $+0.027^{***}$, IFA $+0.015^{***}$), where the conventional constant-MPC ECM is weaker and sample-fragile. Reading the conventional form's insignificant standalone housing coefficient as a failed wealth effect is shown to be a category error. The structural intervals are tight enough to grade the transfer: the IFA m.p.c. matches Williams' 0.022, the NLA m.p.c. is genuinely smaller than his 0.159, and the housing channel is unproven.

2. **The structure-transfers-but-calibrations-don't finding.** Imposing Williams' permanent-income gearing collapses the equilibrium (Spec 12: $\lambda = -0.030$, flipping sign pre-COVID; independently reproduced by Spec 10: $\lambda = -0.048$), because Australia freely estimates a structural gearing of order one — roughly five times Williams' 0.20; this reconciles the companion low-power Wald non-rejection ($\chi^2(6) = 7.55$, $p = 0.27$) as imprecision, not good fit.
3. **The interaction-collinearity diagnosis.** The CCI-interacted regressors are 0.66–0.97 mutually collinear in absolute value (australia_cci_interaction_corr.csv), and the deployed index has no variation before 2007Q3, explaining why the credit channels cannot be separately identified off a single equation and why Williams' identification requires four-equation FIML — corroborated by the Spec 8 reallocation, the calibration collapse, and the placebo battery.
4. **A back-extended Australian master dataset to 1976Q3** ($n = 190$ – 194 quarters) with documented growth-rate splices for house prices, M3, total credit and labour force, plus aggregate and disaggregated wealth proxies — and the direct test it permits, showing that sample length is not the binding constraint.
5. **A structured robustness and placebo suite with honest negative results** — the split credit-conditions placebo verdict, the negligible SUR efficiency gain, the out-of-sample losses beyond one quarter, the universal cointegration-screen failure, the ψ -bound breach, and the look-ahead permanent-income caveat — reported as substantive findings, alongside a multi-equation scaffold (data preparation, house-price and mortgage-stock equations, joint CCI identification, SURs) on which a companion FIML paper can build directly.

1.12.8 11.8 Outstanding work

Two structural routes follow directly from the diagnosis above, and four narrower investigations sit alongside them.

1. **A multi-equation companion paper.** A full LIVES build with FIML and cross-equation parameter restrictions — consumption, house prices, the mortgage stock and home-equity withdrawal — would estimate a single CCI loading across equations under sign restrictions, the route the collinearity diagnosis and the placebo and SUR results identify as the only way to sharpen the credit channels. Custom likelihood code and a sourced HEW series remain the binding obstacles.
2. **Pre-1988 extension of the credit channels, and richer credit data.** Because the financial-liberalisation episode that identifies the credit-conditions index largely predates the 1988Q3 start of ABS sectoral balance-sheet data, extending the disaggregated credit-interaction regressors back into the deregulation window — building on the 1976Q3

back-extension — and bringing in richer direct credit-conditions data (lending standards, securitisation, loan-level series) would, in combination with FIML, give the credit channels the variation they need.

Narrower items: (i) **the ψ -bound breach** — investigating why the structural permanent-income gearing estimates at 1.02–1.13 against an admissibility bound of 0.95, including the role of the unit-income restriction and the measure's non-causal tail; (ii) **a lagged-wealth-timing variant**, entering the wealth ratios at $t-1$ to test whether within-quarter timing inflates the wealth and ECM loadings; (iii) **a real home-equity-withdrawal series**, which both completes the four-equation system and supplies a direct observable for the collateral channel; and (iv) **the FIML estimator itself**, whose cross-equation restrictions are the identification technology everything above points to.

The single-equation framework cannot, on its own, deliver identification of the credit channels that Williams (2010) obtains from joint estimation under parameter restrictions. What it can do — and what this paper establishes — is that the faithful LIVES *form* recovers his error-correction and wealth structure where the conventional constant-MPC ECM is weak, that the recovery survives every COVID treatment, that his Australian *calibrations* do not transfer, and that the credit interactions are collinear and weakly identified off a single equation. Reported honestly, these results point unambiguously to the multi-equation, back-extended programme as the way forward.

1.13 References

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1.14 Appendix A: Data construction

A full data-construction appendix — series sources, splice conventions, deflation, the asset/annualised-income ratio transform, the disaggregated wealth definitions and the 1976Q3 back-extension — is reproduced from the accompanying data documentation. It retains the detail of §§3.1–3.13 of the main paper, together with the master variable coverage table.

The three load-bearing constructions for the faithful LIVES specification (Spec 11) and the conventional baseline (Spec 6) are summarised here for the reader's convenience; the full provenance is in the data appendix.

A.1 Disaggregated wealth (ratios to annualised income). All wealth terms enter as the current quarter's end-of-quarter (closing) stock divided by annualised disposable income, $x_t/4y$ — *contemporaneous* timing, not the $x_{t-1}/4y$ dating of Williams' bracket; the implications for predeterminedness are disclosed in §3.2 and §4.6, with the IV check in §8.1:

- **HA** — housing assets (ABS Balance Sheet 5232035). Enters the faithful LIVES form *only* through the credit interaction ha_x_cci (de-means $CCI \times (HA/4y)$); there is no stand-alone housing-wealth level term, consistent with the theory that the housing marginal propensity to consume is zero at $CCI = 0$ and is unlocked as credit conditions ease.
- **IFA** — illiquid financial assets, equities plus superannuation ($ilfa_y = eq_y + super_y$). Enters as a plain MPC.
- **NLA** — net liquid assets, liquid assets minus total household debt. Enters as a plain MPC. The net-liquid aggregation (constraining the liquid-asset and debt MPCs to be equal and opposite, $\gamma_{LA} + \gamma_{LOANS} = 0$) is *not rejected* in any specification or sample (Appendix C; `australia_nla_restriction_test.csv`).

A.2 Permanent income. The permanent-income term $\log(y^p/y)$ is a discounted ($\eta = 0.05$, $k = 40$ quarters) weighted average of *expected* future income from the Italy-style direct (single-regression) forecast of the discounted future-income aggregate (De Bonis et al. 2020, Appendix A.2). The headline measure uses the full-sample (in-sample, hence look-ahead) forecaster, whose training sample necessarily ends at 2014Q4 (the last quarter with a fully realised 40-quarter horizon), so the final forty quarters — about 27 per cent of the estimation sample — are out-of-training extrapolations; the series also carries a GFC learning ogive that halves it from 2012Q2 onwards (no-ogive robustness: `australia_spec11_ogive_robustness.csv`). The real-time variant is reported as an opera-

tional robustness column (§7.4; `australia_pi_realtime_robustness.csv`). This look-ahead-versus-real-time distinction is flagged wherever a headline number depends on the non-causal measure.

A.3 The 1976Q3 back-extension. Sectoral balance-sheet data from the ABS begin in 1988Q3, which is why the financial-liberalisation episode that identifies the credit channels largely *predates* the estimation sample. A back-extended master dataset to 1976Q3 splices in TRYM house prices, RBA D03 M3 and RBA D02 credit aggregates, historical labour-force series and wealth proxies anchored to 1988Q3 levels. One caveat: the RBA D03 M3 series carries a +14.25 log per cent definitional break at August 1976, inside the opening spine quarters, which makes 1976Q4 a roughly 4.8σ growth outlier in every M3-based proxy (§3.10.1). The back-extended dataset supports the long-history specification (Spec 6b) and the direct sample-length test (§8): back-extending Spec 4 moves λ about a fifth of the way towards Williams' value (from -0.182 to -0.203 , against -0.286 ; `spec46_extended_comparison.csv`) but individual wealth coefficients shrink and the NLA term collapses — sample length is not the binding constraint on identifying the credit channels.

1.15 Appendix B: Coefficient tables

The full per-specification coefficient vectors, with Newey–West HAC standard errors, t -statistics, p -values, the implied error-correction speed λ (the `ecm_lag` coefficient) and the implied structural parameter ($\gamma = \text{OLS}/|\lambda|$), are reproduced in machine form from the committed results files. The current draft regenerates these tables from the committed per-window results files, `australia_full_results.csv` and `australia_precovid_results.csv` (consolidated in `australia_all_results.csv`, with the matched diagnostics in `australia_full_diagnostics.csv` and `australia_precovid_diagnostics.csv` and the four-sample λ vector in `australia_lambda_robustness.csv`), which span all fourteen specifications — Spec 1 through Spec 12, plus the long-history Spec 6b and the RBA-burden Spec 7b — for both the full ($n = 146$; 1988Q3–2024Q4) and pre-COVID ($n = 126$; 1988Q3–2019Q4) windows. In particular they include the faithful LIVES headline (Spec 11) and the Williams-calibration-imposed specification (Spec 12), which were absent from the earlier draft's tables.

Rather than retype the full coefficient matrix, Table B.1 reports the two specifications that carry the paper's central message — the faithful LIVES headline (Spec 11) and the Williams-calibration-imposed specification (Spec 12) — across both samples. The remaining specifications are tabulated in §6 (the specification ladder) and §7 (results), and in full in Tables B.2–B.4 below and the committed CSVs.

Table B.1 — Faithful LIVES (Spec 11) vs Williams-calibration-imposed (Spec 12): selected coefficients (Newey–West HAC). Source: `australia_full_results.csv`, `australia_precovid_results.csv`, `australia_full_diagnostics.csv`, `australia_precovid_diagnostics.csv`. Convention: $\lambda = \text{ecm_lag}$ (negative = error-correction); structural $\gamma = \text{OLS}/|\lambda|$. *** sig 1%, ** sig 5%, * sig 10%.

Term	Spec 11 full ($n = 146$)	Spec 11 pre- COVID ($n = 126$)	Spec 12 full ($n = 146$)	Spec 12 pre- COVID ($n = 126$)
λ (ecm_lag)	-0.448 ($t=-3.57$)***	-0.266 ($t=-4.85$)***	-0.030 ($t=-0.74$)	+0.041 ($t=2.03$)**, sign flip
nla_y	+0.0269 ($t=3.75$)***	+0.0159 ($t=1.81$)*	+0.0011 ($t=0.39$)	-0.0002 ($t=-0.09$, wrong sign)
ilfa_y (=eq+super)	+0.0155 ($t=3.09$)***	+0.0093 ($t=1.74$)*	imposed ($\gamma=0.022$)	imposed ($\gamma=0.022$)
ha_x_cci (γ_1)	+0.0025 ($t=0.71$)	+0.0019 ($t=0.86$)	-0.0012 ($t=-0.88$)	+0.0018 ($t=1.52$)
hp_x_l_minus_cci	+0.0279 ($t=3.08$)***	+0.0137 ($t=1.10$)	—	—
r_x_cci	+0.0028 ($t=4.55$)***	+0.0022 ($t=3.47$)***	—	—
cci_williams (ζ_c)	+0.0001 ($t=0.01$)	+0.0199 ($t=2.77$)***	—	—
ln_yp_over_y	+0.4591 ($t=4.04$)***	+0.2982 ($t=5.81$)***	imposed (ψ_0 =0.20)	imposed (ψ_0 =0.20)
yp_x_cci	-0.5101 ($t=-1.51$)	+0.1869 ($t=1.80$)*	imposed (ψ_1 =0.93)	imposed (ψ_1 =0.93)
Intercept	-0.0139 ($t=-2.13$)**	-0.0099 ($t=-1.54$)	+0.0078 ($t=1.81$)*	+0.0048 ($t=2.13$)**
adj- R^2	0.824	0.239	0.687	0.073
SE (%)	0.683	0.510	0.911	0.565
BIC (Schwarz)	-954.75	-900.15	-893.79	-896.61

Implied structural MPCs for Spec 11 (full sample, $|\lambda| = 0.448$): NLA 0.060, IFA 0.035, housing-collateral γ_1 0.0055 (right-signed, insignificant, $t = 0.71$; cf. Williams' calibrated housing MPC 0.0488). The COVID-controlled cluster $\lambda \approx -0.25$ (pre-COVID -0.266 , $t = -4.85$; Table B.4) is treated as the identified speed of adjustment — about 13 per cent below Williams' -0.286 — because the full-sample value is inflated by the COVID quarters and fails the $|\lambda|$ upper-bound screen. The headline numbers depend on the full-sample (look-ahead) permanent-income measure; see §7.4 and the real-time robustness column.

Table B.2 — Speed of adjustment λ (the ecm_lag coefficient) across all fourteen specifications, full and pre-COVID samples (Newey–West HAC). Source: australia_full_results.csv, australia_precovid_results.csv (point estimates also in australia_lambda_robustness.csv). *** sig 1%, ** sig 5%, * sig 10%.

Specification	λ full ($n = 146$)	λ pre-COVID ($n = 126$)
Spec 1	-0.193 (t=-2.88)***	-0.161 (t=-3.32)***
Spec 2	-0.186 (t=-2.07)**	-0.080 (t=-1.15)
Spec 3	-0.191 (t=-2.90)***	-0.158 (t=-3.17)***
Spec 4	-0.182 (t=-2.45)**	-0.119 (t=-1.95)*
Spec 5	-0.235 (t=-2.47)**	-0.061 (t=-0.62)
Spec 6	-0.239 (t=-2.55)**	-0.087 (t=-0.79)
Spec 6b	-0.248 (t=-3.96)***	-0.240 (t=-4.61)***
Spec 7	-0.341 (t=-2.76)***	-0.054 (t=-0.43)
Spec 7b	-0.381 (t=-2.43)**	-0.061 (t=-0.93)
Spec 8	-0.458 (t=-3.52)***	-0.219 (t=-3.95)***
Spec 9	-0.209 (t=-3.24)***	-0.158 (t=-2.86)***
Spec 10	-0.048 (t=-0.78)	-0.025 (t=-0.57)
Spec 11	-0.448 (t=-3.57)***	-0.266 (t=-4.85)***
Spec 12	-0.030 (t=-0.74)	+0.041 (t=+2.03)**

(Several specifications estimate on different windows from the nominal $n = 146/n = 126$: Specs 2, 5, 6, 7 and 10 on $n = 86$ because a 2002Q3-binding credit term shortens their sample, Spec 7b on $n = 64$, and Spec 6b on the $n = 180$ back-extended sample — see §6.1 and `australia_full_diagnostics.csv`.)

Table B.3 — Long-run coefficient matrix for the disaggregated specifications (full sample): OLS estimate (t -statistic). Source: `australia_full_results.csv`. "—" = term not in that specification; the implied structural parameter is $\gamma = \text{OLS}/|\lambda|$ (Table B.2). *** sig 1%, ** sig 5%, * sig 10%.

Term	Spec 4	Spec 5	Spec 6	Spec 8	Spec 11
nla_y	+0.0045 (+0.39)	−0.0141 (−0.43)	+0.0083 (+0.24)	+0.0337 (+4.39)***	+0.0269 (+3.75)***
eq_y	−0.0079 (−0.41)	−0.0295 (−0.57)	−0.0156 (−0.30)	−0.0014 (−0.11)	—
super_y	+0.0025 (+0.36)	+0.0040 (+0.37)	+0.0060 (+0.66)	+0.0215 (+2.42)**	—
ilfa_y	—	—	—	—	+0.0155 (+3.09)***
ha_y	+0.0035 (+0.53)	−0.0028 (−0.30)	+0.0022 (+0.30)	+0.0030 (+0.48)	—
ha_x_cci	—	—	—	+0.0043 (+0.87)	+0.0025 (+0.71)
ln_hp_over_y	+0.0038 (+0.13)	+0.0447 (+0.83)	+0.0102 (+0.23)	—	—
hp_x_l_minus_cci	—	—	—	+0.0299 (+1.38)	+0.0279 (+3.08)***
real_rate	−0.0002 (−0.33)	−0.0014 (−0.87)	−0.0005 (−0.46)	—	—
r_x_cci	—	—	—	+0.0028 (+2.05)**	+0.0028 (+4.55)***
cci_williams	—	—	—	—	+0.0001 (+0.01)
ln_yp_over_y	+0.2026 (+1.94)*	+0.3316 (+1.43)	+0.3253 (+1.50)	+0.4921 (+3.42)***	+0.4591 (+4.04)***
ln_yp_over_y_post2008	—	—	+0.1704 (+0.87)	—	—
yp_x_cci	—	—	—	−0.5046 (−2.43)**	−0.5101 (−1.51)
ecm_lag	−0.1824 (−2.45)**	−0.2353 (−2.47)**	−0.2386 (−2.55)**	−0.4583 (−3.52)***	−0.4483 (−3.57)***

Table B.4 — Spec 11 across the four sample treatments: the COVID-controlled headline. Source: `australia_spec11_variants.csv` (Newey–West HAC *t*-statistics in parentheses).

Variant	n	λ	nla_y	ilfa_y	ha_x_cci ln_yp_over_y	
Full sample	146	-0.448 (-3.57)	+0.0269 (3.75)	+0.0155 (3.09)	+0.0025 (0.71)	+0.459 (4.04)
Pre-COVID (to 2019Q4)	126	-0.266 (-4.85)	+0.0159 (1.81)	+0.0093 (1.74)	+0.0019 (0.86)	+0.298 (5.81)
COVID quarters dropped	138	-0.248 (-6.66)	+0.0170 (2.11)	+0.0098 (2.00)	+0.0023 (1.05)	+0.281 (8.68)
Quarterly COVID dummies	146	-0.242 (-6.25)	+0.0141 (3.10)	+0.0084 (3.28)	+0.0016 (0.83)	+0.278 (8.18)

The contrast between the Spec 11 and Spec 12 columns of Table B.1 is the structure-transfers-but-calibrations-do-not result: imposing Williams' Australian calibrations ($\psi_0 = 0.20$, $\psi_1 = 0.93$, $\gamma_{\text{IFA}} = 0.022$) via the iterative fixed point collapses the error-correction speed from -0.448 to a statistically insignificant -0.030 , and flips it to the wrong sign and statistically significant $+0.041$ ($t = 2.03$) pre-COVID. Spec 10 (Williams-prior calibrated; `australia_full_results.csv`) independently reproduces the collapse ($\lambda = -0.048$, $t = -0.78$).

The structural-parameter inference (the implied $\gamma = \text{OLS}/|\lambda|$ with delta-method and bootstrap confidence intervals, for both Spec 6 and Spec 11) is in `australia_gamma_inference.csv`. In the Spec 6 frame every Williams wealth value lies inside the (wide) 95 per cent interval, and so does zero – consistency by imprecision, the low-power reading of the companion Wald non-rejection ($\chi^2(6) = 7.55$, $p = 0.27$). In the Spec 11 frame the inference cuts both ways: the intervals exclude Williams' net-liquid m.p.c. (0.159) and peak housing m.p.c. (0.0488) while including his illiquid-financial m.p.c. (0.022); see §9.4.

1.16 Appendix C: Diagnostic battery

The full diagnostic output is reproduced from `australia_full_diagnostics.csv` and `australia_precovid_diagnostics.csv` (consolidated in `australia_all_diagnostics.csv`). For each specification and sample these report the number of observations, the residual standard error (per cent), adjusted R^2 , Durbin–Watson, Breusch–Godfrey serial-correlation tests at lags 1 and 4, the heteroscedasticity test in two forms (full and events-excluded), the Chow break test together with a `chow_method` column recording which test variant produced it (`sctest` where the standard implementation is estimable, `manual_common_coef` where the sub-sample design matrices are rank-deficient and a common-coefficient F-test is substituted, NA where neither applies), the RESET functional-form test, the Schwarz/BIC and the log-likelihood.

C.1 Heteroscedasticity is structural, not event-driven. Every full-sample specification flags `het_diagnosis = "structural"`: the heteroscedasticity test rejects homoscedasticity even after the COVID, GFC and policy-event dummies are excluded (for Spec 11 the full-sample heteroscedasticity $p \approx 1.7 \times 10^{-8}$ and the events-excluded $p \approx 2.0 \times 10^{-9}$). This is why Newey–West HAC standard errors are used for every reported t -statistic throughout the

paper. The pre-COVID Spec 11 residuals are, by contrast, well behaved (heteroscedasticity $p = 0.52$; no serial correlation, $\text{AR}(1)$ $p = 0.32$, $\text{AR}(4)$ $p = 0.47$; RESET $p = 0.75$).

C.2 The multi-break Chow battery is reported in `australia_chow_battery.csv` and `australia_chow_battery_spec11.csv`, with the structural-break tests in `australia_breaks.csv`. On the selector-preferred Spec 3, the 1995Q1, 2000Q1 and 2008Q3 breaks are not rejected (Chow $p = 0.98, 0.73$ and 0.91) while the 2020Q1 break is strongly rejected ($p \approx 8.9 \times 10^{-16}$ – the COVID break). On the faithful Spec 11 the 1995Q1 and 2000Q1 breaks are stable ($p = 0.84, 0.15$), the 2008Q3 break is rejected at 5 per cent ($p = 0.017$), and the 2020Q1 test is not estimable in its standard form (the `sctest` failed note in the CSV). The Bai–Perron supF statistic (169.96, $p = 0$, breakpoint dated 2019Q1 – the algorithm placing the single permitted break just ahead of the COVID collapse) and the recursive-residual CUSUM ($p = 0.97$, stable) together locate the dominant instability at the COVID episode. This break, and not a deeper instability, is what inflates the full-sample λ relative to the COVID-controlled value.

C.3 The net-liquid restriction test (`australia_nla_restriction_test.csv`) records `restriction_accepted = TRUE` in all six rows (Specs 4, 5, 6 across full and pre-COVID samples): the data cannot distinguish separate liquid- asset and debt MPCs, which supports the NLA aggregation used in the faithful LIVES form.

C.4 Cointegration screen (`australia_cointegration.csv`). The Engle–Granger ADF residual test fails to reject the null of no cointegration for *every* estimable specification at the correct MacKinnon critical values (e.g. Spec 11 ADF -3.13 against a -5.47 critical value), while the Johansen rank-1 trace test passes for all estimable forms. No single-equation form clears the Engle–Granger screen. Three specifications are not run through it (coint reported NA): Spec 9, whose Kalman-CCI interaction columns are not committed to the static regression frame, and Specs 10 and 12, whose calibrated-offset long runs make the static EG regression inapplicable.

1.17 Appendix D: Reproducibility

The full reproducibility kit accompanies this paper. The estimation pipeline can be run in three execution modes, all driven from the project root (`/Users/davidstephan/Documents/consmodelling`):

1. Cold rebuild (downloads + estimates).

```
Rscript Australia/R/australia_consumption_model.R
```

Reads the cached ABS workbooks, fetches the RBA series live, rebuilds master, saves `australia_model_dataset.rds`, and then runs estimation. Required when the `data_raw/workbooks` change.

2. Fast re-estimate from the cached RDS (no data work; bit-identical).

```
Rscript Australia/R/run_estimation_from_rds.R
```

Loads the pre-built RDS and runs only the estimation script. This is the canonical reproduction command – it is the path exercised by continuous integration and produces bit-identical results.

3. Offline replay from a portable master CSV (no downloads, hand-editable).

Rscript Australia/R/load_master_from_csv.R

Loads data_raw/master_data.csv, reconstructs master, and runs estimation. Useful without internet access, for a frozen version-controllable snapshot, or to patch a known data error by hand. The portable CSV has been regenerated alongside the current pipeline run (194 rows $\times$ 120 columns, with the earlier vintage's +10 per cent house-price splice step at 2011Q3 eliminated — the boundary now shows a clean -1.78 per cent QoQ log change) and is consistent with the RDS path; an earlier draft's caveat about a stale CSV snapshot no longer applies.

The estimation script writes all fourteen specifications (Spec 1–12, plus the long-history Spec 6b and the RBA-burden Spec 7b) to australia_full_results.csv / australia_precovid_results.csv (consolidated in australia_all_results.csv) and the matching diagnostics files, including the faithful LIVES headline (Spec 11) and the Williams-calibration-imposed specification (Spec 12); all robustness, placebo, counterfactual and comparison outputs cited in the paper are written to the matching CSVs in outputs/.

The current pipeline run adds the following committed artefacts, all cited in the text: the Spec 11 sample-treatment vector (australia_spec11_variants.csv) and ogive-robustness run (australia_spec11_ogive_robustness.csv); the deployed CCI path and its interaction-correlation matrix (australia_cci_williams_series.csv, australia_cci_interaction_corr.csv, australia_cci_williams_path.png); the deployed-protocol knot placebo (australia_williams_knot_placebo_deployed.csv, australia_williams_knot_placebo_deployed_verdict.csv); the IV diagnostics (australia_iv_diagnostics.csv, australia_iv_diagnostics_spec11.csv) alongside the Spec 11 variants of the robustness battery (australia_iv_robustness_spec11.csv, australia_chow_battery_spec11.csv, australia_drehmann_robustness_spec11.csv, australia_joint_pi_robustness_spec11.csv, australia_scaled_income_robustness_spec11.csv, australia_williams_income_robustness_spec11.csv); the Spec 11 long-run decomposition (australia_longrun_contributions_spec11.csv, australia_longrun_decomposition_spec11.png); and the two-specification structural-gamma inference (australia_gamma_inference.csv, Spec 6 and Spec 11 rows). The companion LIVES outputs cited here — the joint calibration Wald test (williams_calibration_wald.csv), the SUR residual correlations (lives_sur_2eq_resid_corr.csv) and the joint knot-survival table (lives_joint_cci_survival.csv) — are committed under LIVES/outputs/.

The environment is pinned with renv (R 4.5.3); the dependency manifest is in DESCRIPTION and the full transitive closure — including the car and systemfit packages used for the Wald and SUR tests — is locked in renv.lock. Restore it with renv::restore(). The kit also ships GitHub Actions continuous integration and a testthat unit-test suite. The repository is hosted publicly at <https://github.com/DavidASephan/ConsModelling>.